Reflection
&
Refraction

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Reflection & Refraction

Reflection of Light

INTRODUCTION

Geometrical optics, or ray optics in a model of optics that describes the light propagation in terms of rays. A ray of light gives the direction of propagation of light. When light meets a surface which separates two media, reflection and refraction takes place. An image may be formed due to this.

'Optics is the branch of physics that studies the behavior and properties of light, including its interactions with matter and construction of instruments that are used to detect it'.

OPTICS

Geometrical or Ray Optics	Wave Optics	Quantum Optics
(Describes light propagation in terms of rays) Geometrical optics, treats light as a collection of rays that travel in straight lines and bend when they pass through another medium or undergoing reflection from a surface.	Deals with wave characteristics of light. Explains phenomena of interference and diffraction etc.  (Fig. 1) (Butterfly, thin-film Interference-wave optics)	Deals with the application of quantum mechanics to optical system. It uses both wave and particle nature of light.  (Fig. 2) Experiment carried out by scientist in quantum optics lab.

Fundamental Laws of Geometrical Optics

Geometrical optics is based on five fundamental laws.

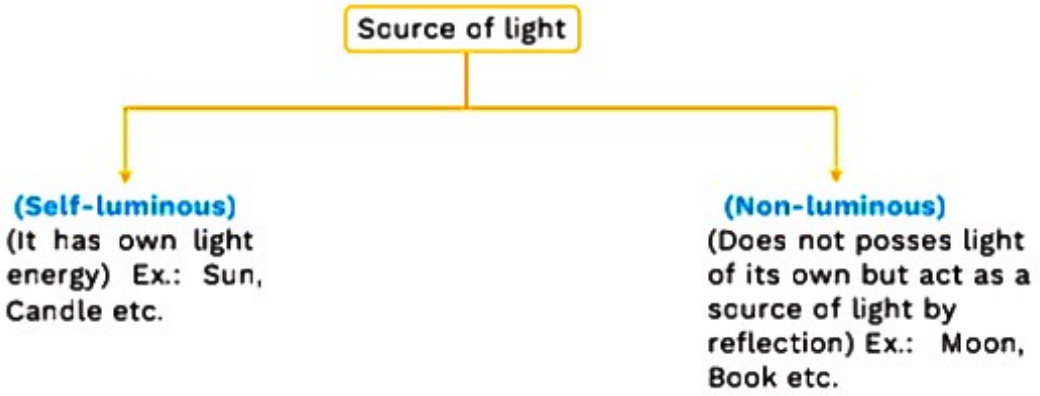
(1) **First Law:** It states that light propagates in straight lines in homogeneous media. It is called rectilinear propagation of light.

(2) **Second Law:** It states that rays do not disturb each other upon interaction. It is calledthe law of independence of light.

(3) **Third Law:** According to this law, a ray of light, after suffering a number of reflections and refraction, has its path back to the source. It is called the law of reversibility.

(4) **Reflection of Light:** The laws of reflection states that the incident ray reflected back to the same medium after striking a surface.

(5) **Refraction of Light:** A light ray refracts whenever it travels at an angle into a medium of different refractive index. The change in speed results in a change in direction.

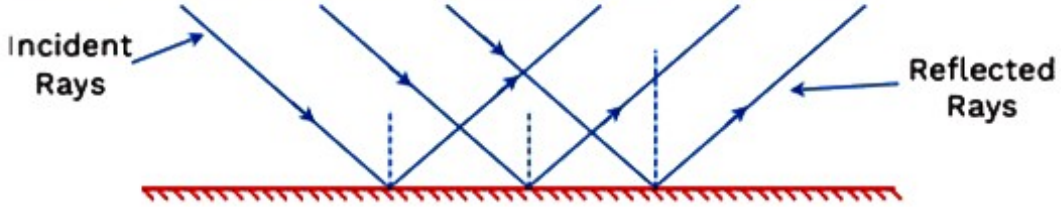


Ray The straight-line path along which the light travels between two points in a homogeneous medium called Ray.



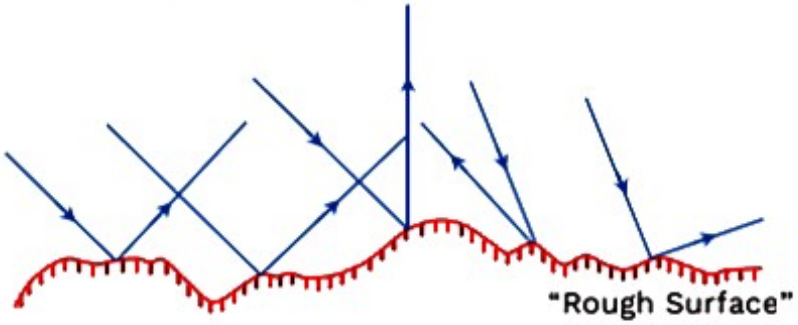
The law of Reflection of Light: When light rays incident at the boundary of two media such as air and glass or glass and water, a part of light bounces back into the same medium. This phenomenon of bouncing back of the light is called a reflection of light. There are two types of reflection.

(1) **Regular Reflection:** When reflection takes place from a perfect plane surface, then rays remain parallel after reflection. It is called regular reflection.



Smooth Surface

(2) **Irregular or Diffused Reflection:** When the surface is rough, light is reflected from the bits of its plane surfaces in different directions. This is called irregular reflection. Diffused reflection enables us to see an object from any position.



Diffuse Reflection

LAWS OF REFLECTION

(1) **First Law:** Incident ray reflected ray and normal at the point of incidence, all lie in the same plane.

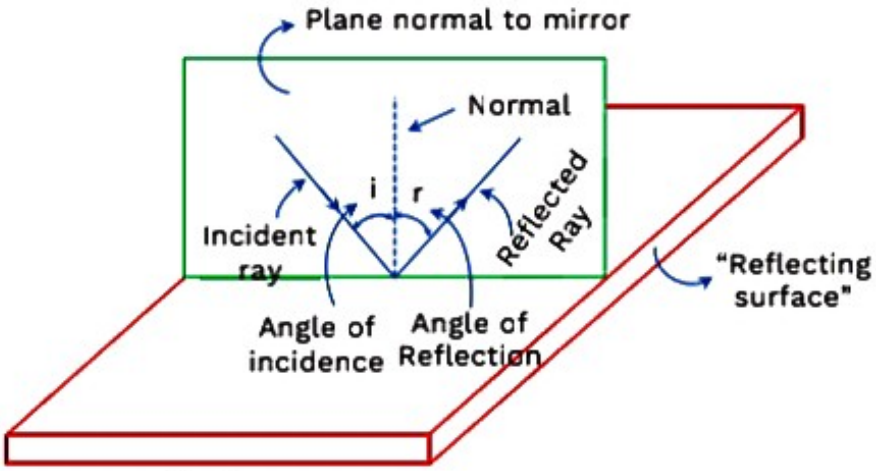
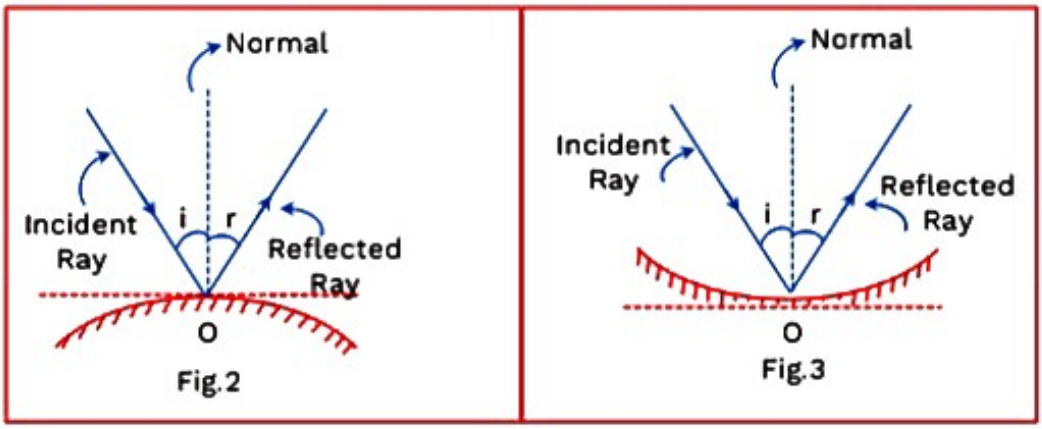
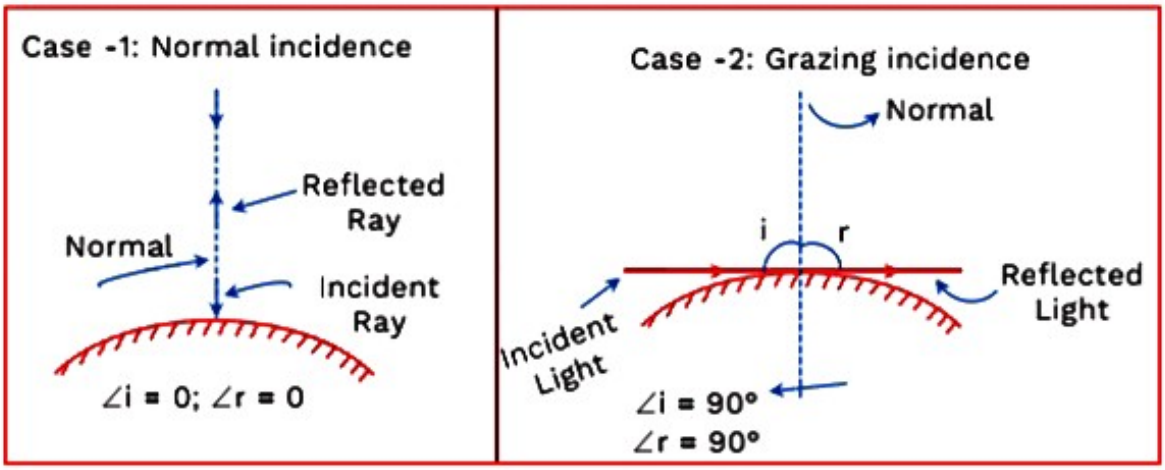


Fig. 1

(2) **Second Law:** The angle of reflection is equal to the angle of incidence.

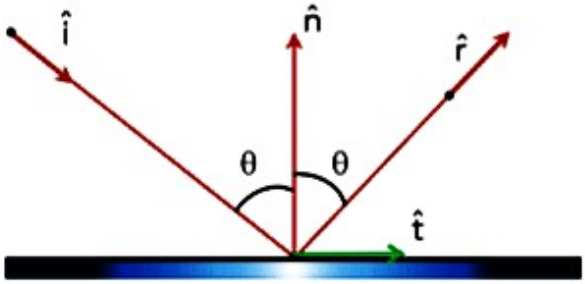


SPECIAL CASES:



Laws of Reflection in Vector Form :

Laws of reflection can be redefined with the help of vector algebra by considering unit vectors in the direction of incident rays, reflected rays and normal to the boundary.



$$\hat{i} = (\sin\theta)\hat{t} - (\cos\theta)\hat{n} \quad \text{... (1)}$$

$$\hat{r} = (\sin\theta)\hat{t} + (\cos\theta)\hat{n} \quad \text{...(2)}$$

$$|\hat{i}| = |\hat{r}| = |\hat{n}| = |\hat{t}| = 1$$

Subtracting equation (1) from equation (2), we get

$$\hat{r} = \hat{i} + (2\cos\theta)\hat{n} \quad \text{...(3)}$$

Also, we know that

$$\hat{i} \cdot \hat{n} = |\hat{i}||\hat{n}|\cos(180 - \theta) = -\cos\theta \quad \text{...(4)}$$

Substituting equation (4) in equations (3), we get

$$\boxed{\hat{r} = \hat{i} - 2(\hat{i} \cdot \hat{n})\hat{n}}$$

Laws of reflection in vector form

Q. A light ray is incident on a plane mirror along a vector $a\hat{i} + b\hat{j} - c\hat{k}$. The normal on incidence point is along $\frac{\hat{i} + \hat{j}}{\sqrt{2}}$. Find a unit vector of the reflected ray.

Sol: The unit vector in the direction of the incident ray

$\hat{e}_i = \dfrac{a\hat{i} + b\hat{j} - c\hat{k}}{\sqrt{a^2 + b^2 + c^2}}$ and we are given the unit vector in the direction to normal

$\hat{n} = \dfrac{\hat{i} + \hat{j}}{\sqrt{2}}$.

The unit vector along the reflected ray,

$$\boxed{\hat{e}_r = \hat{e}_i - 2(\hat{e}_i.\hat{n})\hat{n}}$$

$$\therefore \quad \hat{e}_r = \frac{a\hat{i} + b\hat{j} - c\hat{k}}{\sqrt{a^2 + b^2 + c^2}} - 2\left[\frac{a + b}{\sqrt{a^2 + b^2 + c^2}}\right] \times \frac{1}{\sqrt{2}} \times \left(\frac{\hat{i} + \hat{j}}{\sqrt{2}}\right)$$

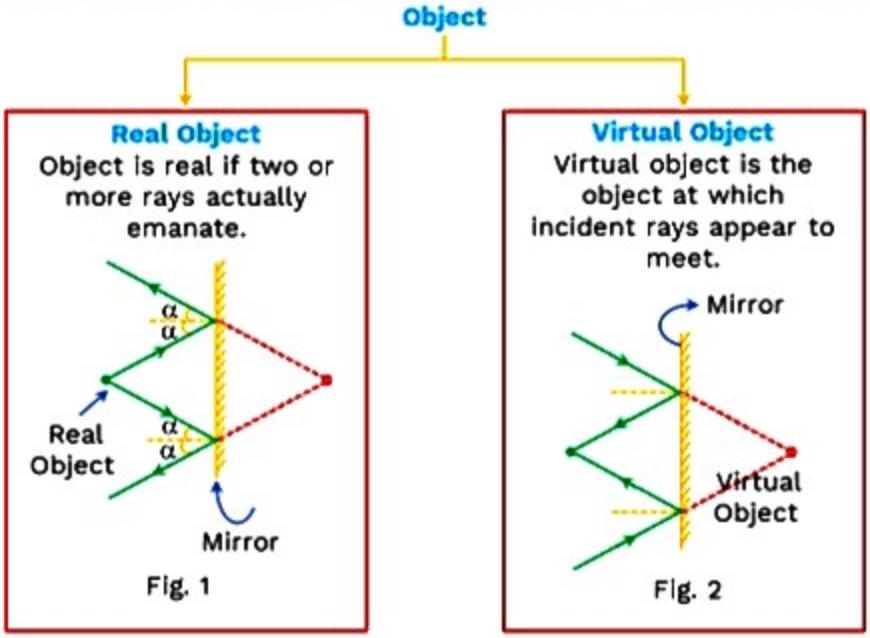
$$\Rightarrow \quad \hat{e}_r = \frac{\left[-b\hat{i} - a\hat{j} - c\hat{k}\right]}{\sqrt{a^2 + b^2 + c^2}}$$

Nature of Object and Image

Ray optics primarily deals with determining the position and nature of the image formed when an object is placed in front of an optical element.

Object :

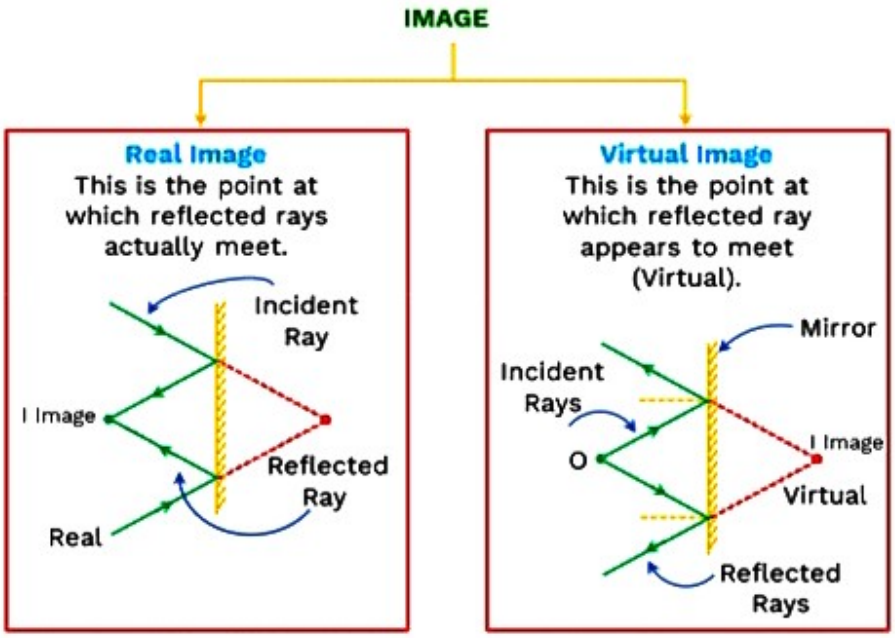
"An object is a source of light rays that are incident on an optical element."



Note In general, and we can say that incident rays are responsible for object formation (This is helpful in ray diagram construction).

Image:

"An image is the point of convergence or apparent point of divergence of rays after they interact with a given optical element."



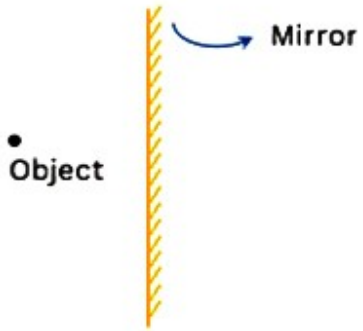
Note: Reflected rays are responsible for image formation.

Plane Mirror - A plane mirror is a mirror with a flat reflecting surface.

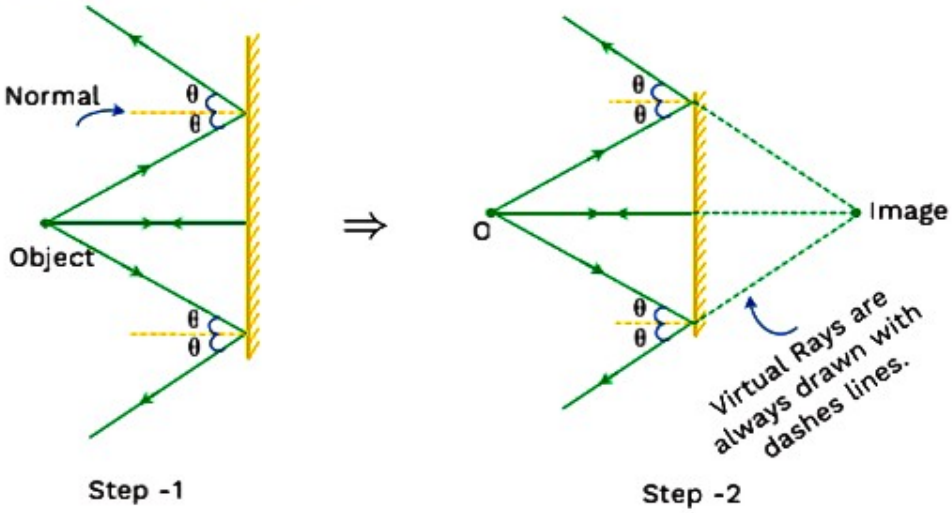
For incident light rays, the angle of reflection equals the angle of incidence.

Image Formation by Plane Mirror
Minimum two rays are required for image formation.
Use laws of reflection for image formation.

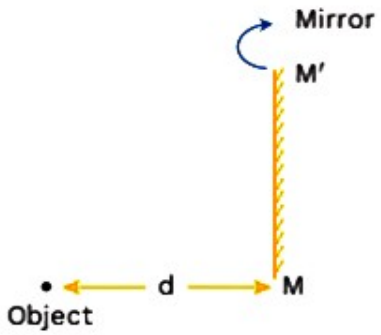
Case-1 Image formation if the object is placed in front of a mirror.



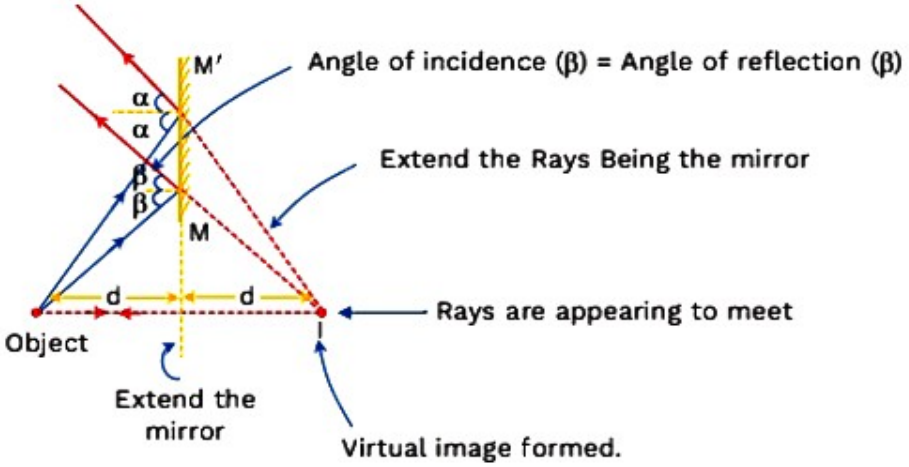
Use the following steps for image formation:



Case-2 Image formation if the object is placed below or above the mirror.



Use the following steps for image formation in this case:



Properties of an image formed by a plane mirror

(1) Image distance is equal to object distance from the mirror.

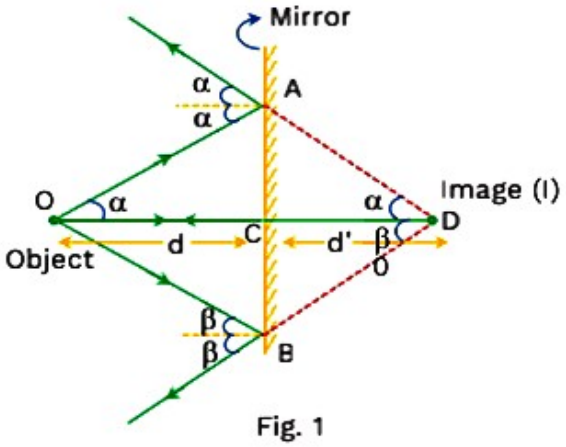
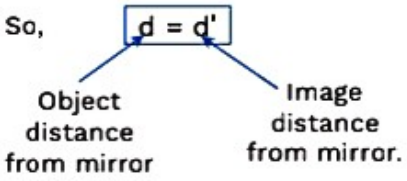


Fig. 1

In Δ OCA and Δ DCA,
∠ AOC = ∠ADC = α.
∠OCA = ∠DCA = 90°.
Side AC is common.
So, Δ OCA ≅ Δ DCA ...(by ASA rule)
So, side, OC = CD ⇒ d = d'



(2) Image is laterally inverted.

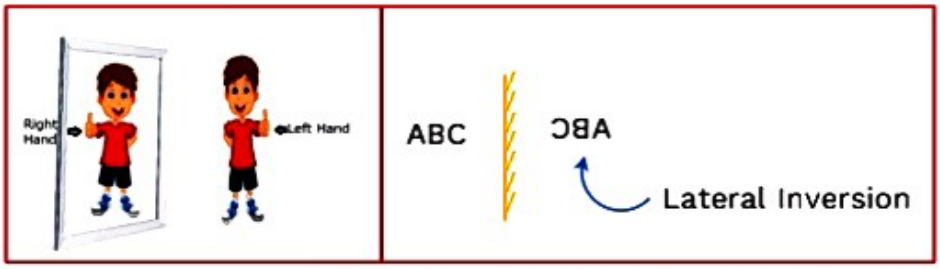
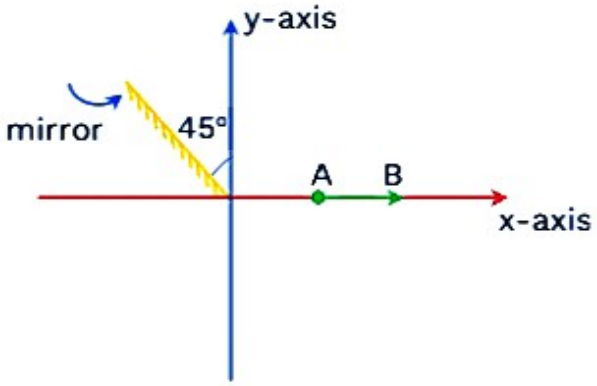


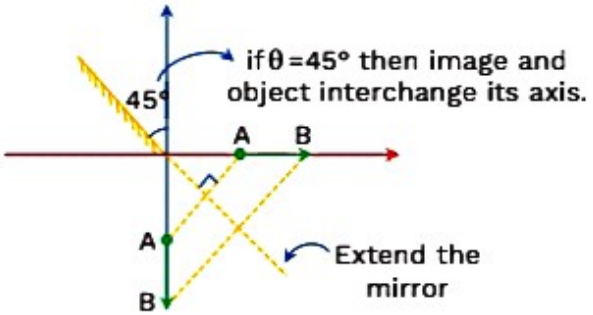
Fig. 2

(3) Nature of Image Nature of an image is always opposite to that of an object.
Real Object → Virtual Image
Virtual Object → Real Image

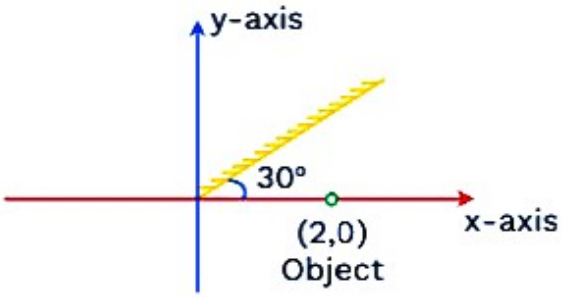
Q. **Draw a ray diagram of an image of given object (AB).**



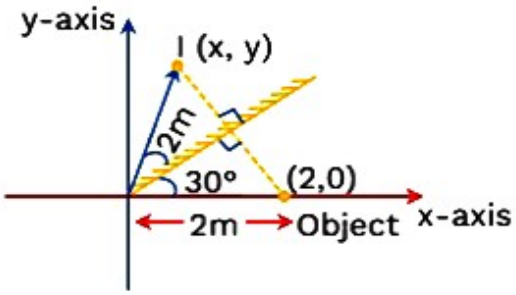
Sol:



Q. **Find the co-ordinate of image.**



Sol:



$$x = 2\cos(30° + 30°) = 2\cos 60° = 2 \times \frac{1}{2} = 1$$

$$y = 2\sin(30 + 30°) = 2\sin 60° = 2 \times \frac{\sqrt{3}}{2} = \sqrt{3}$$

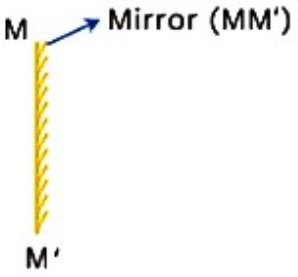
Coordinates of the image are $\left(1, \sqrt{3}\right)$.

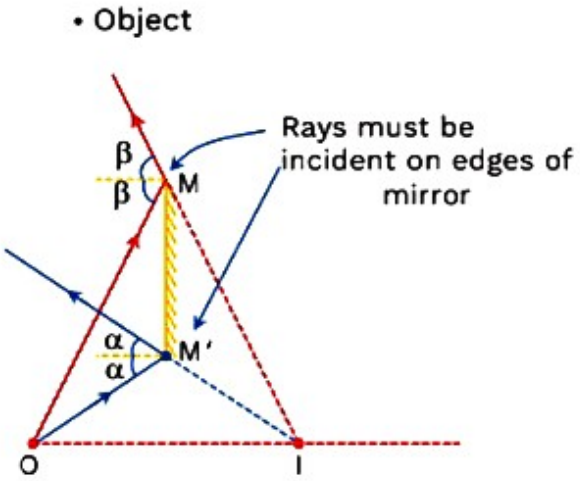
Field of view of an object

It is the region where diverging rays from objects or images are present. If our eyes are present in the field view, then only we can see the image.

Steps to find out field view of an object.

Suppose a point object 'O' is placed in front of a plane mirror as shown.





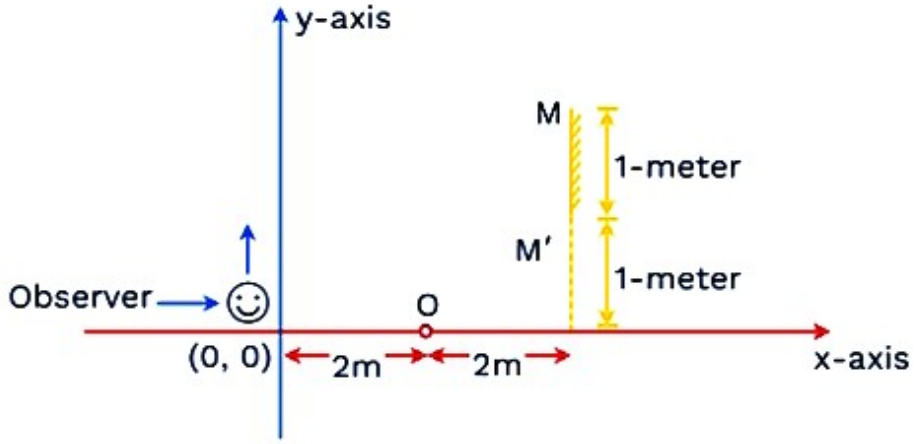
Important Points

(1) Image formation does not depend on the size of the mirror.

(2) Whether we can see the image of an object formed due to mirror or not (field of view) depends on the size of mirror.

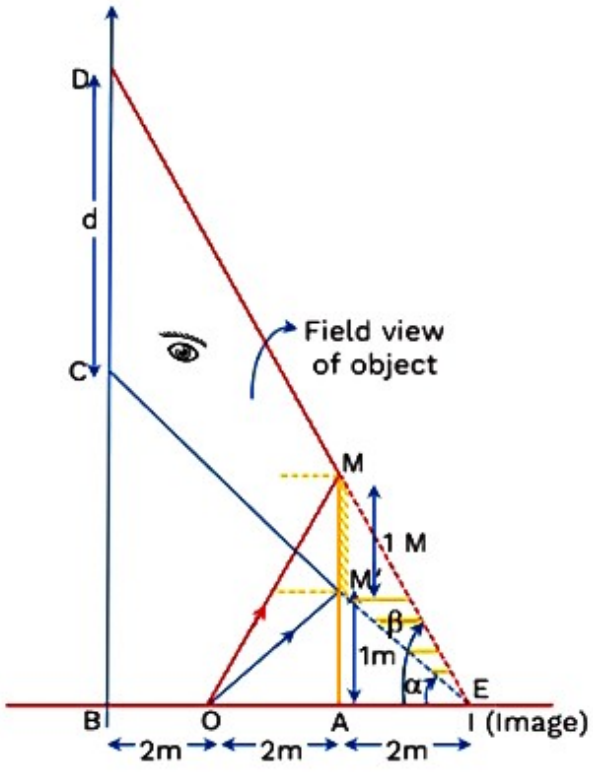
(3) Whatever the size of the mirror, an image is always formed, but visibility depends on the size.

Q.



Find out the distance for which the observer will be able to see the image of an object.

Sol: This problem is based on the field of view concept. Draw the ray diagram and use basic Geometry.



Concept In the field view region, an observer can see an object.

Now,

Δ EAM' $\cong$ EBC

$$\tan\alpha = \frac{AM'}{AE} = \frac{BC}{BE} \Rightarrow \frac{1}{2} = \frac{BC}{6}$$

$$\boxed{BC = 3m}$$

Δ EAM $\cong$ EBD

$$\tan\beta = \frac{AM}{AE} = \frac{BD}{BE} \Rightarrow \frac{2}{2} = \frac{BD}{6}$$

$$BD = 6m$$

So, $d = BD - BC = 6 - 3 = 3$

$$d = 3m$$

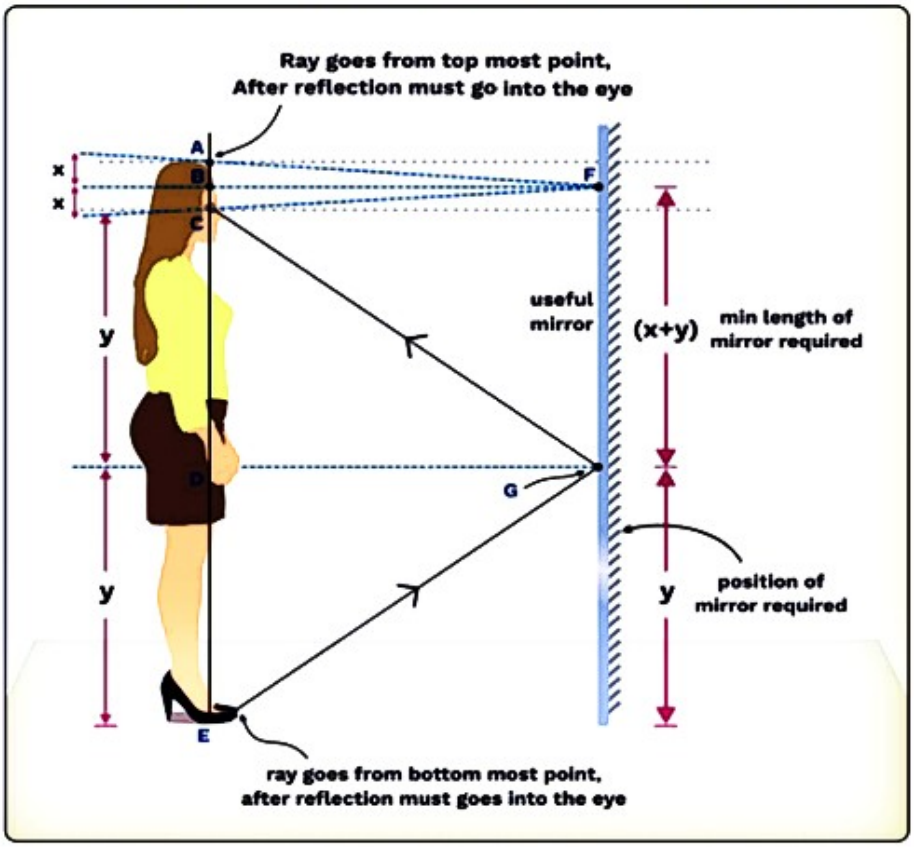
Ans. 3 meter

Minimum Size of a Plane Mirror to See a Complete Image

Case 1:

We know that to see the image of any object. The ray from that object must come to our eye after reflection or refraction from an optical device. (one of the optical devices is a plane mirror). So, if you want to see your complete image, the rays must come from the topmost point of your body (head); also, rays must come from the lowermost of the body (Toe).

From the given figure, use geometry and find the minimum length of the mirror.



In ΔEDG and ΔCDG

$\angle EGA = \angle CGD$ (angle of incidence = angle of reflection)

$\angle GDC = \angle GDE = 90°$

Side DG is common

So; $\Delta EDG \cong \Delta CDG$

This implies that ED = DC = y (let)

Similarly in ΔABF and ΔCBF are similar triangles

So, AB = BC = x (let)

Since total height of object = x + x + y + y = 2 (x + y) = H (let)

From Given figure you are observing that we require that Part of length of mirror from where rays are reflected = length between points G and F = x + y = H/2.

Conclusion:

Minimum length required of a Plane Mirror to See a full Image of an object

$$\frac{H}{2} = \frac{\text{Height of object}}{2}$$

$$\text{Position of mirror required} = \frac{\text{Height of eye from lower most point of object}}{2}$$

Important Points

(a) In order to see the full image of the woman, the mirror is positioned such that the lower edge of the mirror is at half of the height of the eyes of women from the ground.

(b) Minimum size is independent of the distance between women and mirrors.

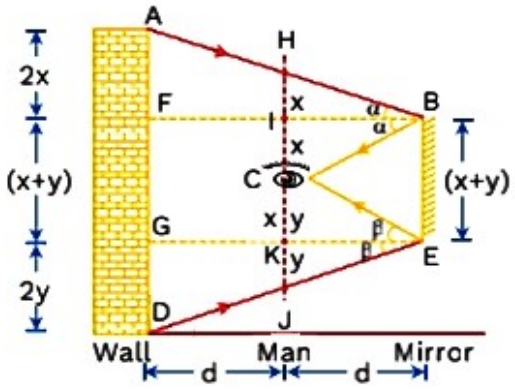
Q. **Let your height is 120 cm, and your eye is 20 cm below the topmost point of the body. Find out the minimum length of mirror required to see the full image, also find the position of the mirror from the surface.**

Ans. (1) Minimum height of the mirror required $= \frac{120}{2} = 60\,\text{cm}$

(2) Position of mirror from surface $= \frac{\text{Height of eye from surface}}{2}$

$= \frac{120 - 20}{2} = \frac{100}{2} = 50\,\text{cm}$ (from the surface)

Case-2: Minimum length of a mirror to see the full image of a wall if the wall is situated behind the man and the man is standing in the middle of the mirror and the wall.



In triangles HBI and IBC, let HI = IC = x. Now, in triangles HBI and ABF, we have

$$\tan\alpha = \frac{AF}{BF} = \frac{HI}{BI} \Rightarrow \frac{AF}{2d} = \frac{x}{d}$$

$\Rightarrow$ AF = 2x

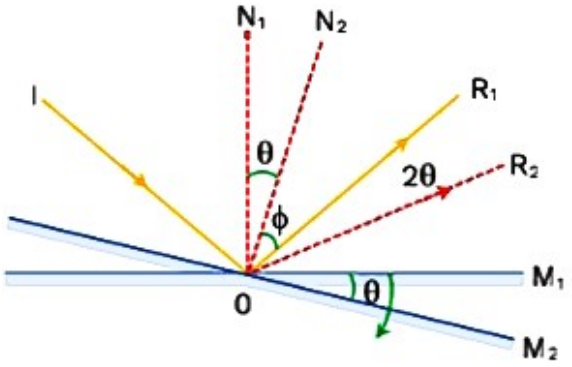
Similarly, if, CK = KJ = y, then DG = 2y. Now, we observe that the height of the wall is 3(x + y) while that of the mirror is (x + y).

Conclusion:

The minimum length of mirror required in the given situation is = H/3 = height of wall /3.

EFFECT OF ROTATION OF PLANE MIRROR

Case-1: Ray fixed mirror rotated.



(a) If the incident ray is fixed and mirror rotates through an angle θ, then the reflected ray rotates through the angle 2θ in the same sense as the mirror rotates.

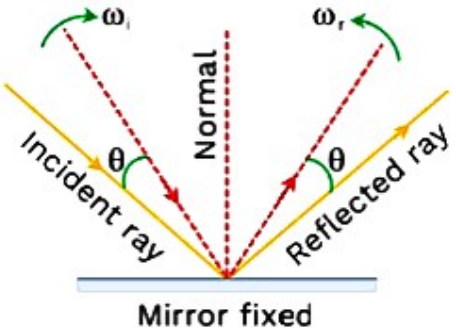
(b) The angular velocity and angular acceleration of the new reflected ray becomes twice as that of the mirror.

$\omega_r = 2\omega_m$ and $\alpha_r = 2\alpha_m$

where ω_r = angular velocity of reflected ray

ω_m = angular velocity of the mirror

Case-2: Mirror fixed ray rotated.



If we fixed the mirror, the incident rays are rotated by some angle, then the reflected rays are also rotated by the same angle but in the opposite sense that of incident rays. Incident ray is fixed, and mirror rotates through an angle θ,

Then, $\omega_i = -\omega_r$

Where,

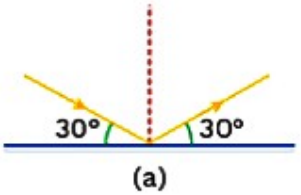
ω_i = angular velocity of the incident ray

ω_r = angular velocity of reflected ray

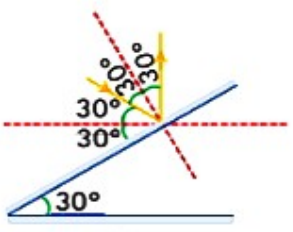
A negative sign represents that the direction of angular velocities is opposite to each other.

Q. **Suppose A light ray is incident on a plane mirror at an angle of 30° with the horizontal direction. Find the angle with horizontal must a plane mirror should be placed in its path so that it becomes vertically upwards after reflection?**

Sol: To make the reflected ray vertical, it should be rotated at an angle of 60°. So, the mirror should be titled by 60°/2 = 30°, as shown in Fig. (b)

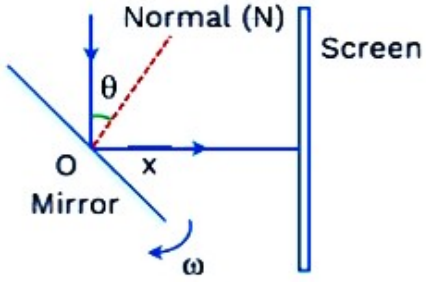


(a)

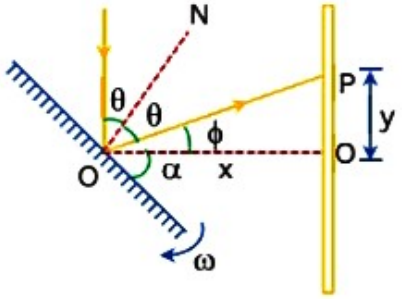


(b)

Q. **A plane mirror hinged at point O, as shown in the figure, is free to rotate in a vertical plane, the point O at a distance x from a long screen placed in front of the mirror. A light ray incident vertically downward is reflected by the mirror at O so that a bright spot is formed at the screen. At the instant, the angle of incidence is θ, and the mirror is rotating clockwise with constant angular velocity ω. Calculate the speed of the spot at this instant of time.**



Sol: Let the bright spot shown in figure with symbol P. Let the distance of point P from O' be y at this instant . Then according to given condition, we need to calculate $\frac{dy}{dt}$



From the figure

$\theta + \theta + \phi = 90°$ ...(1)

$\theta + \phi + \alpha = 90°$ ...(2)

$\Rightarrow \alpha = \theta$

$\Rightarrow \phi + 2\theta = 90°$

$\Rightarrow \phi + 2\alpha = 90°$

$\Rightarrow \frac{d\phi}{dt} + 2\frac{d\alpha}{dt} = 0$

$\Rightarrow \frac{d\phi}{dt} = -2\frac{d\alpha}{dt} \Rightarrow \frac{d\phi}{dt} = 2\omega$

So, the angular speed of the reflected ray is double the angular speed of the mirror.

Since, $y = x\tan\phi$

$\Rightarrow \quad \frac{dy}{dt} = x\sec^2\phi\frac{d\phi}{dt}$

Since $\left|\frac{d\phi}{dt}\right| = 2\omega$

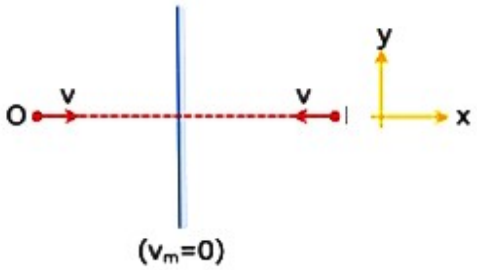
$\Rightarrow \quad \left|\frac{dy}{dt}\right| = \left(x\sec^2\phi\right)(2\omega)$

So, the speed of the spot is $\left|\frac{dy}{dt}\right| = 2x\omega\sec^2\phi$

Since: $\psi = 90 - 2\theta \Rightarrow \boxed{\frac{dy}{dt} = 2x\omega\,\text{cosec}^2(2\theta)}$

VELOCITY OF IMAGE IN A PLANE MIRROR

Case-1: Object moving along the normal to the plane mirror, which is at rest. All velocities are measured w.r.t. ground frame.



The Velocity of object with respect to the mirror

is $\vec{v}_{Om} = v\hat{i}$

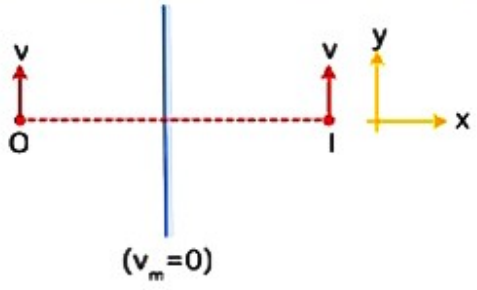
The Velocity of the image with respect to the mirror is

$\vec{v}_{Im} = -v\hat{i}$

The Velocity of the object with respect to the image is

$$\boxed{\vec{v}_{OI} = \vec{v}_{Om} - \vec{v}_{Im} = (2v)\hat{i}}$$

Case-2: Object moving parallel to the plane of mirror (at rest)



The Velocity of object w.r.t. mirror is

$\vec{v}_{Om} = v\hat{j}$

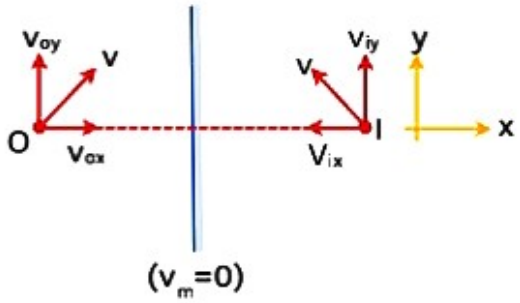
The Velocity of image w.r.t. mirror is

$\vec{v}_{Im} = v\hat{j}$

The Velocity of object w.r.t. image is

$$\vec{v}_{OI} = \vec{0}$$

Case-3: Object moving neither along the normal nor along the parallel to the plane mirror (at rest).



Clearly, we observe this case to be a combination of Case 1 and Case 2. So, here

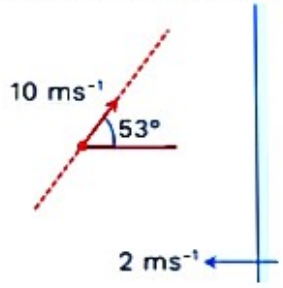
$$(v_{OI})_x = 2v_{ox} \text{ and } (v_{OI})_y = 0$$

The Velocity of the object with respect to image along the x-axis

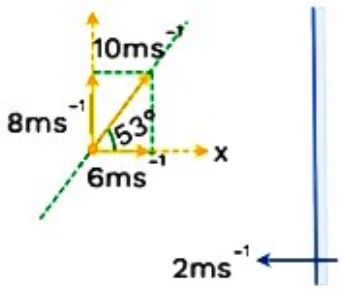
$\Rightarrow \quad (V_{OI})_x = 2V_{ox}$

and along the y-axis $\Rightarrow (V_{OI})_y = 0$

Q. **Find the velocity of the image of a moving particle shown in figure.**



Sol:



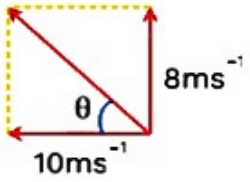
The component of velocity of image perpendicular to mirror is given by,

$$\vec{V}_I = 2\vec{V}_m - \vec{V}_O$$

$$\Rightarrow \quad (V)_{Ix} = 2(-2) - (6) = -10\,ms^{-1}$$

For a component of velocity of image parallel to the mirror,

$$(V)_{Iy} = 8\,ms^{-1}$$

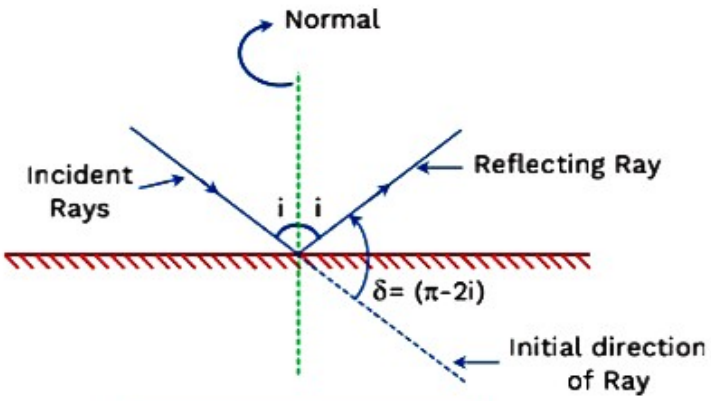


$$\therefore \quad \text{Velocity of image } (V_I) = \sqrt{(V_I)_x^2 + (V_I)_y^2}$$

$$\Rightarrow \quad V_I = \sqrt{100 + 64} = \sqrt{164}\,ms^{-1} \text{ and } \theta = \tan^{-1}\left(\frac{4}{5}\right)$$

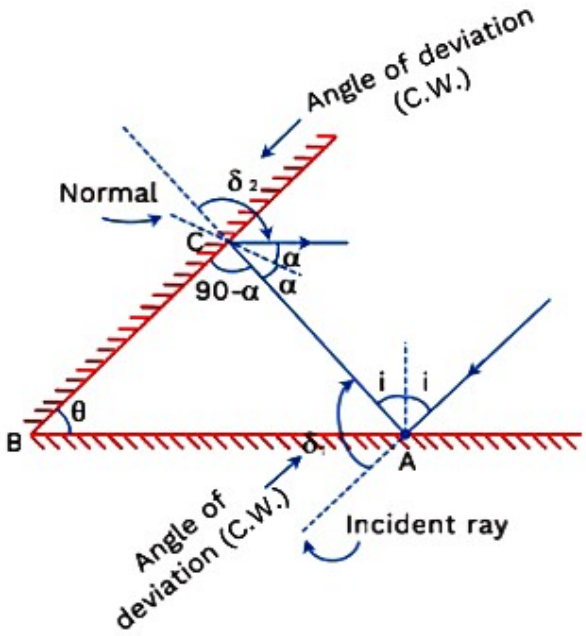
Angle of Deviation

Case-1: Single Mirror



Angle of deviation = (π-2i)

Case-2: Two inclined mirrors



The total angle of deviation,

$$\delta_{net} = (\delta_1 + \delta_2)$$

Clockwise

$\delta_{Net} = (\pi - 2i) + \delta_2$ ...(1)

$\delta_2 = \pi - 2\alpha$

In $\Delta ABC \Rightarrow$ $\theta + 90 - \alpha + 90 - i = 180$

$\Rightarrow \alpha = (\theta - i)$ ...(2)

$\delta_2 = \pi - 2(\theta - i)$

$\delta_2 = \pi - 2\theta + 2i$

$\delta_2 = 2i - 2\theta + \pi$ ...(3)

From equation (1) and (3)

$\delta_{Net} = (\pi - 2i) + 2i - 2\theta + \pi$

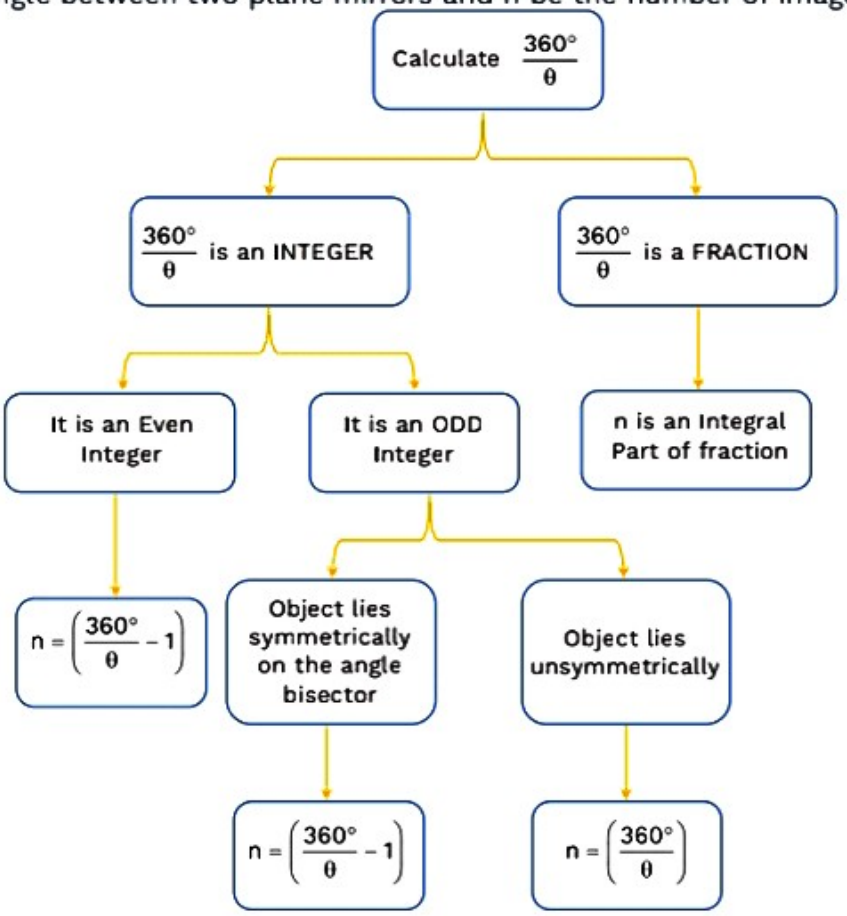
$\delta_{Net} = [(2\pi - 2\theta) \leftarrow \text{clockwise}]$

Conclusion

$$\delta_{Net} = \begin{Bmatrix} (2\pi - 2\theta) & \text{Clock - wise} \\ 2\theta & \text{Anti - clock wise} \end{Bmatrix}$$

NUMBER OF IMAGES FORMED BY TWO INCLINED MIRRORS

Let 0 be the angle between two plane mirrors and n be the number of images formed.



Position of Image formed by two plane mirrors

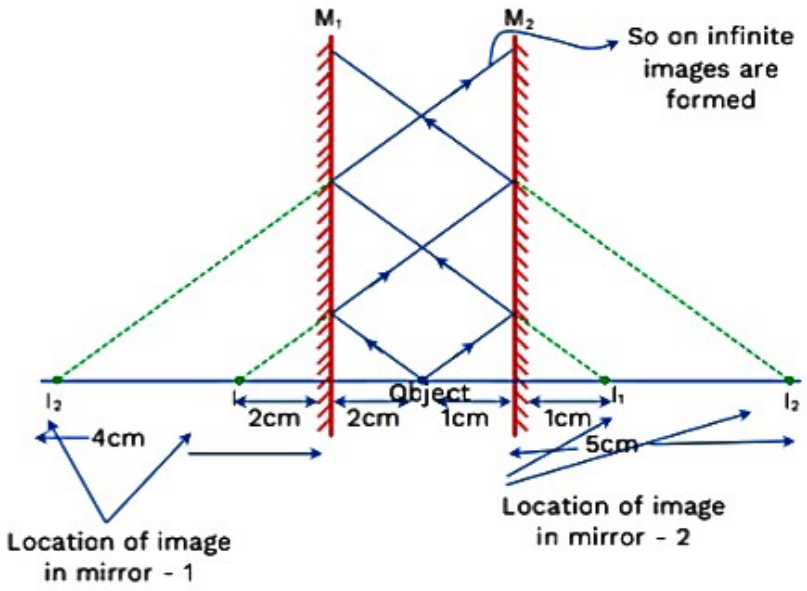
Let object placed of 1 cm from mirror –2.

$$n = \frac{360}{\theta} - 1;\ \theta = 0^\circ$$

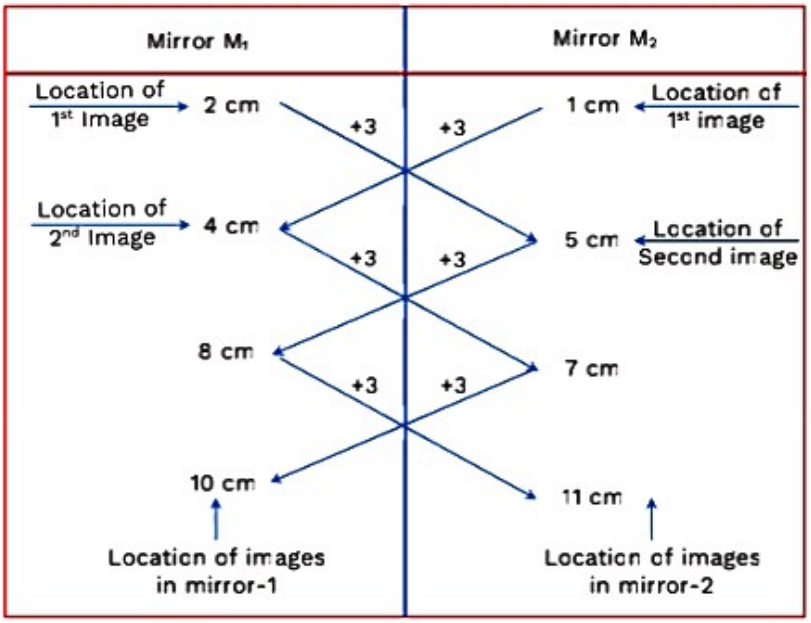
$$n = \infty - 1$$

$$\boxed{n = \infty}$$

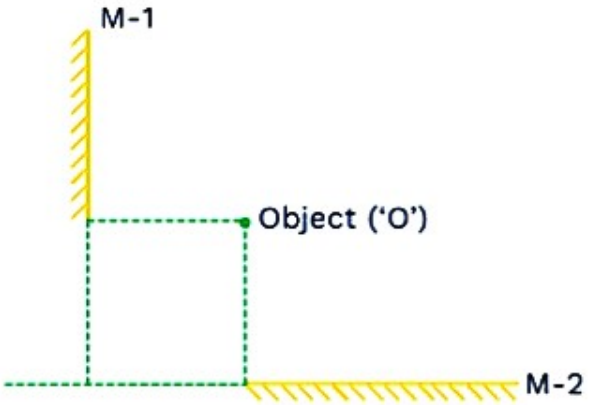
Method-1: By using ray diagram


M1
M2
So on infinite images are formed
I2
I
2cm
2cm
Object
1cm
1cm
I1
I2
4cm
5cm
Location of image in mirror - 1
Location of image in mirror - 2

Method-2

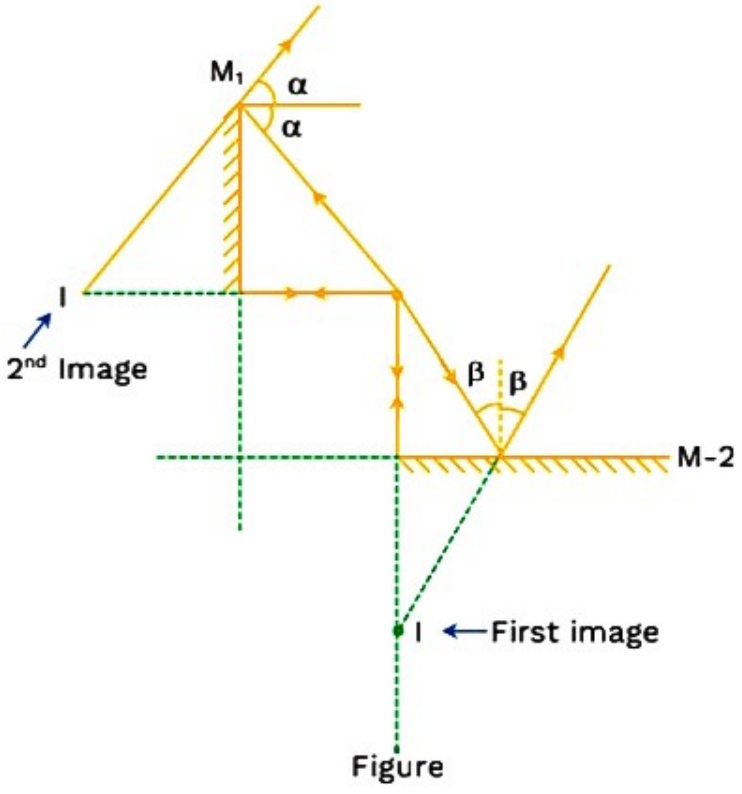

Mirror M1
Mirror M2
Location of 1st Image
2 cm
+3
+3
1 cm
Location of 1st image
Location of 2nd Image
4 cm
+3
+3
5 cm
Location of Second image
8 cm
+3
+3
7 cm
10 cm
11 cm
Location of images in mirror-1
Location of images in mirror-2

Q. **Find the number of images formed by mirror as shown in figure.**



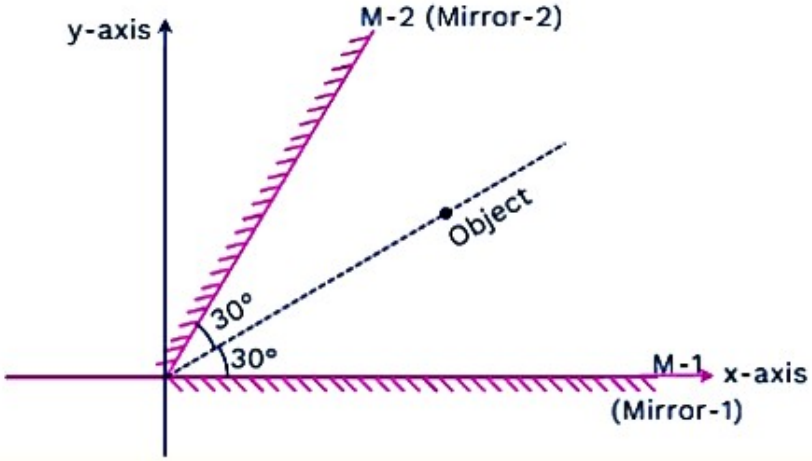
(A) 1 **(B) 2** **(C) 3** **(D) 4**

Sol: Ans. (B)



Figure

Q. **Find number of images formed by two mirrors as shown in figure. Object is placed at angle bisector.**



Sol:

Method-1 By using mathematical formula,

Given angle between two mirrors

$$\theta = 60°$$

$$\Rightarrow \quad \frac{360}{\theta} \Rightarrow \frac{360}{60} = 6 \leftarrow \text{integer value}$$

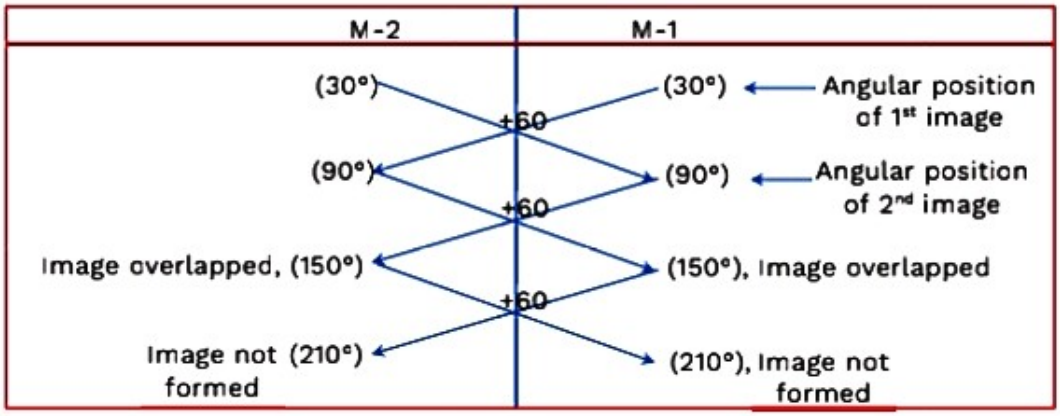
Since it is even integer

So, the number of images formed is equal to

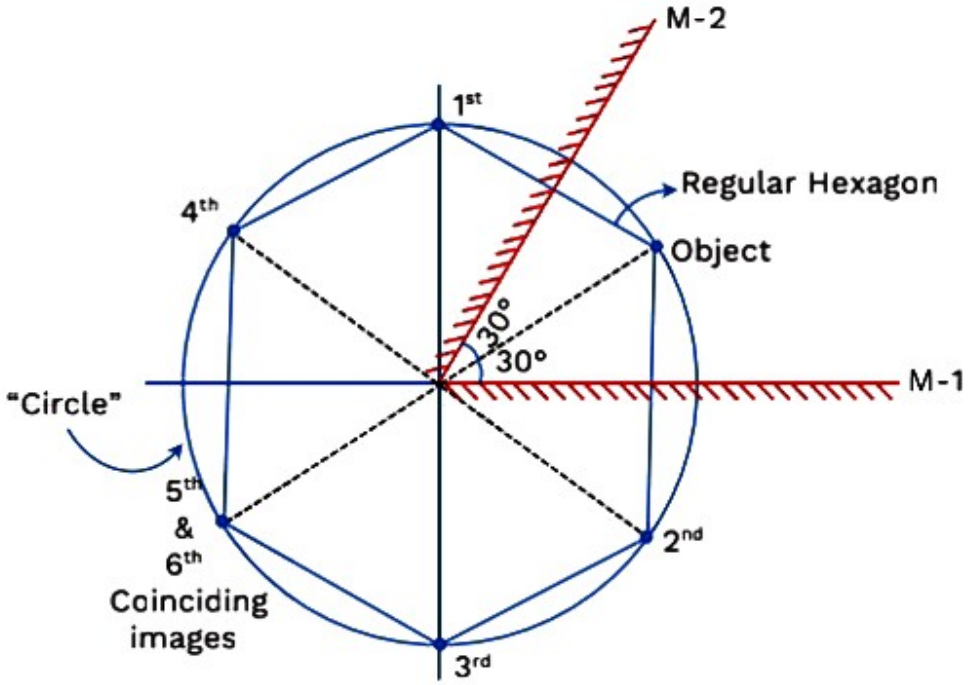
$$n = \frac{360}{\theta} - 1 = 6 - 1$$

$$= 5 \text{ (Images).}$$

Method-2



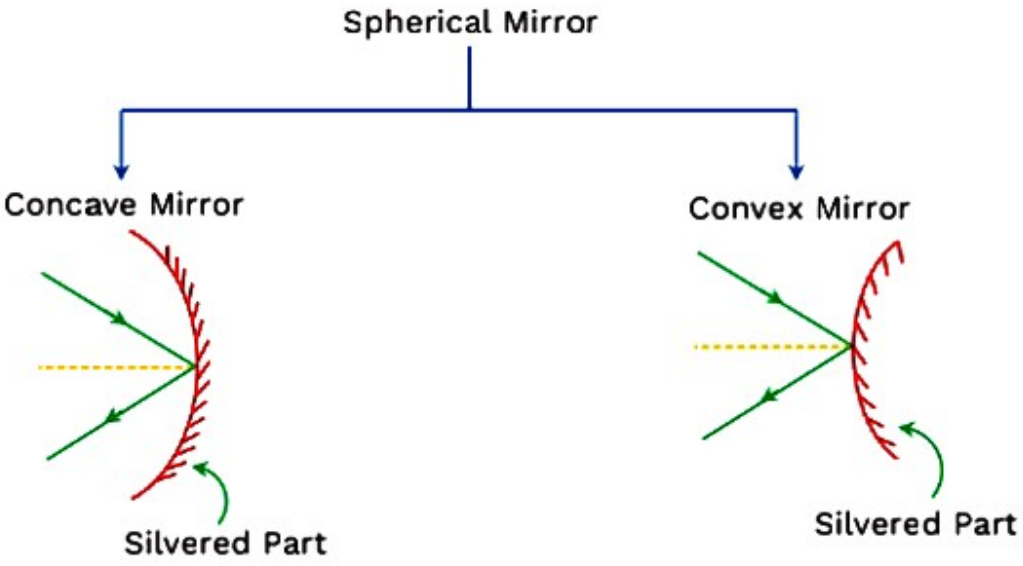
Ray Diagram



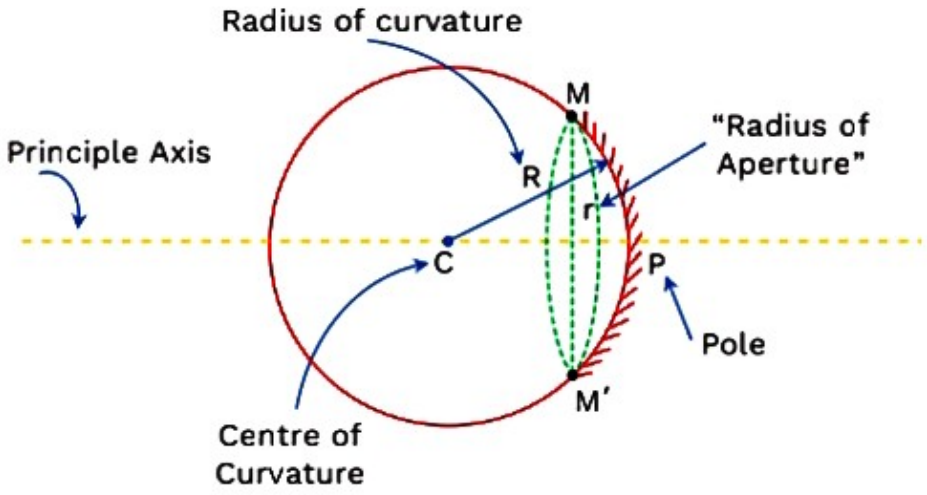
Reflection from Curved Surfaces

Small curved reflecting surfaces can be considered to be a part of a spherical surface. Hence, such surfaces are called spherical mirrors.

Depending upon the surface silvered, these are of two types-



Important Terms and Definition



(1) Pole Centre P of the surface of the mirror

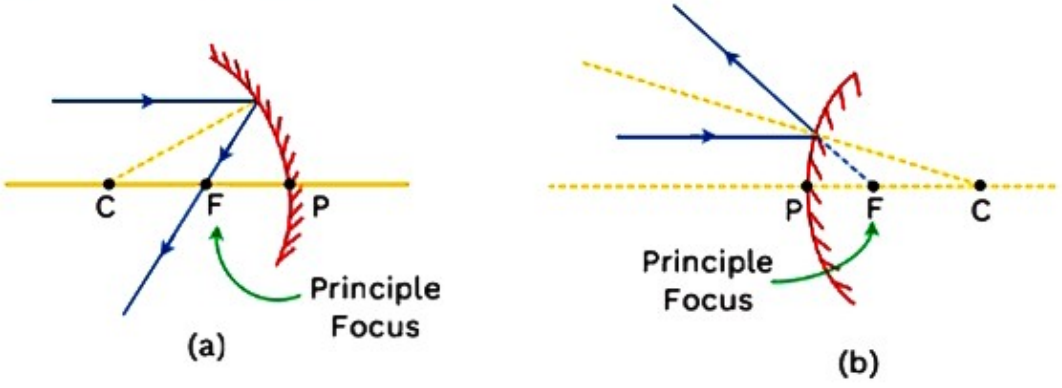
(2) Centre of Curvature Centre C of the sphere

(3) Radius of Curvature Radius 'R' of the sphere.

(4) Principle Axis Line PC, Joining the pole and the center.

(5) Radius of Aperture Half of distance (MM') between the extremities of the mirror surface. This tells up how big the mirror is.

(6) Focus (F) When a narrow beam of rays of light, parallel to the Principal axis and close to it (paraxial rays), is incident on the surface of a mirror, the reflecting beam is found to converge to or appears to diverge from a point on the principal axis. This point is called the focus.

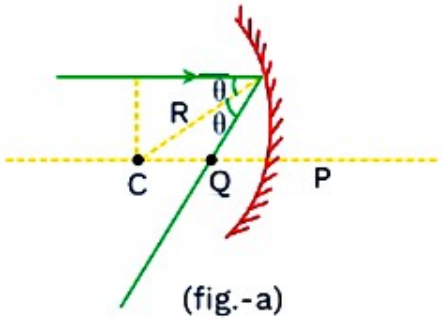


Important Point

(a) Focus is only defined for paraxial rays.

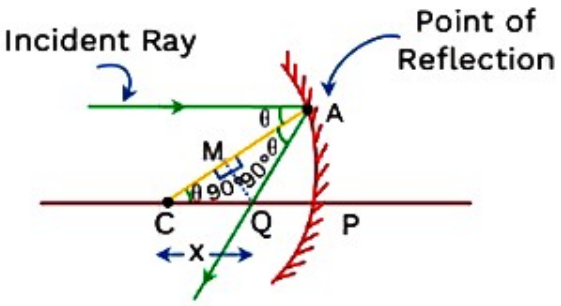
(b) Focal Length of Mirror (PF)- It is the distance between the pole and the principal focus (F).

Q. **Calculate the distance CQ if incident light ray is parallel to principal axis incident at an angle 'θ'. Also, find distance CQ if θ tends to 0.**



(fig.-a)

Sol:



Let CQ = x ; In ΔCMQ and ΔQMA,

∠QCM = θ = ∠QAM

Side MQ is common

∠CMQ = ∠QMA = 90°

So ASA Congruent Triangles.

Δ CMQ ≅ QMA

So CM = MA = R/2

Now, by using Δ CMQ,

$$\cos\theta = \frac{CM}{CQ} = \frac{(R/2)}{x}$$

$$\Rightarrow \boxed{x = \frac{R}{2}\sec\theta}$$

Also $PQ = R - x = R - \frac{R}{2}\sec\theta$

$$\boxed{PQ = R\left(1 - \frac{\sec\theta}{2}\right)}$$

If $\theta \to 0° \Rightarrow \sec\theta \approx 1$

So $$\boxed{x \cong \frac{R}{2} \;\&\; PQ = R\left(1-\frac{1}{2}\right) = \frac{R}{2}}$$

Note

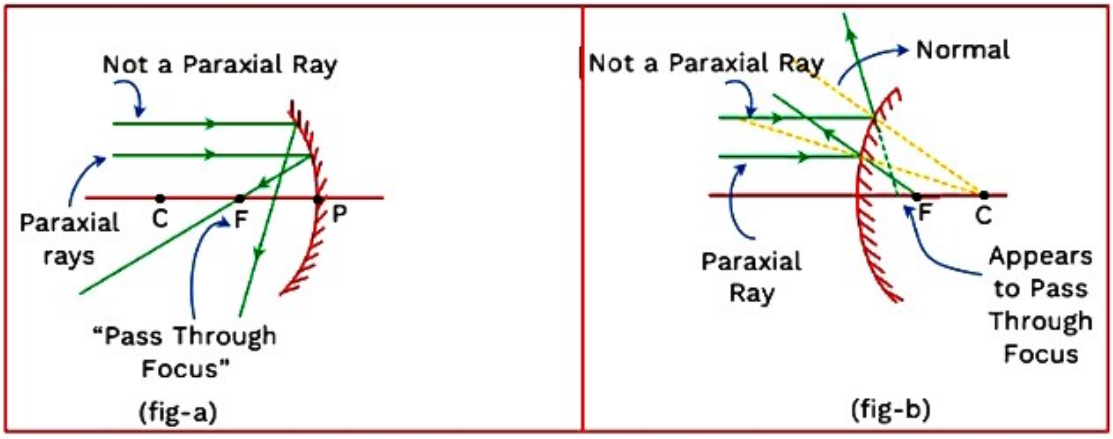
(1) if $\theta \to 0°$ (upto 5°), this is for paraxial ray then PQ = f = focal length of mirror is equal f = R/2.

(2) Focal length of mirror f = R/2 only defined for paraxial rays.

Paraxial Rays

(1) Incident rays that are very close to the principal axis are called paraxial rays.

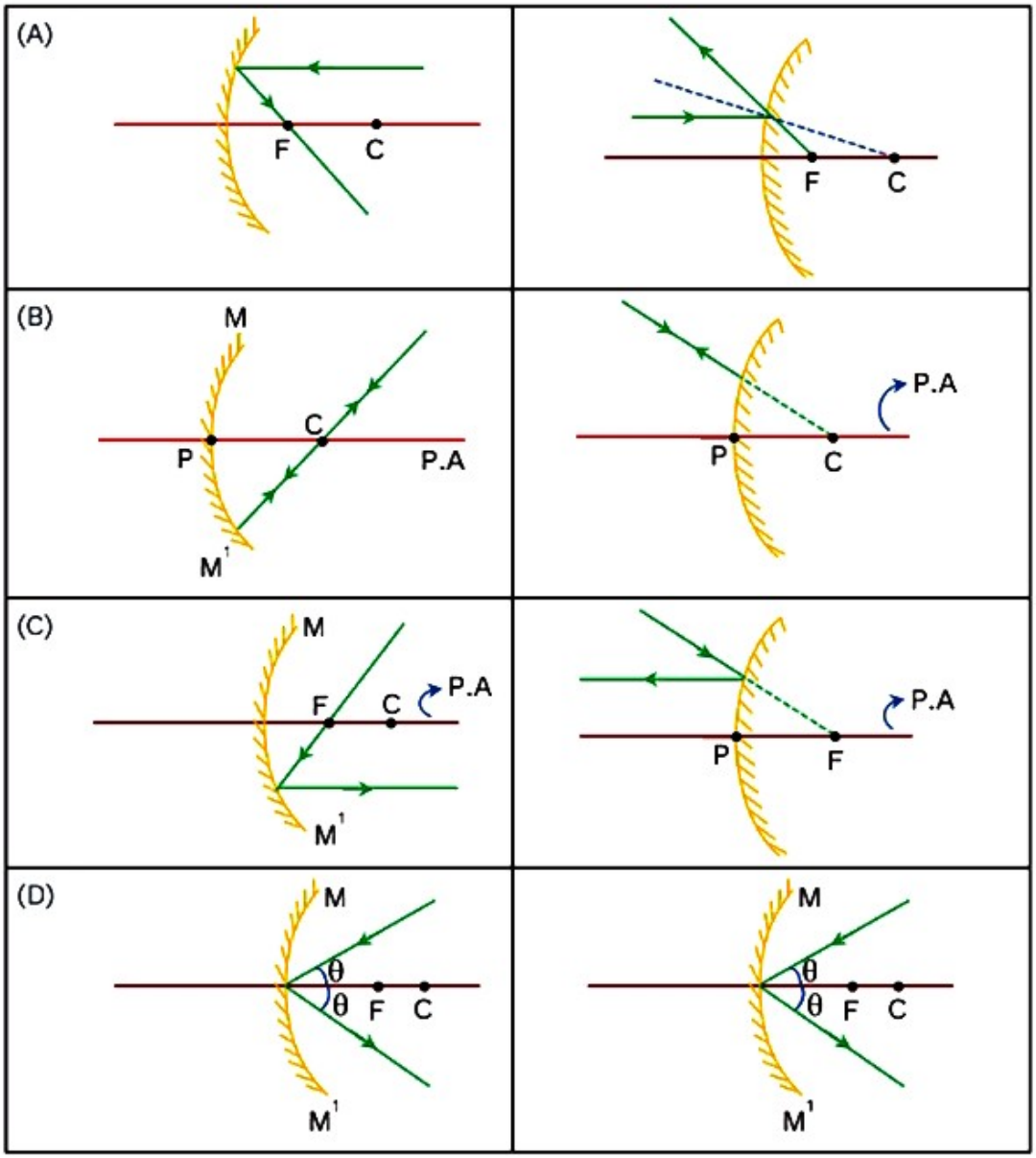
(2) Only paraxial rays incident on spherical mirror parallel to principal axis pass through focus after reflection.



(3) All mathematical calculations are only for paraxial rays.

(4) Focus of a spherical mirror is only defined for paraxial rays.

Standard Paraxial rays which are used for Image Formation



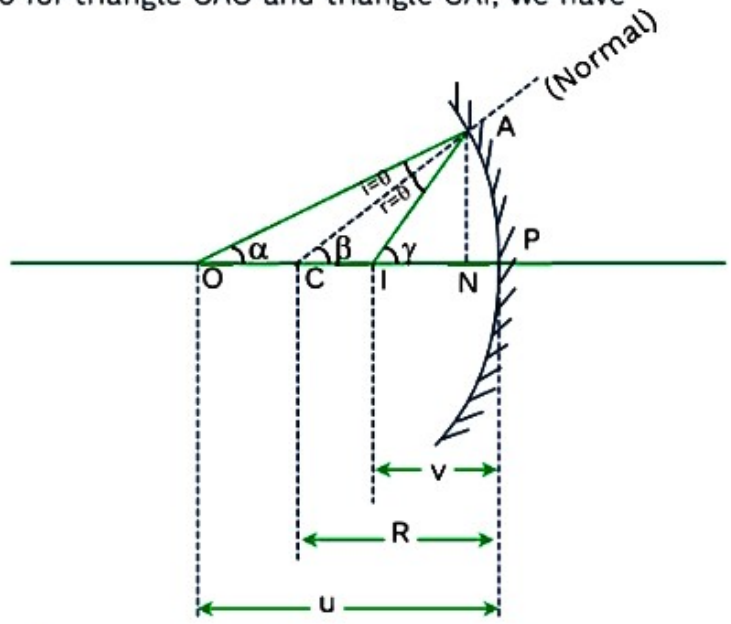
DERIVATION OF MIRROR FORMULA

Case (a) For Concave Mirror

Let's consider a point object 'O' placed on the principal axis of a concave mirror. A ray of light, incident on point 'A' at an angle of incidence 'i' on the mirror makes an angle 'r' with the normal as shown in the figure.

From the laws of Reflection, we know that i = r. Further to find the location of the image, let us take another ray along the principal axis so that it hits the mirror normally at the point 'P' to reverse its path and meet the other ray at point 'I'. This point of intersection of the two rays happens to be the place where the image is formed.

Since from geometry, we know that in a triangle, external angle equals the sum of internal opposite angles, so for triangle CAO and triangle CAI, we have



"If rays are paraxial, θ is very small then NP is negligible."

In ΔOAC →

$\alpha + \theta = \beta$ (Exterior angle) ... (1)

In ΔCAI →

$\beta + \theta = \gamma$ ...(2)

Now subtract equation (2) from equation (1)

$(\alpha + \theta) - (\beta + \theta) = \beta - \gamma$

$\alpha + \theta - \beta - \theta = \beta - \gamma$

$\alpha + \gamma = 2\beta$ ...(3)

Since α, β, γ are very small →

So $\alpha \approx \tan\alpha$

$\beta \approx \tan\beta$

$\gamma \approx \tan\gamma$

From equation – (3)

$$\tan\alpha + \tan\gamma = 2\tan\beta$$

Now use ΔOAN , ΔCAN & ΔIAN

$$\frac{AN}{NO} + \frac{AN}{NI} = 2 \times \frac{AN}{NC}$$

Since rays are paraxial

So point N and P are very close to each other that why we can take approximation →

$NO \approx PO$

$NC \approx PC$

$NI \approx PI$

$$\frac{1}{PO} + \frac{1}{PI} = 2 \times \frac{1}{PC}$$

$$\left.\begin{array}{l} PO = u \\ PC = R \\ PI = v \end{array}\right\} ; \boxed{\frac{1}{u} + \frac{1}{v} = \frac{2}{R}}$$

Mirror Formula →

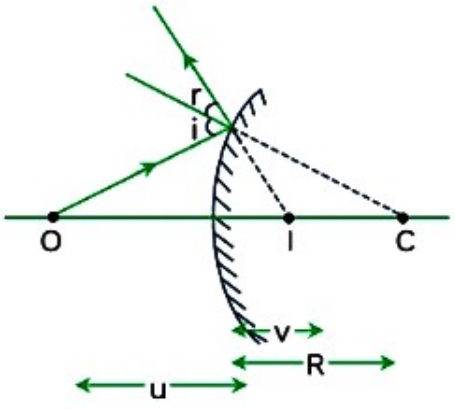
$$\Rightarrow \quad \boxed{\frac{1}{v} + \frac{1}{u} = \frac{2}{R}} \quad \rightarrow \text{Mirror formula in terms of u, v, R.}$$

$$\Rightarrow \quad \frac{1}{v} + \frac{1}{u} = \frac{1}{\frac{R}{2}}$$

$$\boxed{\frac{1}{v} + \frac{1}{u} = \frac{1}{f}} \quad \rightarrow \text{Mirror formula in terms of u, v, f.}$$

Note → Put u, v, R, f with the proper sign.

Case (b) Convex Mirror

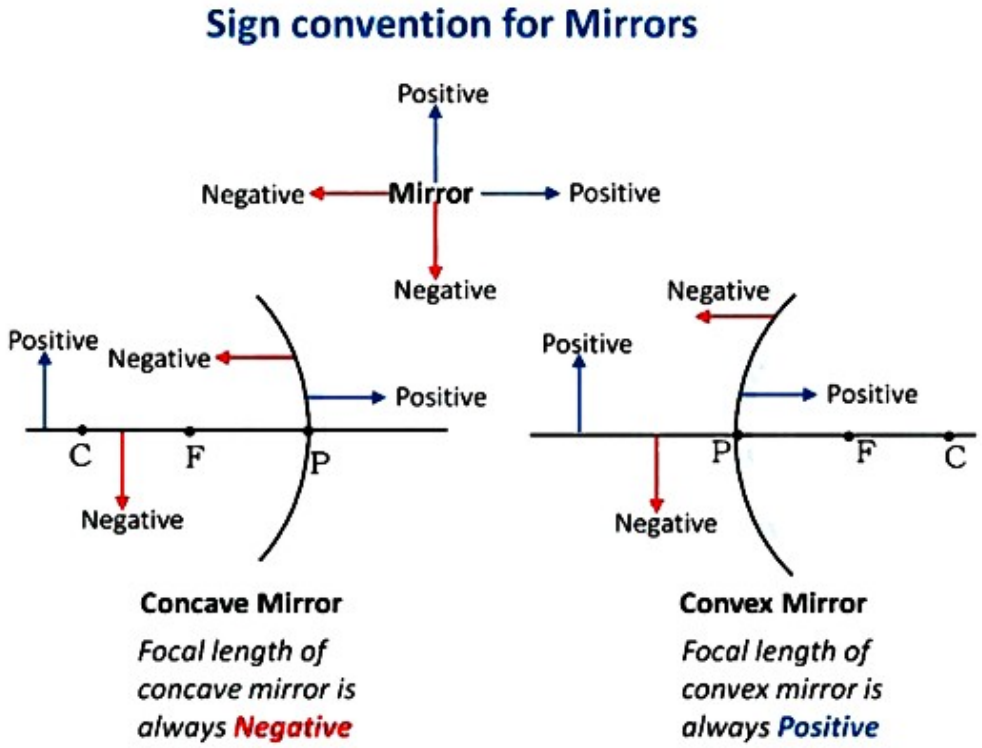


$$\boxed{\frac{1}{v}+\frac{1}{u}=\frac{2}{R}}$$

$\Rightarrow$ $\boxed{\frac{1}{v}=\frac{1}{u}=\frac{1}{f}}$ $\qquad f=\frac{R}{2}$ Put u, v, r, f with the proper sign.

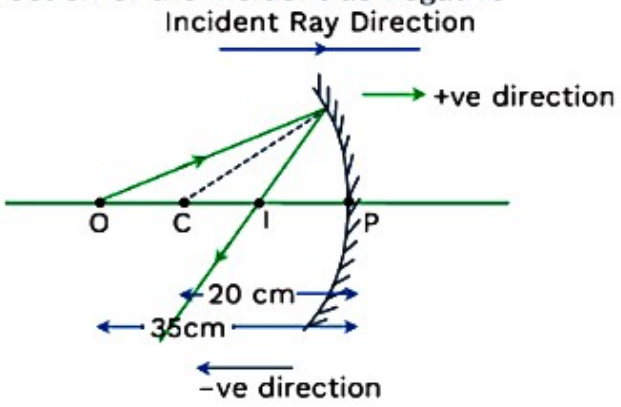
Note Interestingly, and the mirror formula is the same irrespective of the mirror used.

Sign Convection



(a) Origin is placed at the pole (P). All distances are to be measured from the pole (P).

(b) Distance measured in the direction of the incident ray are taken as positive and opposite of the direction of the incident as negative



Suppose Given data for mirror (Concave in this case)

R = 20 cm Obj. distance(u) = 35 cm	By using sign convection we can write u = – 35 cm f = R/2 = –10 cm.

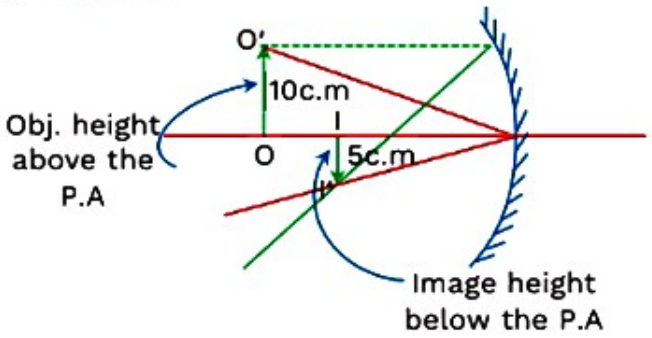
Suppose in the problem we ask to find the location of the image then use mirror formula put u, f with the proper sign but do not use the sign of v (Since this variable we have asked to find)

so; $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$$\boxed{\frac{1}{v}+\frac{1}{-35}=\frac{1}{10}} \qquad ...(1)$$

Now we can easily calculate v with a sign.

(c) Distances above the principal axis are taken as positive, below the principal axis are taken as negative.



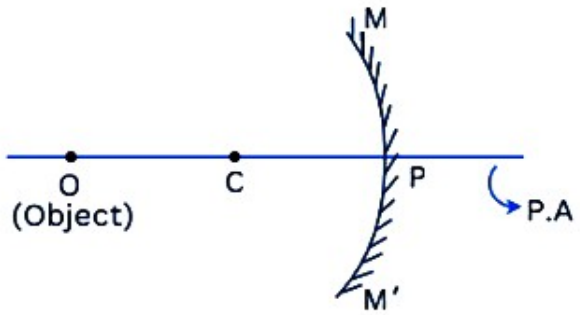
By using sign convection Height of Object (OO') = 10 c.m

Height of Image (II') = – 5 cm

(d) This sign convection is used to find the position and nature (virtual or real, erect, or inverted) of the image formed by the mirror (or Lens).

(e) Object distance is denoted by u, image distance by 'v,' focal length by 'f,' and radius of curvature 'R.'

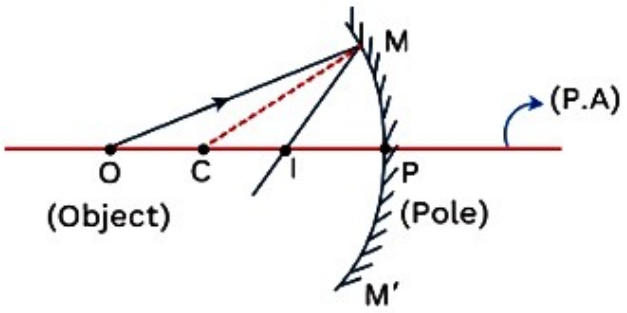
Q



Given
u = 30 cm
R = 20 cm

Find the location of the image. Also, draw a ray diagram.

Sol:



Mirror Formula $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

$f=\frac{R}{2}=\frac{20}{2}=10\text{cm}$

$u = -30$ cm

$f = -10$ cm

$\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \Rightarrow \frac{1}{v}+\frac{1}{-30}=\frac{1}{-10}$

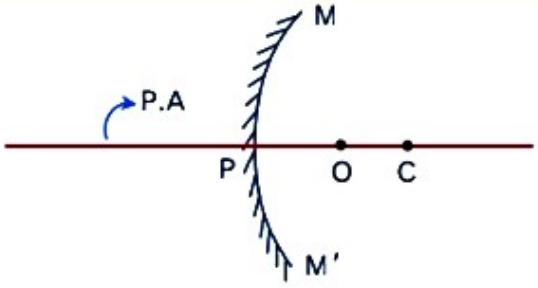
$\frac{1}{u}=\frac{1}{30}-\frac{1}{10}=\frac{1-3}{30}=\frac{-2}{30}=\frac{1}{-15\text{cm}}$

$\boxed{V = -15\,\text{cm}}$

A negative sign indicates an image is formed on the left side of the pole. Also, it is a real image.

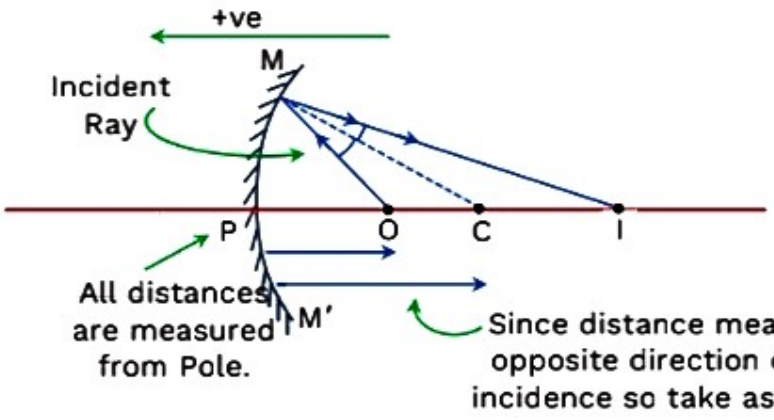
Ans. Location of image 15 cm left of the pole and its real image.

Q



Given **u = 30 cm**
f = 20 cm
Find the location of image also draw ray diagram.

Sol:



$u = -30$ cm

$f = -20$ cm

From Mirror Formula →

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} + \frac{1}{-30} = \frac{1}{-20}$$

We should not use a sign since it is an unknown variable.

$$\frac{1}{v} = \frac{1}{30} - \frac{1}{20} = \frac{2-3}{60} = \frac{-1}{60}$$

$v = -60$ cm

Sign indicates it's real image also formed right side of pole (Opp. to direction of ray incident)

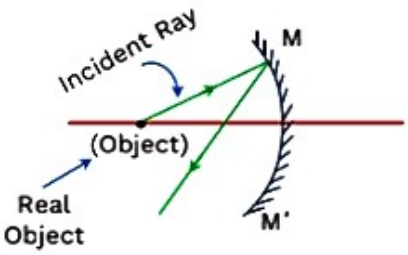
$|v| = 60\,\text{cm}$

Type of Image and Object

Object

Real object

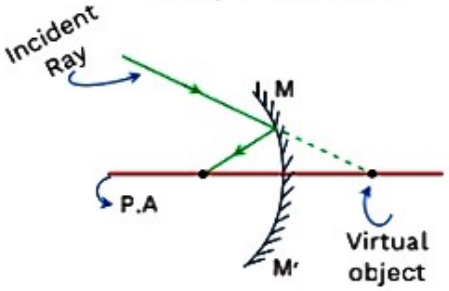
Incident rays are responsible for object formation.



- A Real object is one in which light rays physically emanate from the object

Virtual object

Incident rays are responsible for object formation.

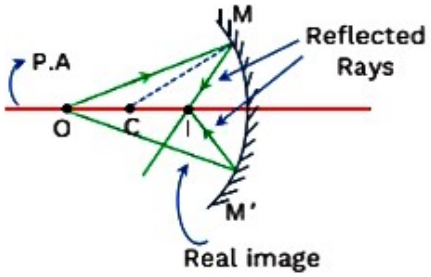


A Virtual object is one from which light rays appear to be emanating but physically do not.

Image

Real Image

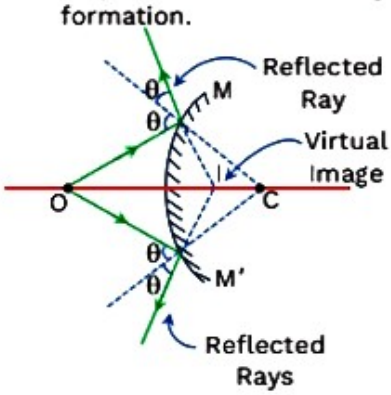
- A real image is one in which light rays physically intersect at the image location.
- Reflected Rays are responsible for image formation.



(Real rays are meeting at this point)

Virtual Image

- A Virtual image is one in which light rays do not physically intersect at the image point but appear to diverge from that point.
- Reflected Rays are responsible for image formation.



Important Point →

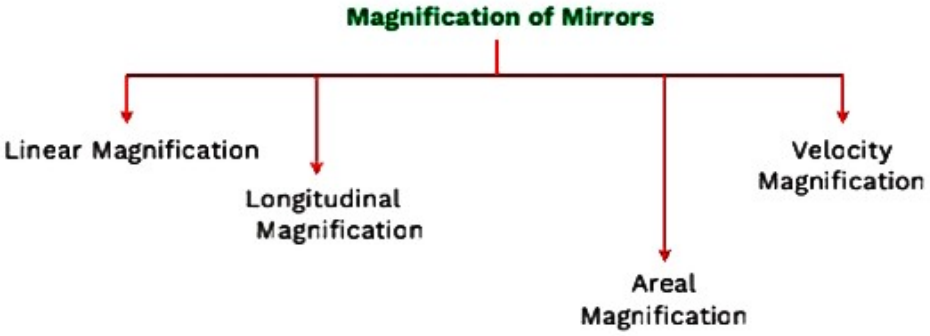
Summary -1

1.	Real object	Can form →	Real Image
2.	Real Object	Can form →	Virtual Image
3.	Virtual Object	Can form →	Real Image
4.	Virtual Object	Can form →	Virtual Image

Summary -2

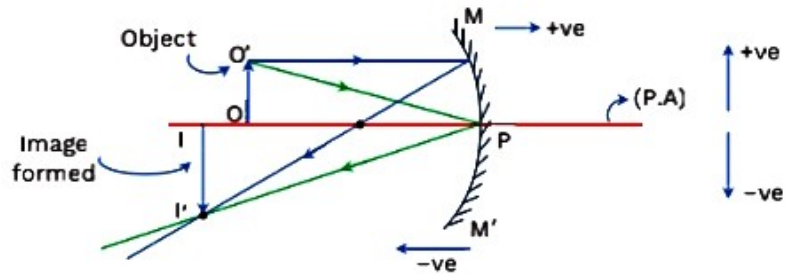
1.	f = +ve	for Convex Mirror
2.	f = –ve	for Concave Mirror
3.	for Real Object	u must be –ve
4.	for Virtual Object	u must be +ve
5.	for Real Image	v must be –ve
6.	for Virtual image	v must be +ve

Note → In short, we can say –ve means real And +ve means virtual in the case of mirrors.



Linear Magnification It is also called lateral or transverse magnification.

Derivation of formula



Now in $\Delta POO'$; $\tan\theta = \frac{OO'}{PO}$ ...(1)

in $\Delta PII'$; $\tan\theta = \frac{II'}{PI}$ ...(2)

from equation (1) and equation (2)

$$\frac{OO'}{PO} = \frac{II'}{PI}$$

$$\frac{II'}{OO'} = \frac{PI}{PO} \quad \text{...(3)}$$

By using sign convection

$II' = -h_i$; $OO' = h_o$

$PI = -v$; $PO = -u$

By definition of Linear magnification

$$m = \frac{\text{height of image}}{\text{height of object}}$$

$$m = \frac{II'}{OO'} = \frac{PI}{PO}$$

$$m = \frac{-h_i}{+h_o} = -\frac{v}{u}$$

Linear or Transverse Magnification $\boxed{m = \frac{h_1}{h_0} = -\frac{v}{u}}$

Important Point Put u, v, h_1, h_0 with the proper sign.

Note → from mirror formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u}$$

$$\frac{1}{v} = \frac{u - f}{uf}$$

$$m = -\frac{v}{u} = -\frac{1}{u} \times \left\{\frac{uf}{-f + u}\right\}$$

$$m = \frac{-f}{u - f}$$

$$\boxed{\text{Linear magnification } m = \frac{f}{f - u} = \frac{h_1}{h_0} = \frac{-v}{u}}.$$

Q **A concave mirror of the focal length of 10cm. An object is placed at a distance of 15 cm from the pole of the mirror. Calculate the size, nature, and position of the image formed.**

Sol: The rough figure indicating the pole of the mirror, focus, and the given distances are known. The sign convention is also given.

Since u = –15 cm (negative since it lies to the left of O)

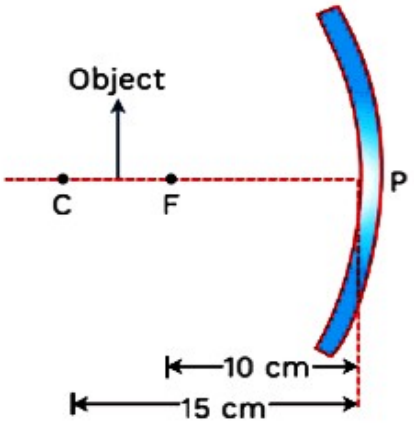
f = –10 cm (negative since it lies to the left of O)

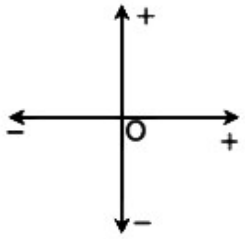
Since we have, from mirror formula that

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

$$\Rightarrow \quad \frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{(-10)} - \frac{1}{(-15)} = \frac{1}{-10} + \frac{1}{15} = \frac{-15 + 10}{150} = \frac{-5}{150}$$

$$\Rightarrow \quad v = -\frac{150}{5} = -30\,\text{cm}$$



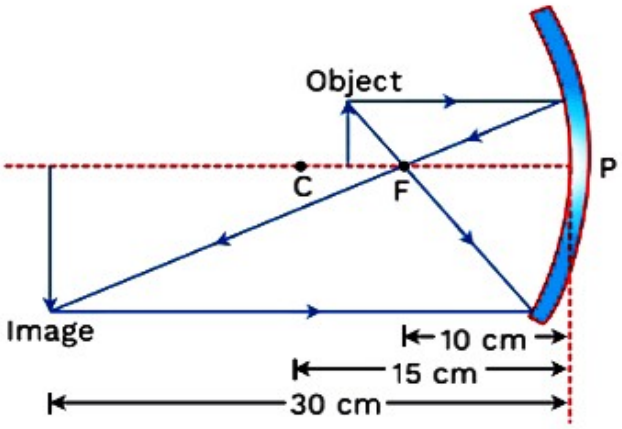


Sign convention

The negative sign for v shows that the image lies to the left of O.
Now, the magnification is given by

$$m = -\frac{v}{u} = -\frac{-30}{-15} = -2$$

The negative sign for m indicates that the image is inverted, and hence real and is double the size of the object.
Thus, we find that the image is real, inverted, twice the size of the object, and is formed 30 cm in front of the mirror. The ray diagram is shown in figure.



Q **A beam of light converges towards a point O, behind a convex mirror of focal length 20 cm. Find the nature and position of image if the point O is**
(A) 10 cm behind the mirror
(B) 30 cm behind the mirror

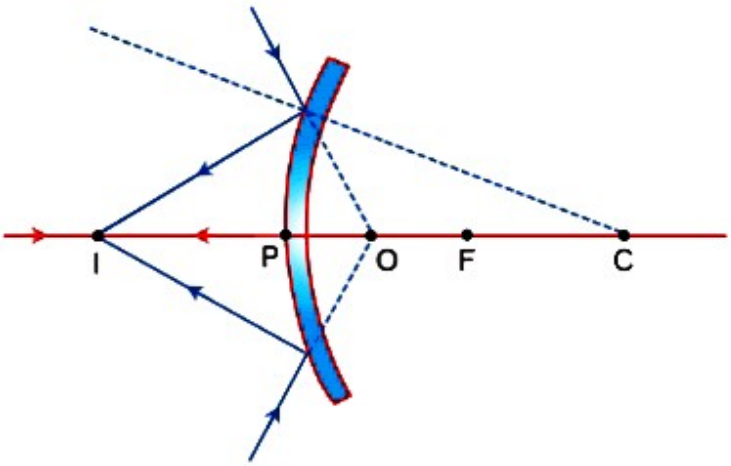
Sol: **(A)** Here, in this case the object is virtual. So, for this we have
$u = +10$ cm, $f = +20$ cm

Using mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v} = \frac{1}{f} - \frac{1}{u} = \frac{1}{20} - \frac{1}{+10} = \frac{1-2}{20} = \frac{-1}{20}$$

$\Rightarrow \quad v = -20$ cm

Magnification, $m = \frac{v}{-u} = -\frac{(-20)}{10} = 2$



Hence, the image formed will be real, erect and enlarged, and at a distance of 20 cm in from of the mirror.

(B) Again, in this case to the object is virtual. So, we have

$u = +30$ cm, $f = +20$ cm

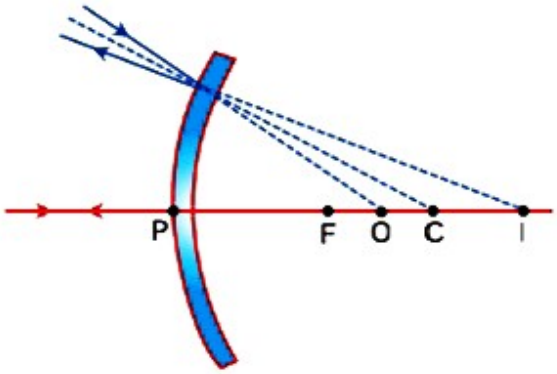
Using mirror formula $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$, we get

$$\frac{1}{v}=\frac{1}{f}-\frac{1}{u}=\frac{1}{20}-\frac{1}{30}=\frac{3-2}{60}=\frac{1}{60}$$

$$\Rightarrow \quad v = 60 \text{ cm}$$

Magnification, $m=-\frac{v}{u}=-\frac{60}{30}=-2$

Hence, the image formed will be virtual, inverted and enlarged, and at a distance of 60 cm behind the mirror.



Important Points:

Note that for the real objects, a convex mirror always gives a virtual and diminished image, but for virtual objects, it gives a real image if $u < f$ and a virtual image if $u > f$.

LONGITUDINAL MAGNIFICATION OR AXIAL MAGNIFICATION

When an object is placed with its length along the principal axis, then instead of defining the linear magnification, we define the axial magnification. Mathematically we define axial magnification for small objects as.

$$m_{axial}=\frac{\text{Size of image along principal axis}}{\text{Size of object along principal axis}}$$

$$m_{axial}=-\left(\frac{v_2-v_1}{u_2-u_1}\right)=-\frac{dv}{du}$$

Further, since we know that $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$ i.e. $v^{-1}+u^{-1}=f^{-1}$.

Now differentiate the above expression with respect to u, we get

$$\frac{d}{du}\left(v^{-1}\right)+\frac{d}{du}\left(u^{-1}\right)=\frac{d}{du}\left(f^{-1}\right)$$

$$\Rightarrow \quad -v^{-2}\frac{dv}{du}-u^{-2}=0$$

$$\Rightarrow \quad \boxed{m_{axial}=\frac{dv}{du}=-\frac{v^2}{u^2}=-\left(\frac{f}{f-u}\right)^2=-\left(\frac{f-v}{f}\right)^2}$$

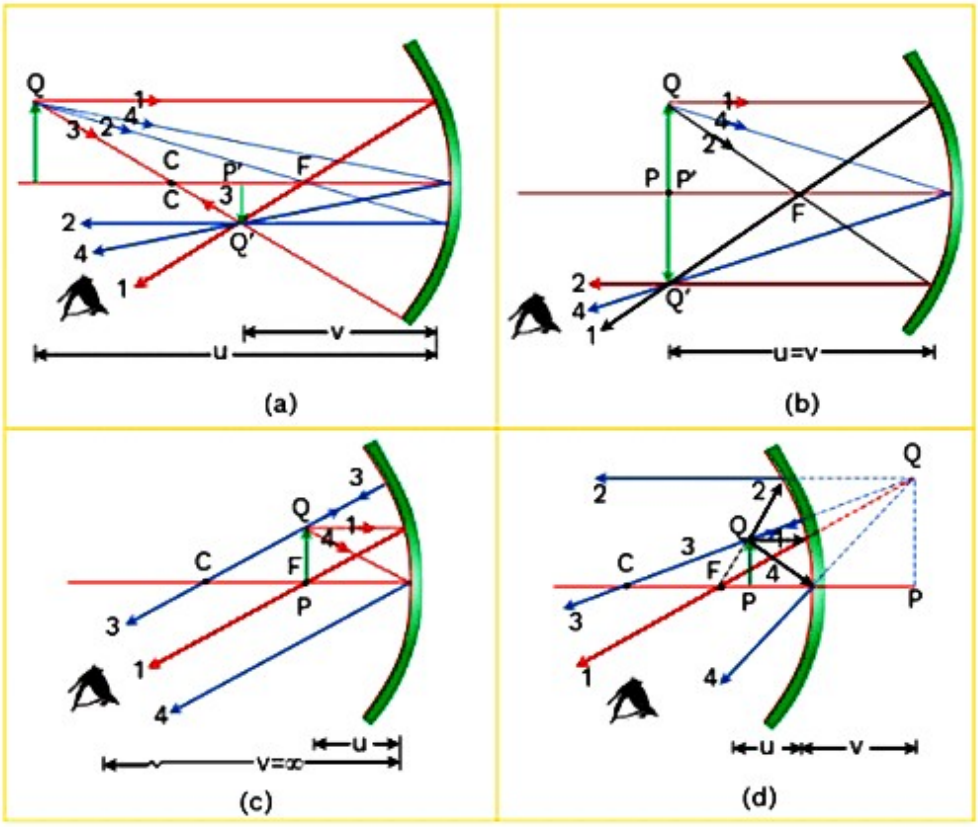
AREAL MAGNIFICATION

Areal magnification is simply defined as the ratio of the area of the image to the area of the object, so

$$\boxed{m_{areal}=\frac{\text{Area of image}}{\text{Area of object}}=\frac{A_i}{A_o}=\frac{v^2}{u^2}=\left(\frac{f}{f-u}\right)^2=\left(\frac{f-v}{f}\right)^2}$$

RULES FOR RAY DIAGRAMS

The position, size, and nature of the images formed by mirrors are conventionally expressed by ray diagrams (see figure).



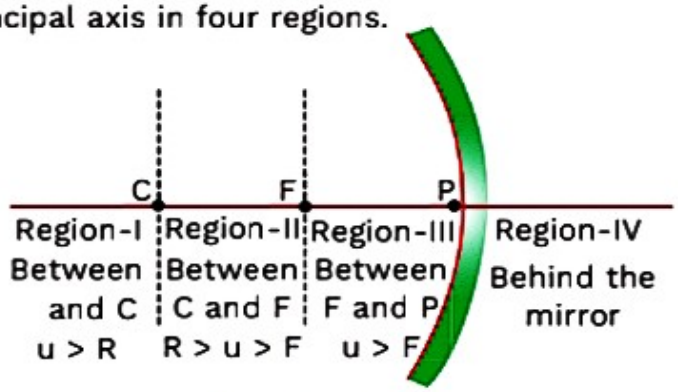
We can locate the image of any extended object graphically by drawing any two of the following four special rays.

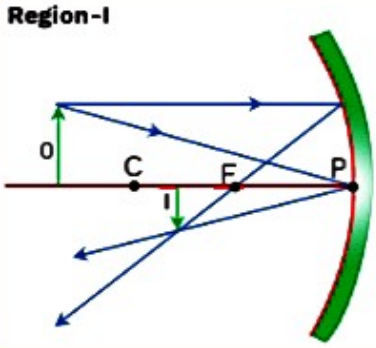
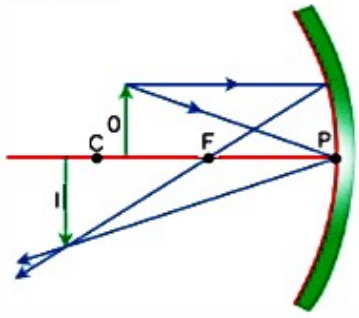
1. A ray initially parallel to the principal axis is reflected through the focus of the mirror (1).
2. A ray passing through the center of curvature is reflected back along with itself (3).
3. A ray initially passing through the focus is reflected parallel to the principal axis (2).
4. A ray incident at the pole is reflected symmetrically (4).

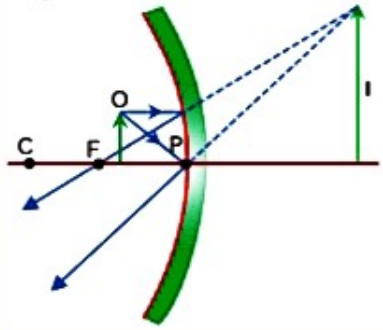
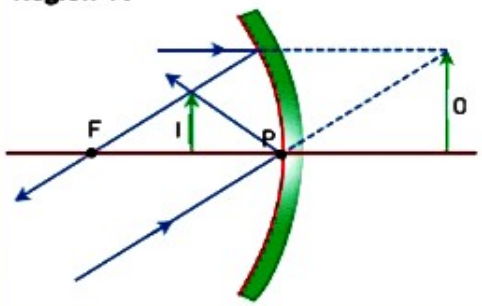
Image Formation

Nature of Image Formed by Concave Mirror

We can divide principal axis in four regions.

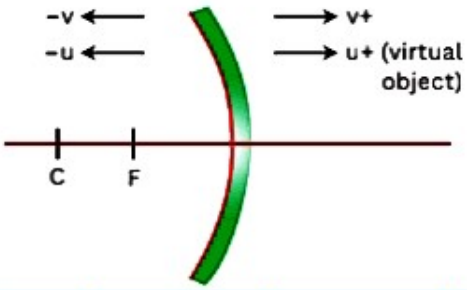


Region-I 	• Object placed between centre of curvature and infinity. Image real, inverted, smaller (diminished) • If object is moved away from center of curvature towards infinity, magnification goes on decreasing.
Region-II 	• The object is placed between and center of curvature. Image is real, inverted, magnified. • If the object is moved from focus towards center of curvature, magnification goes on decreasing at the center of curvature it becomes unity.

Region-III 	• The object is placed between pole and focus. Image is virtual, erect, magnified. • If the object is moved towards the pole, magnification goes on decreasing at pole magnification m tends to unity.
Region-IV 	• Object is virtual point object. Image real, erect, smaller (diminished) • As the object moves away from the pole, magnification goes on decreasing.

OBJECT DISTANCE (U) VS IMAGE DISTANCE (V) GRAPH

Concave mirror $v = \frac{uf}{u-f}$, f is negative. Let $f = -F_0$.



u	$-\infty$	$-2F_0$	$-F_0$	0	F_0	∞
v	$-F_0$	$-2F_0$	$\pm\infty$	0	$\frac{-F_0}{2}$	$-F_0$

$$v = \frac{-uF_0}{u+F_0}$$

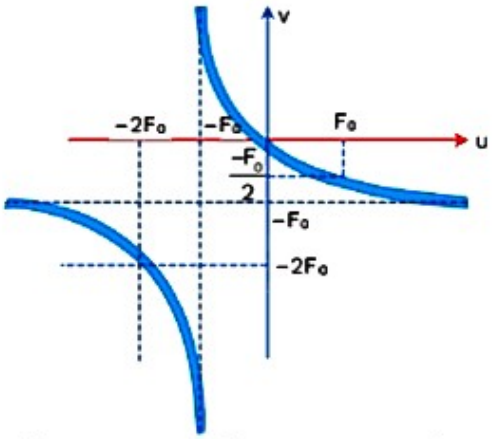
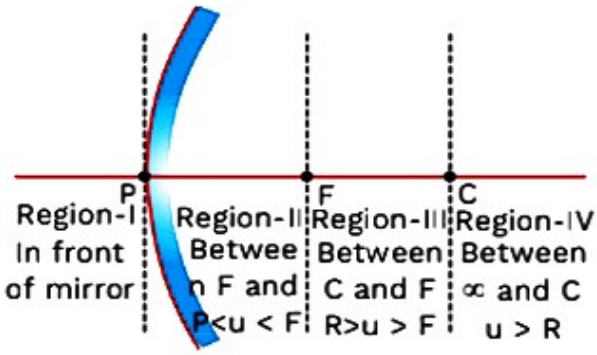


Fig. u-v curve for concave mirror

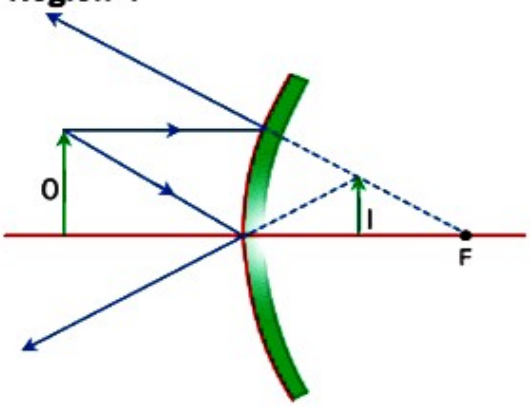
NATURE OF IMAGE IMAGE BY CONVEX MIRROR

The figure shows a convex mirror with four different object positions, as shown in the figure.

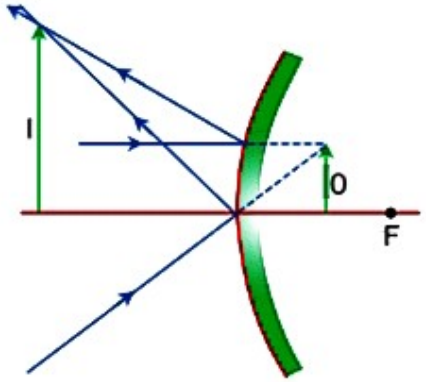
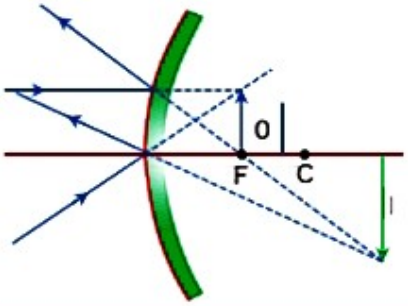
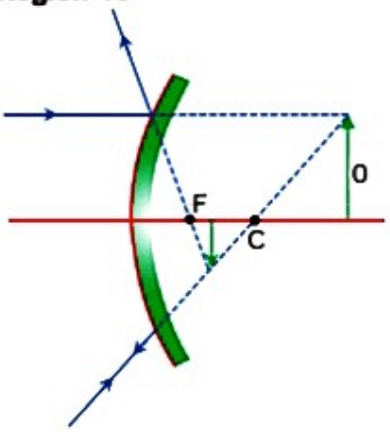
Now we analyze the corresponding image formed.



Region-I

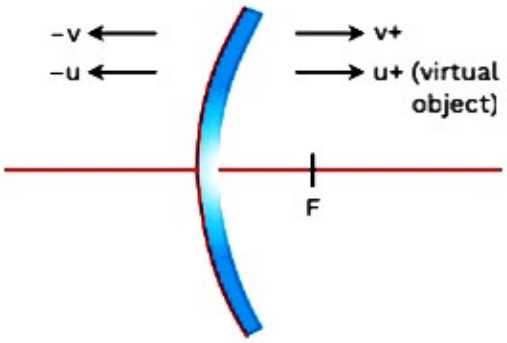


- Object placed in front of mirror. For all positions of object in front of mirror. Image is virtual, erect, smaller in size.
- As object moved towards pole. Magnification increases and tends to unity at pole.

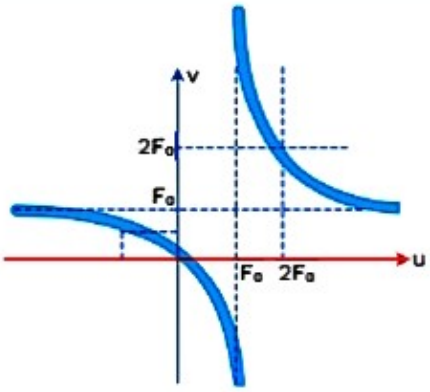
Region-II 	• The virtual object is placed between pole and focus. An image formed is real, erect, enlarged. • As object is moved away from pole. Magnification increases.
Region-III 	• Virtual object placed between F and C. Image formed virtual, inverted, enlarged. • As the object is moved toward C, magnification decreases.
Region-IV 	• The virtual object is placed beyond C. Image is virtual inverted smaller. • As the object is moved away from C, image size further decrease magnification decreases.

Convex mirror $v = \dfrac{uf}{u - f}$

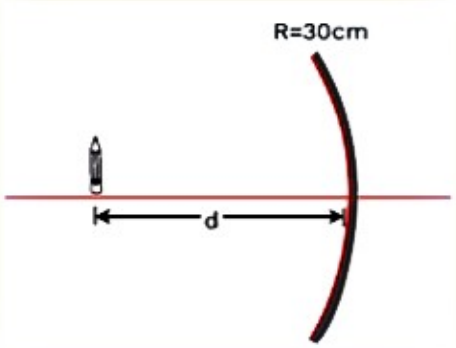
F is positive. Let $f = F_0 \Rightarrow v = \dfrac{uf_0}{u - f_0}$



u	$-\infty$	$-F_0$	0	F_0	$2F_0$	∞
v	F_0	$\frac{F_0}{2}$	0	$\pm\infty$	2F	F_0



Q **The radius of curvature of the concave mirror is 30cm if we put a pen perpendicular to the principal axis at each of the following distances.**



(A) d = 40 cm **(B) d = 30 cm** **(C) d = 15 cm** **(D) d = 5 cm**

Find the image of the pen, magnification of each image.

Sol: The incident ray is traveling towards the right, all the distance measured in the rightward direction should be positive,

The focal length of the mirror is $f = \frac{R}{2} = -15\,cm$. From mirror equation, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

(A) In this case, d = 40 cm, the object is placed on the left side of the center of curvature and u = −40 cm

$$\frac{1}{v} + \frac{1}{(-40)} = \frac{1}{(-15)} \Rightarrow v = -24\,cm$$

The minus sign indicates that the image is in front of the mirror, real.

$$\text{Magnification, } m = -\frac{v}{u} = -\left(\frac{-24}{-40}\right) = -0.6$$

As m < 1, the image is diminished in size, minus sign shows that the image is inverted.

(B) In this case, d = 30 cm, the object is placed at the center of the curvature of the mirror. In this case, the image will form at the object position. The image is real, formed in front of the mirror.

$$\text{Magnification, } m = -\frac{v}{u} = -\left(\frac{-30}{-30}\right) = -1$$

m = −1, the size of the object and the image are equal minus sign shows that the image is inverted.

(C) In this case, d = 15 cm, the object is placed at the focus. So the image will form at infinity in this case.

$$\text{Magnification, } m = -\frac{v}{u} = -\frac{-\infty}{(-15)} = -\infty$$

The image is enlarged real, formed in front of the mirror, and will form at infinity.

(D) In this case, d = 5 cm, the object is placed on the right side of the focus u = −5 cm

$$\frac{1}{v} + \frac{1}{(-5)} = \frac{1}{(-15)} \Rightarrow \quad = +7.5\,cm$$

The plus sign indicates that the image is behind the mirror virtual. It is a general observation that an object placed inside the focal point of a concave mirror produces a virtual image.

$$m = -\left(\frac{v}{u}\right) = -\left(\frac{+7.5}{-5}\right) = +1.5$$

As m > 1, the image is magnified; plus, sign shows that it is erect.

Q **A concave mirror of focal length 30 cm. Calculate the position of the object in front of the mirror so that the size of the image is three times the size of an object.**

Sol: As the object is in front of the mirror, this means an object is real. Also, the magnified image is formed, so there are two possibilities-

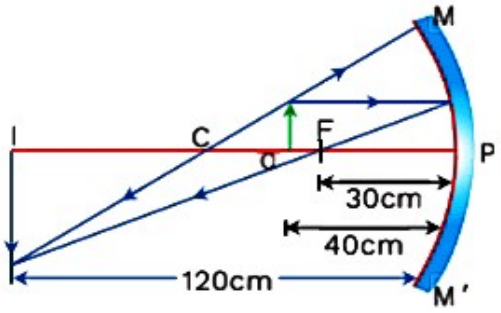
(A) If the image is real and inverted

$$m = -\left(\frac{v}{u}\right) = -3 \text{ i.e., } v = 3u$$

So from mirror formula, $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

We get, $\frac{1}{3u} + \frac{1}{u} = \frac{1}{(-30)}$ i.e., $u = -40\,\text{cm}$

i.e., object must be at a distance of 40 cm in front of the mirror (i.e., between C and F) as shown in the figure.



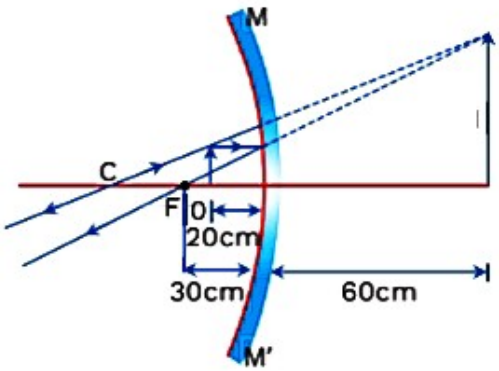
(B) If the image is virtual and erect

$m = -(v/u) = 3$ i.e., $v = -3u$

So from mirror formula,

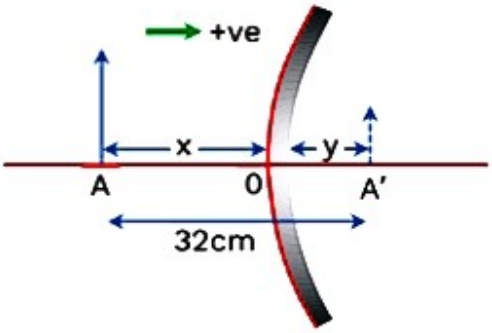
$\frac{1}{(-3u)} + \frac{1}{u} = \frac{1}{(-30)}$ i.e., $= -20\,\text{cm}$

i.e., object must be at a distance of 20 cm in front of the mirror (i.e., between F and P) as shown in the figure.



Q **Given that the distance between a real object and its image in a convex mirror of focal length 12 cm is 32 cm. Find the size of the image if the object size is 1 cm.**

Sol Let x and y be the magnitudes of object and image distance.



We have AA′ = 32 cm

$\Rightarrow$ AO + A′O = 32

$\Rightarrow$ (x + y) = 32 ...(i)

And also, u = −x, v = +y

$$\frac{1}{-x}+\frac{1}{y}=\frac{1}{+12} \quad \text{...(ii)}$$

Solving Eqs. (i) and (ii) simultaneously, we can get u and v. The relevant answers are u = −24 cm, v = +8 cm

Using $I=-1\left(\frac{+8}{-24}\right)=+\frac{1}{3}$

So, the image size is 1/3 cm.

Cutting of a Mirror:

A mirror is cut along its principal axis, and two parts are separated perpendicular to the principal axis.

1. For the first part, object O will be below the principal axis, and for the second part, object O will be above the principal axis.

2. For the first part, the image will form above its principal axis, and for the second part, the image will form below the principal axis at a distance of $m_1 y_1$ and $m_2 y_2$, respectively.

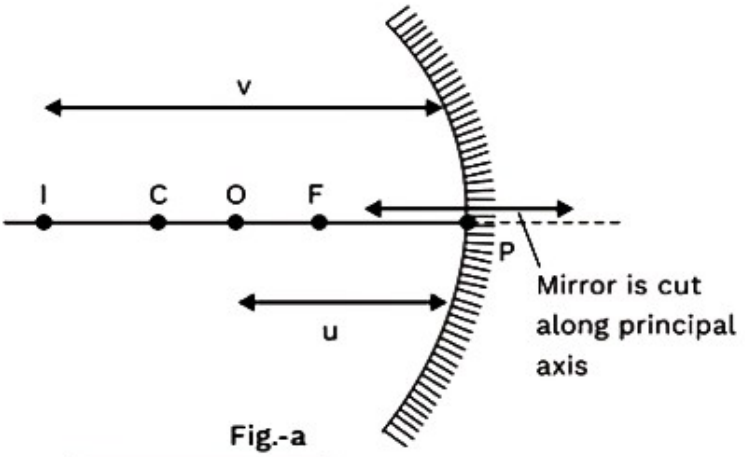


Fig.-a

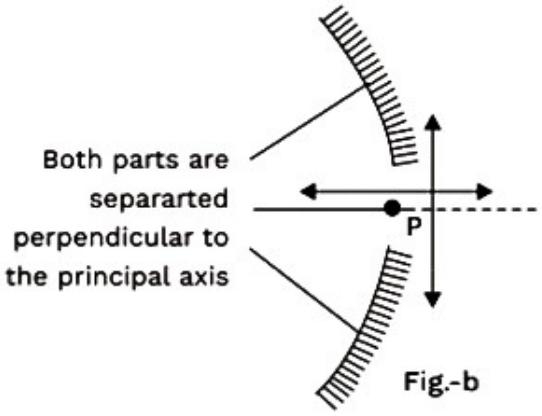
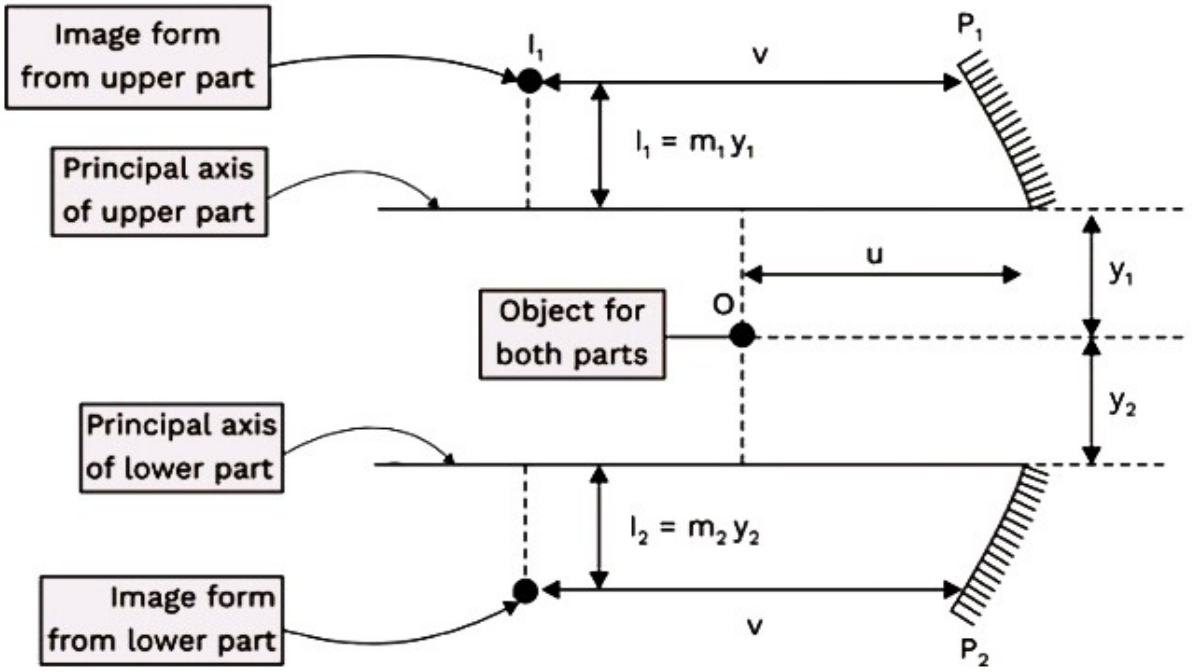


Fig.-b



For image partition from principal axis,

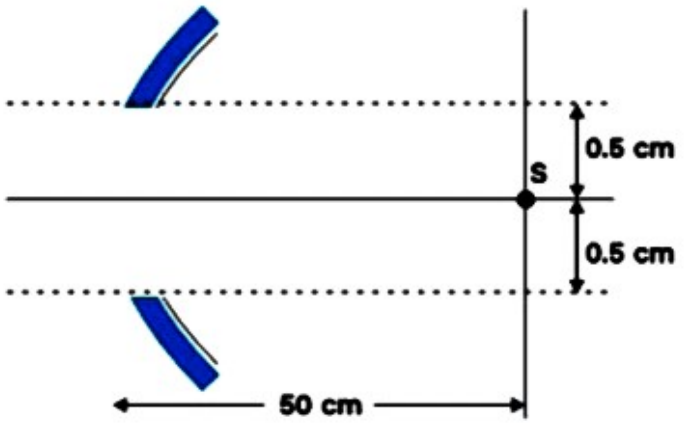
$$m = \frac{-v}{u} = \frac{I_1}{O} \Rightarrow |I_1| = my_1$$

Thus, $s = my_1 + y_1 + my_2 + y_2$: $s = (m + 1)(y_1 + y_2)$

Where x s is the separation distance between two images I^1 and I_2 of the same object O formed due to two halves of the mirror.

Read the following paragraph and answer the following three questions:

A concave mirror forms a real image of a point source lying on the optical axis at a distance of 50 cm from the mirror. The focal length of the mirror is 25 cm. The mirror is cut in two, and its halves are drawn at a distance of 1 cm apart in a direction perpendicular to the optical axis.

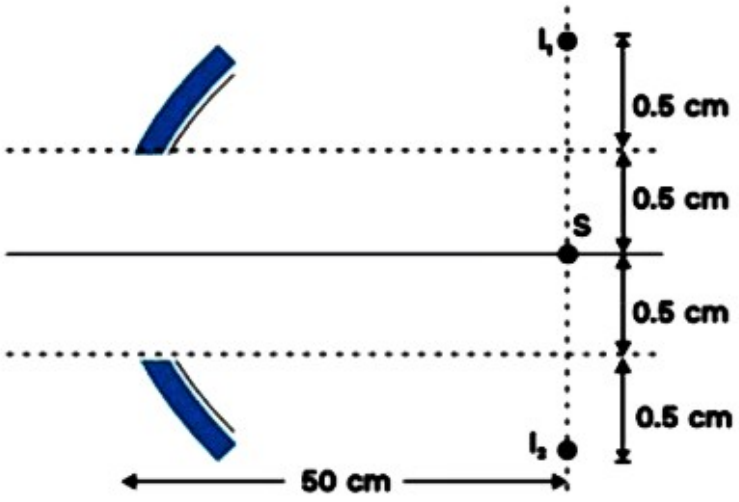


Q. **The distance between the two images formed by the two halves of the mirror is**

(A) 0.5 cm **(B) 1 cm** **(C) 1.5 cm** **(D) 2 cm**

Sol. Ans. (D)

Using $\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \Rightarrow v = -50\,cm$



and magnification $= \frac{-v}{u} = -1$

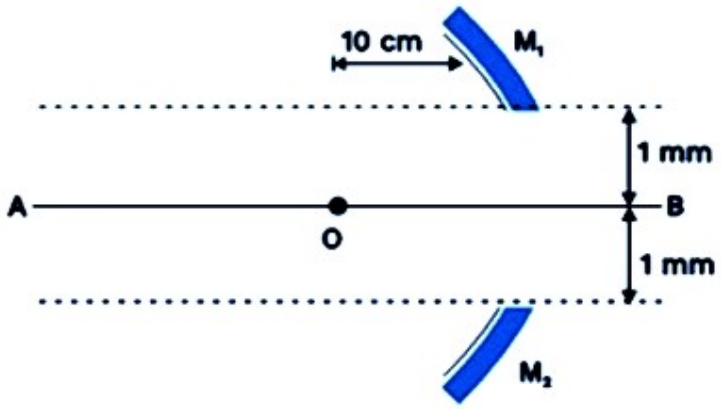
So distance between two images = 2 cm

Q. **Two statements are provided. Choose which is true (T) and which is false (F).**
Statement I: Image formed by a half-cut mirror has just half the intensity as that of the intensity of the image formed by the complete mirror.
Statement II: More the lateral separation between the two halves of the mirror, less will be the separation between the two images, as shown in the given situation.
(A) T, T **(B) T, F** **(C) F, T** **(D) F, F**

Sol. Ans (B)

For half the mirror, due to reflection being half of the incoming rays, so the intensity of the image is obviously half. More the lateral separation, the more will be the gap between the two images formed.

Q. **A concave mirror of focal length 20 cm is cut into two parts from the middle, and the two parts are moved perpendicularly by a distance of 1 mm from the previous principal axis AB. Find the distance between the images formed by the two parts in mm.**



Sol. Using mirror formula, $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$

Here u = –10 cm, f = –20 cm

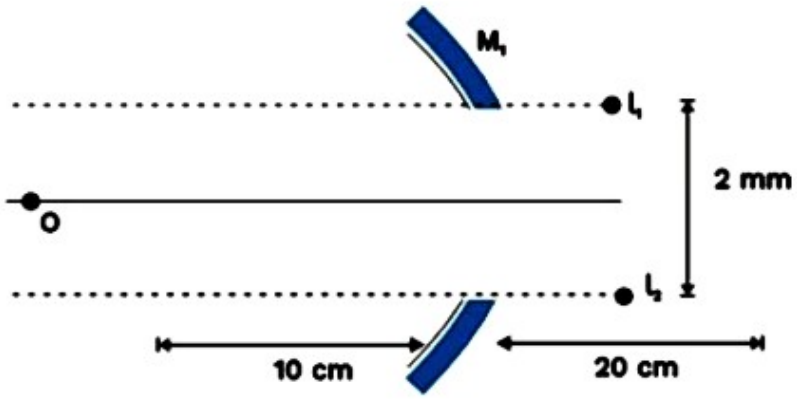
Hence $\frac{1}{v}+\frac{1}{(-10)}=\frac{1}{(-20)} \Rightarrow v = 20\,cm$

The magnification, $m=\frac{I}{O}=-\frac{v}{u}$

From M_1O = –1 mm, u = –10 cm, v = 20 cm

$$\frac{I_1}{(-1mm)} = \frac{(+20cm)}{(-10cm)} \Rightarrow I_1 = -2cm$$

Hence the image from M_1 will form 2 m above the principal axis behind the mirror



Hence separation between the images will be 2 mm.

RELATION BETWEEN OBJECT AND IMAGE VELOCITY

Case (i) – When an object is moving along the principal axis (velocity magnification)

Let a point object is moving along the principal axis of a concave mirror of focal length f. We can relate the object and image positions by the mirror formula

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f} \qquad ...(i)$$

Differentiate Eq. (i) with respect to time.

$$\Rightarrow \quad -\frac{1}{v^2}\frac{dv}{dt} - \frac{1}{u^2}\frac{du}{dt} = 0$$

or $$\boxed{\frac{dv}{dt} = -\left(\frac{v^2}{u^2}\right)\frac{du}{dt}} \qquad ...(ii)$$

Let $\frac{dv}{dt} = V_{i,m}$ = velocity of image w.r.t. mirror

and, $\frac{du}{dt} = V_{o,m}$ = velocity of object w.r.t. mirror

Also, $\frac{v^2}{u^2} = m^2$

Then Eq. (ii) becomes

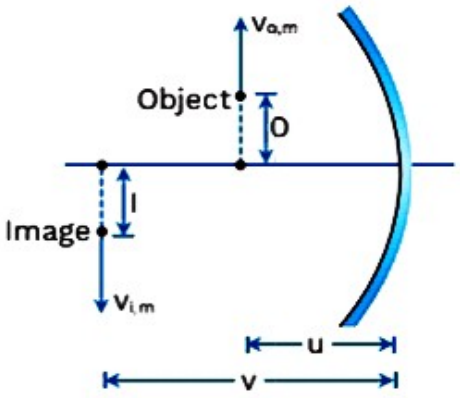
$$\boxed{V_{i,m} = -m^2 V_{o,m}} \qquad ...(iii)$$

Important Points

1. Image speed = m^2 × Object speed
2. In vector notation, Eq. (iii) can be written as
   $\bar{V}_{i,m} = -m^2\bar{V}_{o,m} \Rightarrow \left(\bar{V}_i - \bar{V}_m\right) = -m^2\left(\bar{V}_o - \bar{V}_m\right)$
3. An object moves between infinite and centre of curvature.
   In this case, $|m| < 1$, hence $V_{i,m} < V_{o,m}$.
   The speed of the image is less than the speed of an object
4. When the object is at the centre of curvature $|m| = 1$,
   hence $V_{i,m} = V_{o,m}$
5. An object moves between the center of curvature and focus.
   In this case, $|m| > 1$, hence $V_{i,m} > V_{o,m}$
   The speed of an image is more than the speed of an object.

Case (ii)- When object is moving normal to the principal axis

In this case, a point object is moving perpendicular to the principal axis of a concave mirror, as shown in the figure. In this situation, the object distance will not change; it makes image distance also constant.



For constant values of v and u, the value of the transverse magnification is also constant or $m = \frac{I}{O} = -\frac{v}{u}$ is a constant. We can write,

$I = mO$ ...(i)

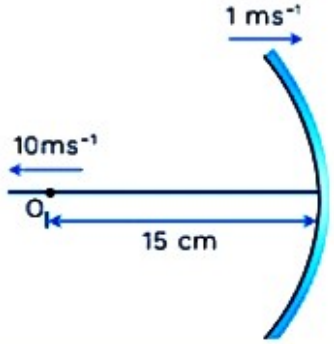
Differentiate Eq. (i) with respect to time, we get

$$\boxed{\frac{dI}{dt} = m\frac{dO}{dt} \Rightarrow \left(V_{i,m}\right)_y = m\left(V_{o,m}\right)_y}$$

Here $\frac{dI}{dt} = (V_{I,m})_y$ = velocity of image w.r.t. mirror in the direction perpendicular to the principal axis.

And, $\frac{dO}{dt} = (V_{O,m})_y$ = velocity of object w.r.t. mirror in the direction perpendicular to the principal axis.

Q **A concave mirror of focal length 10 cm and a point object both are moving away from each other with velocities 1 ms–1 and 10 ms–1 respectively, as shown in the figure. Find the velocity of the image when the object is at a distance of 15 cm from the mirror.**



Sol: Using $\frac{1}{v} + \frac{1}{u} = \frac{2}{R}$, we get

$$\frac{1}{v} - \frac{1}{15} = -\frac{1}{10} \Rightarrow v = -30\,cm$$

Now, using $\bar{V}_{I,m} = -m^2 \bar{V}_{O,m}$

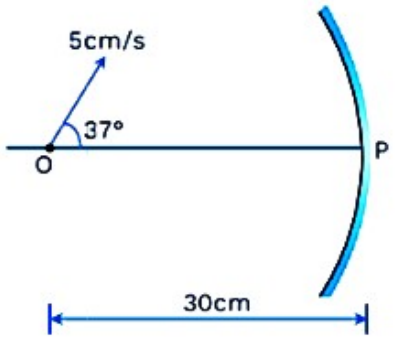
Or $(\bar{V}_I - \bar{V}_m) = -m^2 (\bar{V}_O - \bar{V}_m)$

$$\Rightarrow \left[\bar{V} - (1)\right] = -\left(\frac{(-30)}{(-15)}\right)^2 \left[(-10) - (1)\right]$$

$\Rightarrow$ V_I = 45 cm s^{-1} towards right

So, the image will move with velocity 45 cm s^{-1} towards right.

Q **A point object is moving with velocity 5 cm/s at angle 37° towards a stationary concave mirror of focal length 20 cm as shown in figure. Find the image position and its velocity when the object is at a distance 30 cm from the mirror.**



Sol We have mirror formula

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f} \qquad ...(i)$$

Substituting the values, u = −30 cm and f = −20 cm in Eq. (i), we have

$$\frac{1}{v}+\frac{1}{-30}=\frac{1}{-20}$$

Solving this equation, we get v = −60 cm

Further, $m=-\frac{v}{u}=-\frac{(-60)}{(-30)}=-2$ and $m^2=4$

Let us make the component of the object velocity along the principal axis and along the normal to the principal axis.

$$\bar{V}_{o,m}=5\cos 37°\hat{i}+5\sin 37°\hat{j}(cm/s)$$

$$\Rightarrow \quad \bar{V}_{o,m}=4\hat{i}+3\hat{j}(cm/s)$$

Now, using $\bar{V}_{i,m}=-m^2\bar{V}_{o,m}$ for finding the component of the image velocity along the principal axis

Or $\left(\bar{V}_i-\bar{V}_m\right)=-m^2\left(\bar{V}_o-\bar{V}_m\right)$

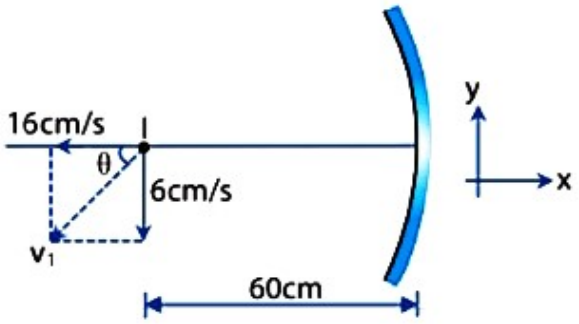
$$\left(\bar{V}_i-0\right)=-4\left(4\hat{i}-0\right)\Rightarrow\left(\bar{V}_i\right)_x=-16\hat{i}\,cm/s$$

The velocity of the image in the direction perpendicular to the principal axis,

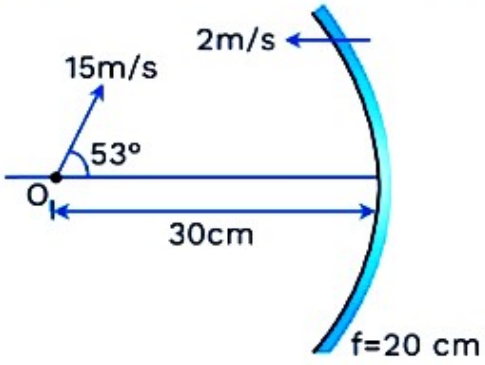
$$\left(V_{i,m}\right)_y=-m\left(V_{o,m}\right)_y=-4\times 3\hat{j}=-12\hat{j}\ cm/s$$

Hence the velocity of image, $\bar{V}_{image}=-16\hat{i}-12\hat{j}(cm/s)$

The position and velocity of the image are shown below.



Q **Find the velocity of image in a situation as shown in the figure.**



Sol Given Object distance u = −30 cm, f = −20 cm

The velocity of the object $\vec{v}_o = \left(9\hat{i} + 12\hat{j}\right) m/s$ and velocity of the mirror

$\vec{v}_m = -2\hat{i}\, m/s$

Using mirror formula $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

Magnification $m = -\frac{v}{u} = -\frac{(-60)}{(-30)} = -2$

The velocity of the image with respect to mirror in the direction along the principal axis (x-direction)

$$(\vec{V}_{i,m})_x = -m^2(\vec{V}_{o,m})_x \Rightarrow (\vec{v}_i - \vec{v}_m)_x = -m^2(\vec{v}_0 - \vec{v}_m)_x$$

$$\left[\vec{v}_i - (-2\hat{i})\right] = -(-2)^2\left[(9\hat{i}) - (-2\hat{i})\right]$$

$$\left(\vec{v}_i\right)_x = -46\hat{i}\ m/s$$

Velocity of image w.r.t. mirror in y-direction

$$(\vec{V}_{i,m})_y = m(\vec{V}_{o,m})_y \Rightarrow (\vec{v}_i - \vec{v}_m)_y = m(\vec{v}_0 - \vec{v}_m)_y$$

$$\left[\vec{v}_i - 0\right] = (-2)\left[(12\hat{i}) - 0\right]$$

$$\Rightarrow \quad (\vec{v}_i)_y = -24\hat{j}\,(m/s)$$

Hence velocity of the image $\vec{v}_i = \left(-46\hat{i} - 24\hat{j}\right)m/s$.

Refraction of Light

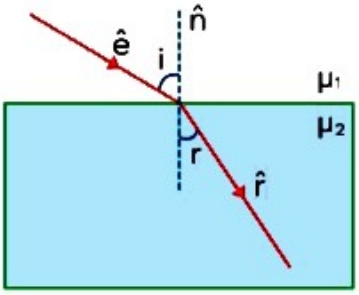
REFRACTION

Refraction is the phenomenon in which direction of propagation of light changes at the boundary when it passes from one medium to the other. In case of refraction, frequency of light does not change but wavelength of light changes.

- **Laws of refraction**

(i) Incident ray, refracted ray and normal always lie in the same plane. In vector form $\boxed{(\hat{e} \times \hat{n}).\hat{r} = 0}$

(ii) The product of refractive index and sine of angle made by the ray with the normal at incident is constant for a given ray in both media. $\mu_1 \sin i = \mu_2 \sin r$ (Snell's law).



.Absolute refractive index

It is defined as the velocity of light in a vacume to the velocity of light in a medium. It is denoted by μ or n.

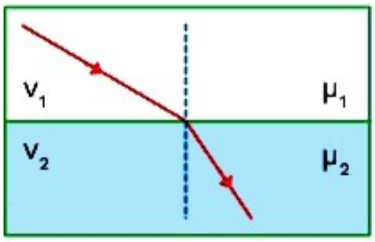
$$\mu \text{ or } n = \frac{c}{v}.$$

In denser medium velocity of light is less so refractive index of that medium is high.

$$\because v_{glass} < v_{water}, \quad \therefore \mu_G > \mu_w$$

- **Relative refractive index of two medium**

When a ray of light passes from one medium to the other , The relative refractive index of medium 2 with respect to medium 1 is written as $_1\mu_2$ and is defined as the speed of light in medium 1 to the speed of light in medium 2.

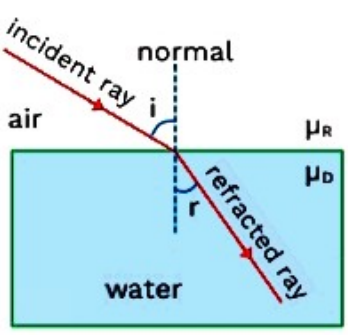


$$_1\mu_2 = \frac{\mu_2}{\mu_1} = \frac{(c/v_2)}{(c/v_1)} = \frac{v_1}{v_2}$$

- **Bending of light ray when travel from one medium to another medium**

According to Snell's law, $\mu_1 \sin i = \mu_2 \sin r$

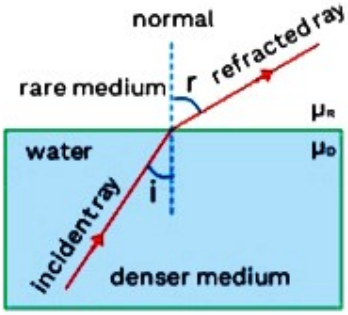
(i) When light passes from rarer to denser medium $\mu_1 = \mu_R$ and $\mu_2 = \mu_D$



$$\text{so that } \frac{\sin i}{\sin r} = \frac{\mu_D}{\mu_R} > 1 \Rightarrow \angle i > \angle r$$

In passing from rarer medium to the denser medium then the ray bends towards the normal.

(ii) If light passes from denser medium to rarer medium then the ray bends away from the normal direction



IMPORTANT POINTS

(a) Whenever light goes from one medium to another, the frequency of light (f) remains unchanged. Since

$$\mu = \frac{c}{v} = \frac{\text{Speed of light in vacuum}}{\text{Speed of light in medium}}$$

$$\Rightarrow \quad \mu = \frac{f\lambda_{air}}{f\lambda_{medium}} = \frac{\lambda_{air}}{\lambda_{medium}}$$

where λ_{air} and λ_{medium} being wavelengths of light in air and medium respectively.

$$\Rightarrow \quad \frac{\sin i}{\sin r} = \frac{\mu_2}{\mu_1} = \frac{\frac{c}{v_2}}{\frac{c}{v_1}} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$$

From above we conclude that

$$\mu_1\lambda_1 = \mu_2\lambda_2$$

$$\Rightarrow \quad \mu\lambda = \text{constant}$$

Also, we conclude that

$$\mu_1 v_1 = \mu_2 v_2$$

$$\Rightarrow \quad \mu v = \text{constant}$$

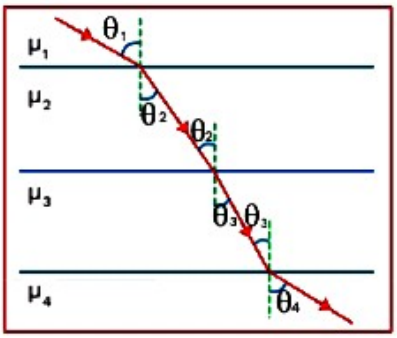
(b) $^2\mu_1 \times {}^1\mu_2 = 1$

$$\Rightarrow \quad {}^2\mu_1 = \frac{1}{{}^1\mu_2}$$

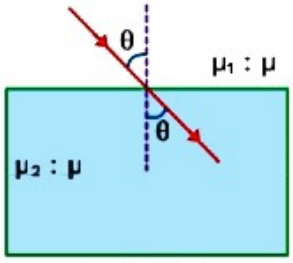
(c) When a ray of light travels through multiple layers of different medium then snell's law can be written as

$\mu_1 \sin\theta_1 = \mu_2 \sin\theta_2 = \mu_3 \sin\theta_3 = \mu_4 \sin\theta_4 =$ constant

In general, $\mu\sin\theta$ = constant.



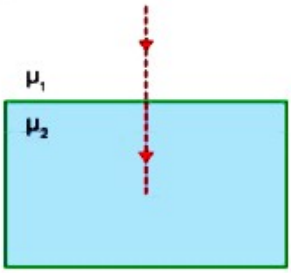
(d) The light ray is not refracted if the refractive indices of the two media are equal as shown in figure, also the boundary between the two media is not visible. This is why a transparent solid is invisible in a liquid of same refractive index.



$\mu_2 = \mu_1 = \mu$

Condition for no refraction

(e) If light ray is normally incident (i.e. i= 0), then, it passes undeviated from the boundary as shown in the figure.



Q **The refractive index of water with respect to air is 4/3 and the refractive index of glass with respect to air is 3/2. Then the refractive index of water with respect to glass is**

(A) $\frac{9}{8}$ **(B)** $\frac{8}{9}$ **(C)** $\frac{1}{2}$ **(D) 2**

Sol: Given : ${}^{a}\mu_{w} = \frac{4}{3}$; ${}^{a}\mu_{g} = \frac{3}{2}$; ${}^{g}\mu_{w} = ?$

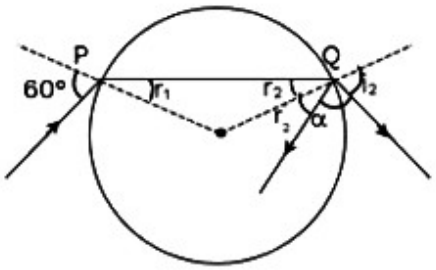
$${}^{a}\mu_{w} \times {}^{w}\mu_{g} = {}^{a}\mu_{g}$$

$${}^{w}\mu_{g} = \frac{{}^{a}\mu_{g}}{{}^{a}\mu_{w}} = \frac{3/2}{4/3} = \frac{9}{8}; \quad {}^{g}\mu_{w} = \frac{1}{{}^{w}\mu_{g}} = \frac{1}{(9/8)} = \frac{8}{9}.$$

Q **A ray incident at a point at an angle of incidence of 60° enters a glass sphere of refractive index $\sqrt{3}$ and is reflected and refracted at the farther surface of the sphere. The angle between the reflected and refracted rays at this surface is**
(A) 50° **(B) 60°** **(C) 90°** **(D) 40°**

Sol: Reflraction at P,

$$\frac{\sin 60^{\circ}}{\sin r_1} = \sqrt{3}$$



$$\sin r_1 = \frac{1}{2}$$

or $r_2 = r_1$; $\quad \therefore r_2 = 30°$

Refraction at Q, $\frac{\sin r_2}{\sin i_2} = \frac{1}{\sqrt{3}}$ or $\frac{\sin 30^{\circ}}{\sin i_2} = \frac{1}{\sqrt{3}}$

or $i_2 = 60°$

At point Q, $r_2' = r_2 = 30^{\circ}$

$$\therefore \quad \alpha = 180^{\circ} - \left(r_2' + i_2\right) = 180^{\circ} - \left(30^{\circ} + 60^{\circ}\right) = 90^{\circ}$$

Q **A light ray passes through a medium whose refractive index varies with distance as $n = n_0\left(1 + \frac{x}{2a}\right)$. If ray enters the medium parallel to x-axis, what will be the time taken for ray to travel between x = 0 and x = a?**

Sol: Since, we know that $\mu = \frac{c}{v}$

$$\Rightarrow \quad v = \frac{c}{\mu}$$

So, if v be the speed at a distance x from y-axis, then

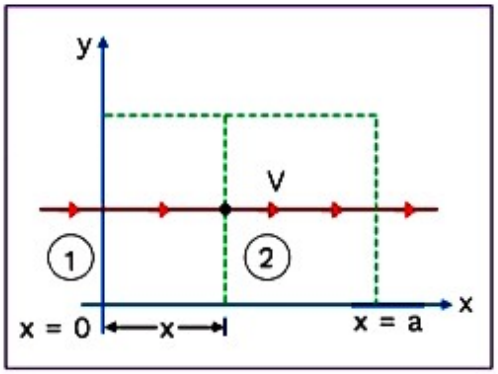
$$v = \frac{c}{n_0\left(1+\frac{x}{2a}\right)}$$

Since, $v = \frac{dx}{dt}$, so we have

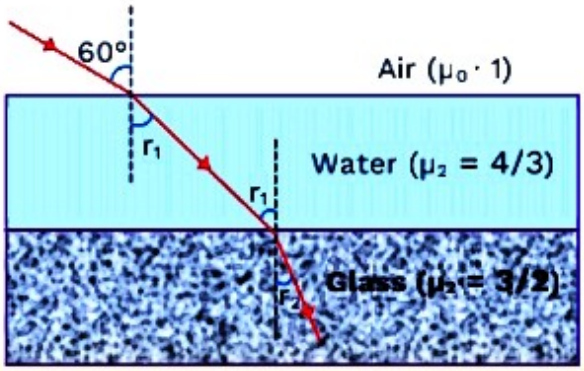
$$\frac{dx}{dt} = \frac{c}{n_0\left(1+\frac{x}{2a}\right)}$$

$$\Rightarrow \int_0^t dt = \frac{n_0}{c}\int_0^a \left(1+\frac{x}{2a}\right)dx$$

$$\Rightarrow t = \frac{5n_0a}{4c}$$



Q **A ray of light is incident at an angle of 60° on the surface of water as shown in figure. calculate the angle between this ray and the normal to the glass.**



Sol According to Snell's Law, we have
$\mu_0 \sin(60°) = \mu_1 \sin r_1 = \mu_2 \sin r_2$

$$\Rightarrow \sin r_2 = \frac{\mu_0 \sin(60°)}{\mu_2} = \frac{1\left(\frac{\sqrt{3}}{2}\right)}{\left(\frac{3}{2}\right)} = \frac{1}{\sqrt{3}}$$

$$\Rightarrow r_2 = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right) = 35°$$

Q **A light beam passes from a parallel plate glass slab of refractive index μ_2 placed in a medium of refractive index μ_1. Show that the emerging beam is parallel to the incident beam.**

Sol: Applying Snell's Law at A we get

$\mu_1 \sin i = \mu_2 \sin r_1$

$\Rightarrow \frac{\mu_1}{\mu_2} = \frac{\sin r_1}{\sin i}$ .... (1)

Applying Snell's Law at B, we get

$\mu_2 \sin r_1 = \mu_1 \sin e$

$\Rightarrow \frac{\mu_1}{\mu_2} = \frac{\sin r_1}{\sin e}$ ... (2)

From equation (1) and (2), we get.

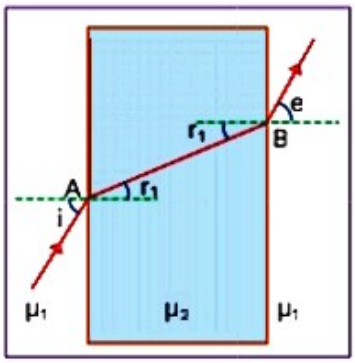
$i = e$

From equation (1) and (2), we get

$\frac{\sin r_1}{\sin i} = \frac{\sin r_2}{\sin e}$

Since $\angle r_1 = \angle r_2$

$i = e$



Q **Given that refractive index of glass with respect to water is 9/8. Also Refractive index of glass with respect to air is 3/2. calculate the refractive index of water with respect to air.**

Sol: Given, ${}^{w}\mu_g = \frac{9}{8}$ and ${}^{2}\mu_g = \frac{3}{2}$

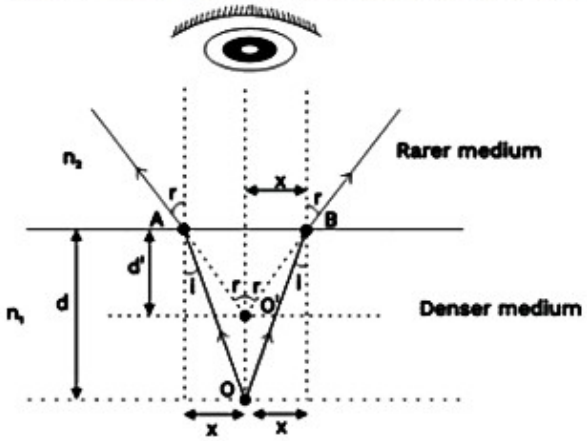
Since, ${}^{a}\mu_g \times {}^{g}{}_{w} = {}^{a}\mu_w$

$\Rightarrow {}^{g}\mu_w = \frac{{}^{a}\mu_g}{{}^{w}\mu_g}$ $\left\{\because {}^{g}{}_{w} = \frac{1}{{}^{w}\mu_g}\right\}$

$\Rightarrow {}^{a}\mu_w = \frac{\left(\frac{3}{2}\right)}{\left(\frac{9}{8}\right)} = \frac{4}{3}$

Apparent Depth and Normal Shift

Case1: When object in a denser medium is seen from a rarer medium.



Suppose a point object is situated in a denser medium, observing it from a rarer medium. Then the object appears to be a shift toward us. The depth seemed to us called apparent depth. Observe ray diagram of this situation

Now, Apply Snell's Law at the Boundary of two media.

$n_1 \sin i = n_2 \sin r$ .....(1)

Since rays are paraxial so i and r are very small.

Use approximation.

$n_1 i = n_2 r$

$n_1 \times \tan_i = n_2 \tan r$

Use ray diagram and put the value of tan i and tan r

$$n_1 \frac{x}{d} = n_2 \frac{x}{d'}$$

$d' = \frac{n_2}{n_1} \times d$

Real depth

This formula known as refraction at plane surface

Apparent depth

Important point:

1. This formula is only valid for paraxial rays.
2. d = Real depth.
3. d′ = Apparent depth.
4. $n_2 = n_2$ is the refractive index of that medium in which rays are going
4. $n_1 = n_1$ is the refractive index of that medium from where rays is incoming.

Normal shift:

The distance between image and object is called normal shift

$S = d - d'$

$$= d - d \times \frac{n_2}{n_1}$$

$$\text{Normal shift (s)} = \left(1 - \frac{n_2}{n_1}\right) d$$

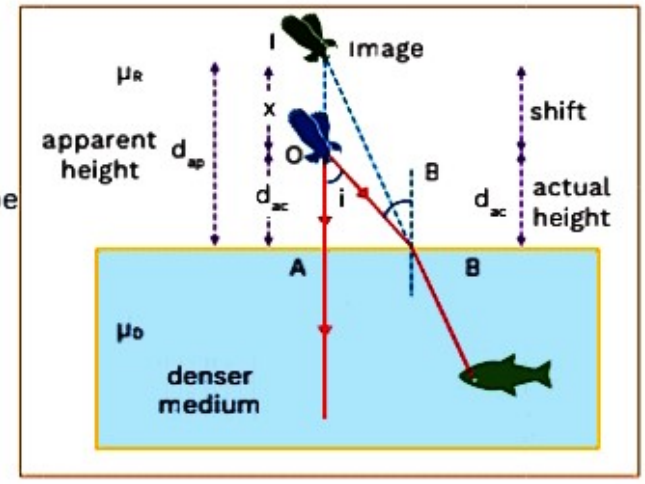
Case 2: When an object in a rarer medium is seen from a denser medium.

$$\frac{d_{ac}}{d_{ap}} = \frac{\mu_1}{\mu_2} = \frac{\mu_R}{\mu_D} = \frac{1}{\mu} (< 1)$$

$$d_{ap} = \mu d_{ac} \quad \text{i.e.,} \quad d_{ap} > d_{ac}$$

So; A high flying object (Eagle) appears to be higher than in reality, and it is given by

$$x = d_{ap} - d_{ac} \Rightarrow x = [\mu - 1] d_{ac}$$



LATERAL SHIFT

Lateral shift is defined as the perpendicular distance between incident and emergent ray. In the given figure, BM is the lateral shift.

$$\text{In } \Delta M'B; \cos r = \frac{(t)}{AB}$$

$$AB = t \sec r$$

$$\text{In } \Delta AMB; \sin(i - r) = \frac{BM}{AB}$$

$$BM = AB \sin(i - r)$$

$$BM = (t \sec r) \sin(i - r)$$

$$BM = (t \sec r) \sin(i - r).$$

in.

Problem Based on Lateral Shift :

Q **Light ray, while it is, passes through a parallel glass slab of thickness 10 cm placed in air. The refractive index of the material is $\sqrt{3}$. Find the lateral shift.**

Sol: Apply Snell's Law :-

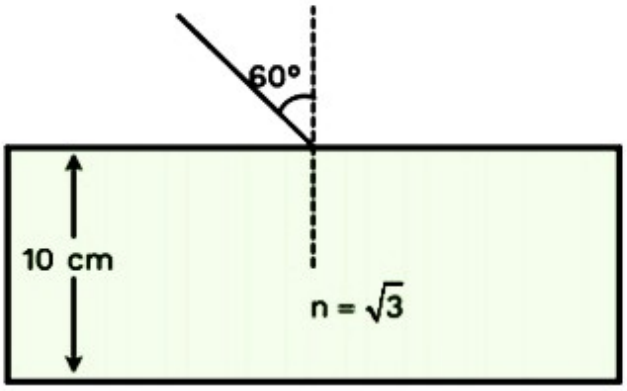
$n_1 \sin 60° = n_2 \sin r$

$$\sin r = \frac{1 \times \sqrt{3}}{2 \times \sqrt{3}}$$

$r = 30°$

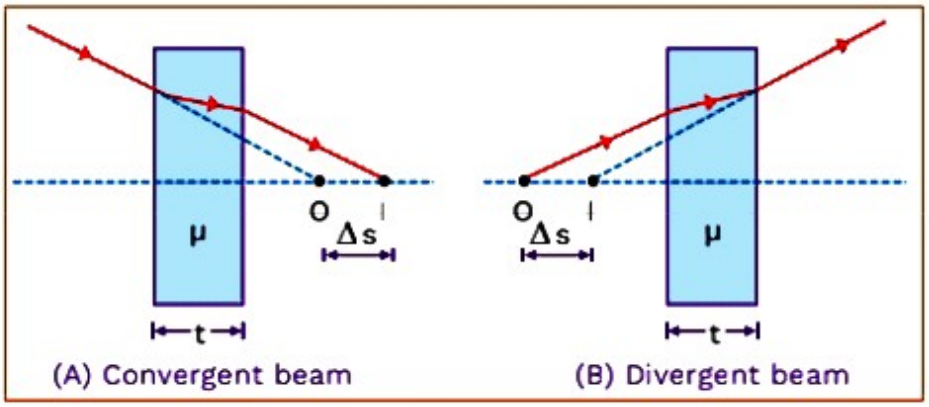
Lateral shift = 10 sec 30° sin(60 – 30°)

$$= \frac{10}{\sqrt{3}} \times 2 \times \frac{1}{2} = \frac{10}{\sqrt{3}}$$



The shift from the slab

When a glass slab is placed in between object and observer, then the position of an object is shifted from the original position.



$$\Delta s = t\left(1 - \frac{1}{\mu}\right)$$

MULTISLABS

Suppose If a number of slabs or immiscible liquids of depth d_1, d_2, d_3..... and refractive index μ_1, μ_2, μ_3, ... are placed one over the other then.

$d = d_1 + d_2 + d_3 + ...$

Now The apparent depth can be written as.

$$d' = \frac{d_1}{\mu_1} + \frac{d_2}{\mu_2} + \frac{d_3}{\mu_3} + ...$$

Therefore, an equivalent refractive index can be calculated as

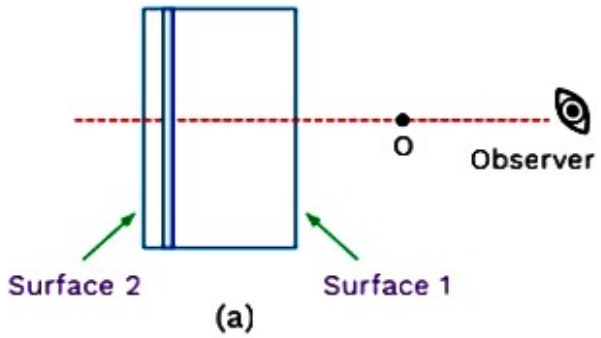
$$\mu = \frac{d}{d'} = \frac{d_1 + d_2 + d_3 + \ldots}{\left(\frac{d_1}{\mu_1}\right) + \left(\frac{d_2}{\mu_2}\right) + \left(\frac{d_3}{\mu_3}\right) + \ldots} = \frac{\Sigma d_i}{\Sigma\left(\frac{d_i}{\mu_i}\right)}$$

Special Case:
If there are only two slabs of equal thickness

$$\mu = \frac{d + d}{\left(\frac{d}{\mu_1}\right) + \left(\frac{d}{\mu_2}\right)} = \frac{2\mu_1\mu_2}{\mu_2 + \mu_2} = \text{harmonic mean of } \mu_1 \text{ and } \mu_2$$

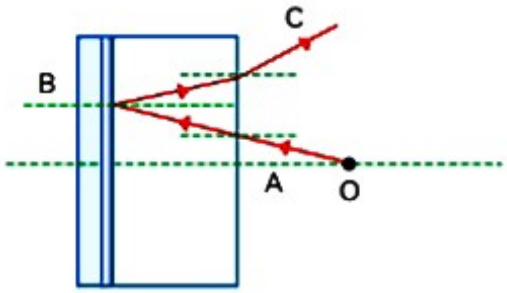


Combination of mirror and Slab:
Let us observe what happens if one surface of the slab is silvered. Consider the silvered slab shown in Fig. (a). suppose an object is placed in front of a silvered glass slab.



(a)

Here, a ray of light from the object first refracts at surface 1. It is then reflected from surface 2 before refracting again at surface 1 and emerging [see Fig. (b)]. So, we can consider a silvered slab as a combination of



(b)

1. A refracting surface,
2. A reflecting surface, and
3. A refracting surface again.

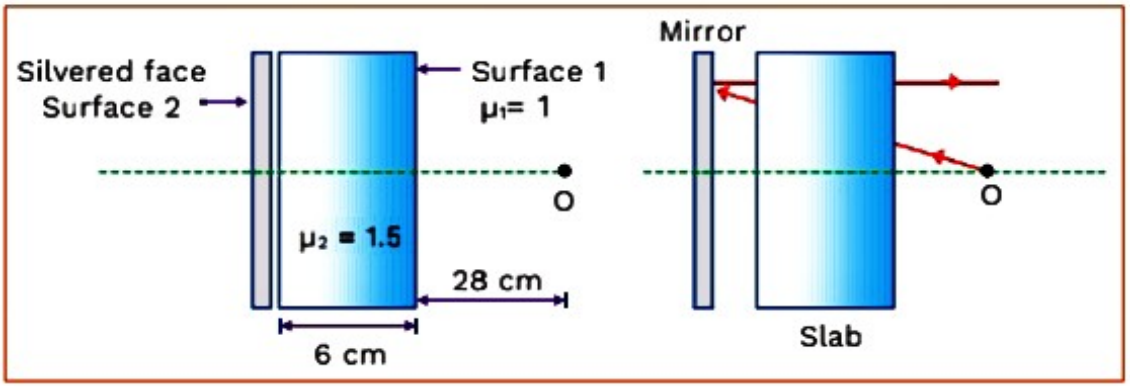
The above situation can be considered as a combination of a slab and a plane mirror placed together. Thus, a silvered slab is a combination of

1. A glass slab,
2. A plane mirror, and
3. A glass slab again.

Q **Let an object is placed in front of a slab of thickness 6 cm at a distance 28 cm from it. The other face of the slab is silvered. Find the postion of the final image. Given refractive index is 1.5**

Sol: **Use the method of interface:** A ray of light from the object O undergoes under following phenomenon 1. refraction, 2. reflection, and then 3. refraction.

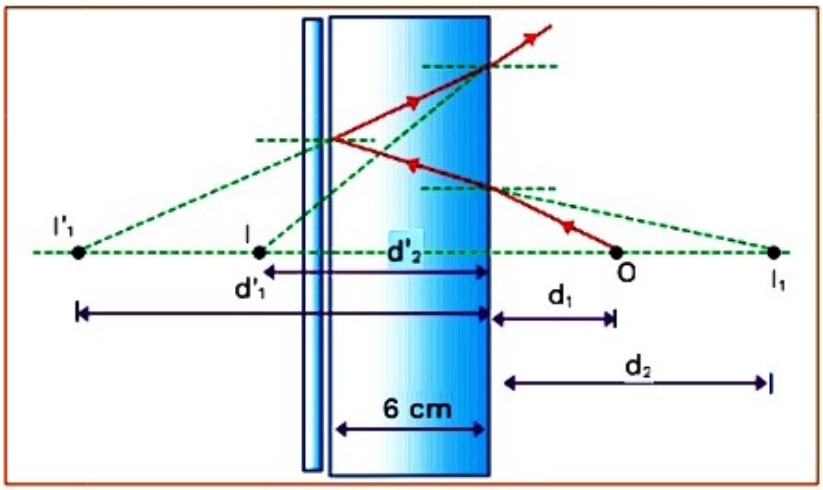
Refraction at surface 1



Here, $\mu_1 = 1$, $\mu_2 = 1.5$, $d_1 = 28$ cm, $d_2 = ?$

Since $d_2 = \frac{d_1}{n_{relative}} = \frac{d_1}{(1/\mu)}; \left[n_{rel} = \frac{n_{incident}}{n_{refracted}} = \frac{1}{\mu}\right]$

$\Rightarrow \quad d_2 = \mu d_1 = 1.5 \times 28 = 42$ cm



Therefore, d_2 = 42 cm from the first interface. The first image I_1 is formed 42 cm in front of the slab.

Reflection at surface 2

The object for reflection at the second surface is the image from refraction at the first.

Therefore, object distance from the mirror is = 42 + 6 = 48 cm. As a result of reflection, the image will be formed as far behind the mirror as the object is in front of it. Therefore, the second image I'_1 is formed 48 cm behind the mirror.

Second refraction at surface 1

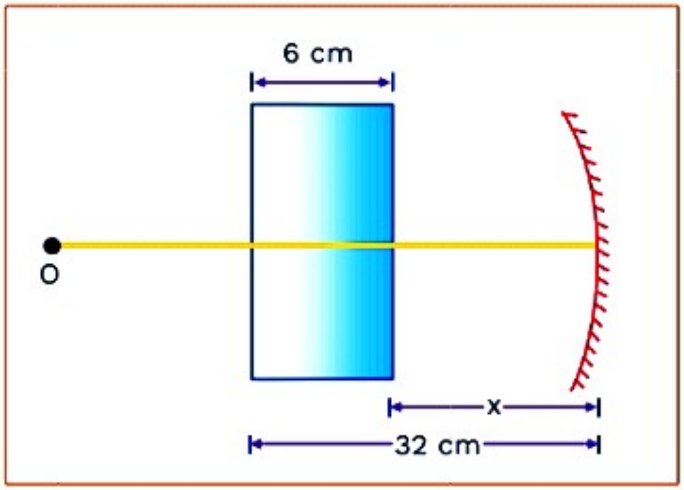
Object distance from surface 1.

$d_1 = 48 + 6 = 54$ cm

$$d_2' = \frac{d_1'}{n_{relative}} = \frac{d_1'}{\left(\frac{n_{incident}}{n_{refraction}}\right)} = \frac{54}{(1.5/1)} = 36\text{cm}$$

So, the final image is at a distance of d_2 = 36 cm behind the first interface.
Hence, the final image is formed 36 cm behind surface 1 or 30 cm behind surface 2.

Q **A point object O is placed in front of a concave mirror of focal length 10 cm. A glass slab of refractive index $\mu = \frac{3}{2}$ and thickness of 6 cm is inserted between object and mirror.**



Find the position of final image when the distance x shown in the figure is (a) 5 cm (b) 20 cm

Sol: The normal shift produced y a glass slab is given by

$$\Delta x = \left(1 - \frac{1}{\mu}\right) t = \left(1 - \frac{2}{3}\right)(6) = 2\text{cm}$$

i.e., for the mirror the object is placed at a distance $(32 - \Delta x) = 30$ cm from it.

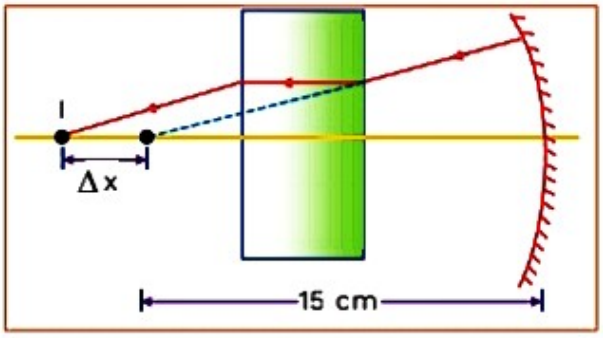
Applying mirror formula i.e $\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$, we get

$$\frac{1}{v} - \frac{1}{30} = -\frac{1}{10}$$

$\Rightarrow$ $v = -15$ cm

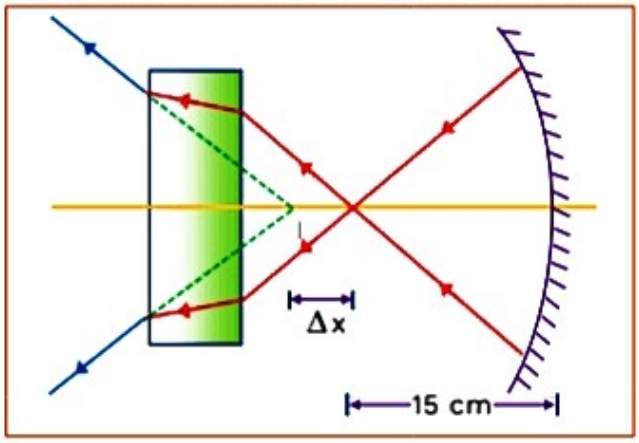
(a) When $x = 5$ cm

The light falls on the slab after being reflected from the mirror, as shown. But the slab will again shift it by a distance $\Delta x = 2$ cm. Hence, the final real image is formed at a distance $(15 + 2) = 17$ cm from the mirror.

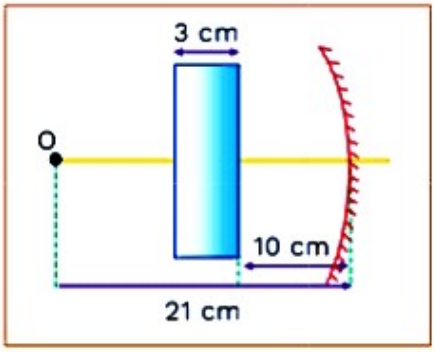


(b) When x = 20 cm

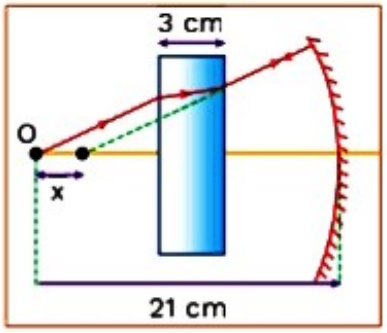
This time also the final image is at a distance of 17 cm from the mirror, but it is virtual, as shown in figure.



Q **An object is placed 21 cm in front of a concave mirror with a radius of curvature of 20 cm. A glass slab of thickness 3 cm and refractive index 1.5 is placed closed to the mirror in space between the object and the mirror. Find the position of the final image formed if the distance of the nearer surface of the slab from the mirror is 10 cm.**



Sol: Shift by slab $x = d\left(1-\frac{1}{\mu}\right) = 3\left(1-\frac{1}{1.5}\right) = 1\,cm$



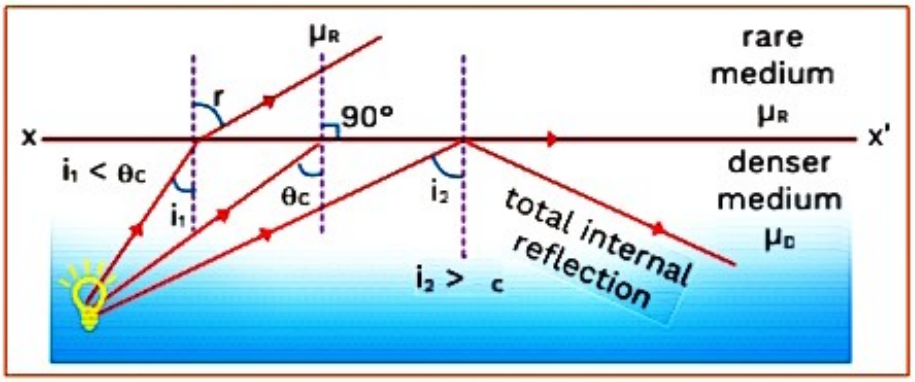
for image formed by mirror u = – (21 – 1) cm=–20 cm.

$\frac{1}{u}+\frac{1}{v}=\frac{1}{f} \Rightarrow \frac{1}{-20}+\frac{1}{v}=\frac{1}{-10} \Rightarrow v = -20cm$

shift in the direction of light v = – (20 + 1) = –21 cm.

TOTAL INTERNAL REFLECTION

If the incident ray falls from a denser medium to a rarer medium, then the angle of incidence for which the angle of refraction becomes 90° is known as the critical angle. If the angle of incidence is greater than the critical angle, then the surface starts acting as a reflecting surface. This phenomenon knows as total angle reflection.



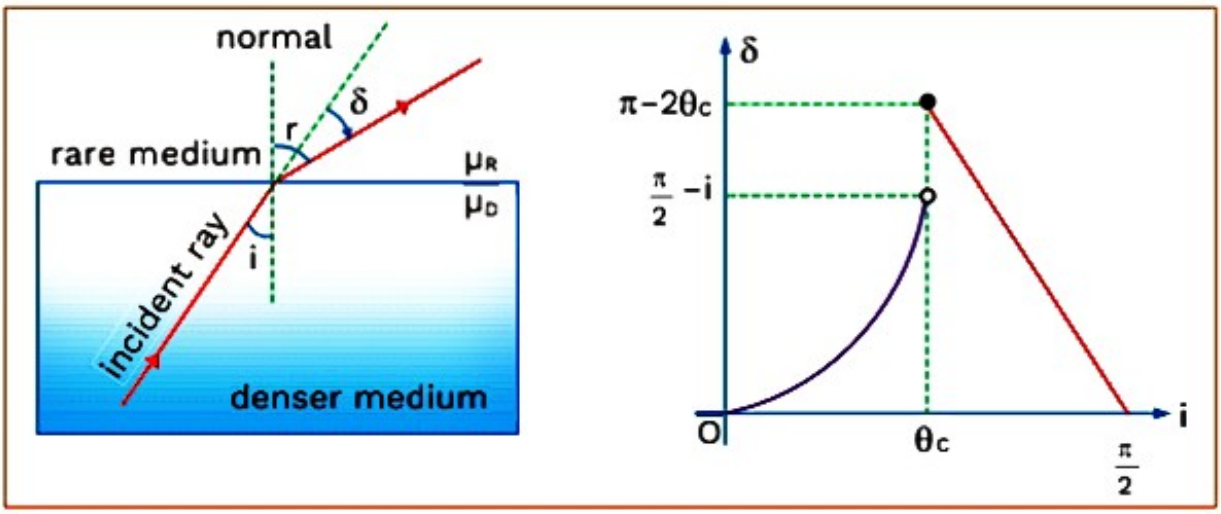
CONDITIONS

- Angle of incidence > critical angle [$i > \theta_c$]
- Light rays should travel from denser medium to rare medium, for example, Glass to air or water to air, etc.

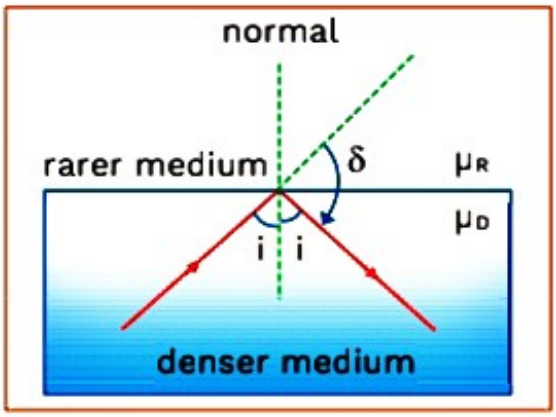
Snell's law at boundary xx', $\mu_D \sin\theta_c = \mu_R \sin 90 \Rightarrow \sin\theta_c = \frac{\mu_R}{\mu_D}$

The graph between angle of deviation (δ) and angle of incidence (i) as rays goes from denser to rare medium

- If $i < \theta_c$, $\mu_D \sin i = \mu_R \sin r$; $r = \sin^{-1}\left(\frac{\mu_D}{\mu_R}\sin i\right)$ so $\delta = r - i = \sin^{-1}\left(\frac{\mu_D}{\mu_R}\sin i\right) - i$



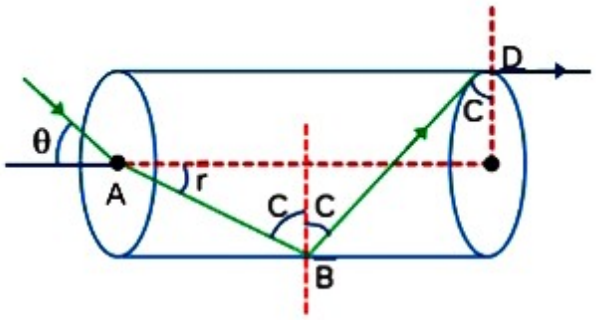
- If $i > \theta_c$; $\delta = \pi - 2i$



Q **Light is incident making an angle θ with the axis of a transparent cylindrical fiber of refractive index $n = \sqrt{\frac{5}{4}}$ as shown in the figure. Determine the maximum value of θ so that the light entering the cylinder does not come out of the curved surface.**



Sol: The ray of light is incident at A and it just gets reflected totally at B. Therefore incident angle at B is equal to the critical angle given as $C = \sin^{-1}\left(\frac{1}{n}\right)$



Snell's law of refraction at A gives.

$$\frac{\sin\theta}{\sin r} = n$$

$$\Rightarrow \quad \sin r = \frac{\sin\theta}{n} \qquad \text{....(1)}$$

Since $r + C = 90°$

$$\Rightarrow \quad \sin r = \sin(90° - C) = \cos C$$

For a ray not to come through the curved surface,

$$r \leq 90 - C$$

$$\Rightarrow \quad \sin r \leq \sqrt{1 - \sin^2 C} \leq \sqrt{1 - \frac{1}{n^2}} \qquad \text{....(2)}$$

Eliminating sin r from (1) and (2), we get

$$\frac{\sin\theta}{n} \leq \sqrt{1 - \frac{1}{n^2}}$$

$$\Rightarrow \quad \sin\theta \leq \sqrt{n^2 - 1}$$

$$\Rightarrow \quad \sin^2\theta \leq 1.25 - 1$$

$$\Rightarrow \quad \sin^2\theta \leq 0.25$$

$$\Rightarrow \quad \sin\theta \leq \frac{1}{2}$$

$$\Rightarrow \quad \theta \leq 30°$$

$$\Rightarrow \quad \theta_{max} = 30°$$

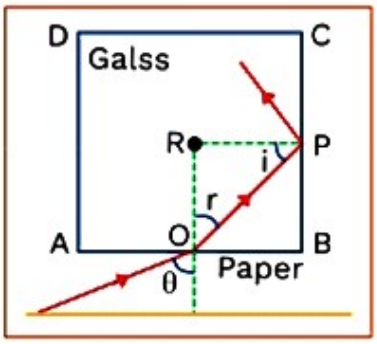
Q **A rectangular block of glass is placed on a printed page lying on a horizontal surface. Find the minimum value of the refractive index of glass for which the letters on the page are not visible from any of the vertical faces of the block.**

Sol: Light will not emerge out from the vertical face BC,
when

$i >$ Critical Angle (C)

$\Rightarrow \sin i > \sin C$

$\Rightarrow \sin i > \frac{1}{\mu}$ $\quad \left\{\because \sin C = \frac{1}{\mu}\right\}$



Applying Snell's law at O, we get

$1 \sin\theta = \mu \sin r$

$\Rightarrow \sin\theta = \mu \sin(90° - i) = \mu \cos i$

$\Rightarrow \cos i = \frac{\sin\theta}{\mu}$

$\Rightarrow \sin i = \sqrt{1 - \cos^2 i} = \sqrt{1 - \frac{\sin^2\theta}{\mu^2}}$

Therefore, the condition for no light to emerge from vertical face BC becomes,

$$\sqrt{\frac{\mu^2 - \sin^2\theta}{\mu^2}} > \frac{1}{\mu}$$

$\Rightarrow \mu^2 > 1 + \sin^2\theta$

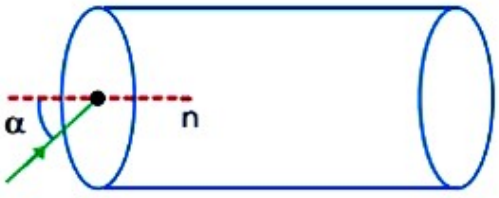
$\Rightarrow \mu^2 > 1 + 1$ $\quad [\because$ maximum value of θ can be $90°]$

$\Rightarrow \mu > \sqrt{2}$

So, the minimum value of refractive index is.

$\mu_{min} = \sqrt{2}$

Q **There is a transparent rod as shown in the figure. A Light ray is incident at an angle α on one planar end. The refractive index of the rod is n. Calculate the least value of n so that the light entering the rod does not emerge out from the curved surface of the rod irrespective of the value of α.**



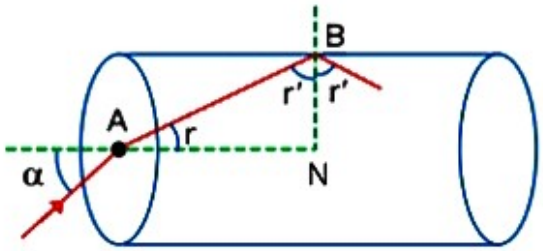
Sol: Since, $\sin C = \frac{1}{n}$

In triangle ABN, $r' + r + 90° = 180°$

$\Rightarrow r' = 90 - r$

$\Rightarrow (r')_{min} = 90° - (r)_{max}$

and $n = \frac{\sin i_{max}}{\sin r_{max}} = \frac{\sin 90°}{\sin (r)_{max}}$; $(i_{max} = 90°)$



Then, $\sin(r)_{max} = \frac{1}{n} = \sin C$

$(r')_{max} = C$

$\Rightarrow (r')_{min} = (90° - C)$

Now, if minimum value of r' i.e., $90° - \theta_c$ is greater than θ_c, then obviously all values of r' will be greater than θ_c i.e., total internal reflection will take place at face AB in all conditions. Therefore, the necessary condition is.

$(r')_{min} \geq C$

$\Rightarrow (90° - C) \geq C$

$\Rightarrow \sin(90° - C) \geq \sin C$

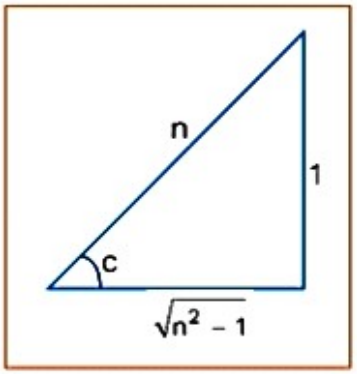
$\Rightarrow \cot C \geq 1$

$\Rightarrow \sqrt{n^2 - 1} \geq 1$

$\Rightarrow n^2 \geq 2$

$\Rightarrow n \geq \sqrt{2}$

Therefore, minimum value of n is $\sqrt{2}$

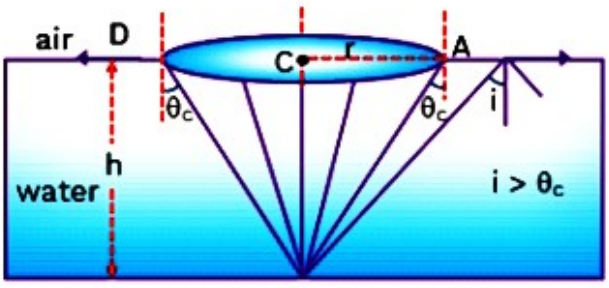


LIGHT SOURCE INSIDE A MEDIUM (WATER):

Suppose a light source is situated at the bottom index μ.

The height of the liquid is h.

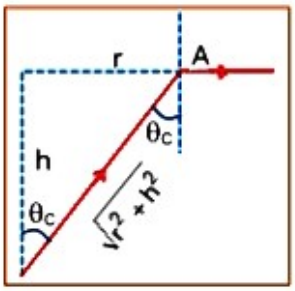
Now use the phenomenon of total internal reflection; you can see that light emerging out from liquid in a circle. You can see a circular patch of light.



$$\sin\theta_c = \frac{r}{\sqrt{r^2+h^2}} \;\&\; \sin\theta_c = \frac{1}{\mu} \Rightarrow \frac{1}{\mu} = \frac{r}{\sqrt{r^2+h^2}} \Rightarrow \frac{1}{\mu^2} = \frac{r^2}{r^2+h^2}$$

$$\Rightarrow \mu^2 r^2 = r^2 + h^2 \Rightarrow \left(\mu^2 - 1\right) r^2 = h^2 \Rightarrow \text{radius of circular portion}$$

$$r = \frac{h}{\sqrt{\mu^2 - 1}} \text{ and area } = \pi r^2$$



Important Points

A diver in water at a depth d sees the world outside through a horizontal circle of radius. r = d tan θ_c.

In case of total internal reflection, as all (i.e. 100%) incident light is reflected back into the same medium, there is no loss of intensity, while in case of reflection from mirror or refraction from lenses, there is some loss of Intensity as all light can never be reflected or refracted. This is why images formed by TIR are much brighter than those formed by mirrors or lenses.

REFRACTION IN A MEDIUM WITH VARIABLE REFRACTIVE INDEX

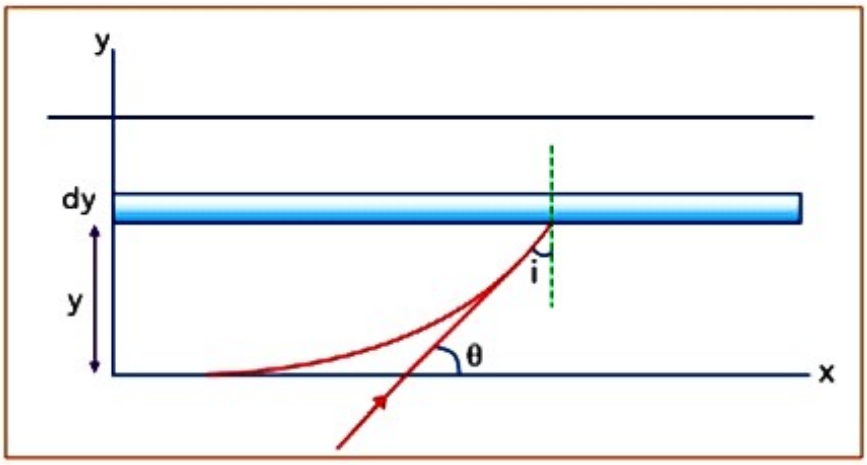
So far, we have assumed that the refractive index of the slab is a constant. This need not be necessarily true. An example of such a situation is the atmosphere. The atmosphere becomes thinner as we go up. Hence, the refractive index of air is highest close to the surface of the earth and decreases as we move upward. How do we analyze such a problem?

Divide the medium into different layers. This indicates that each layer has a different refractive index having value according to a given expression. If μ is a function of x, then a differential layer will be a thin layer of thickness dx. If μ is a function of y, then a differential layer will be a thin layer of thickness dy.

At a given layer, let the angle of incidence be i. Relate this angle of incidence to the initial condition and the refractive index at that point using the relation.

μ sin (i) = constant ...(1)

From the given ray diagram (figure), draw a tangent to the path taken by the ray of light. Geometrically, relate the slope of this tangent to the angle of incidence.

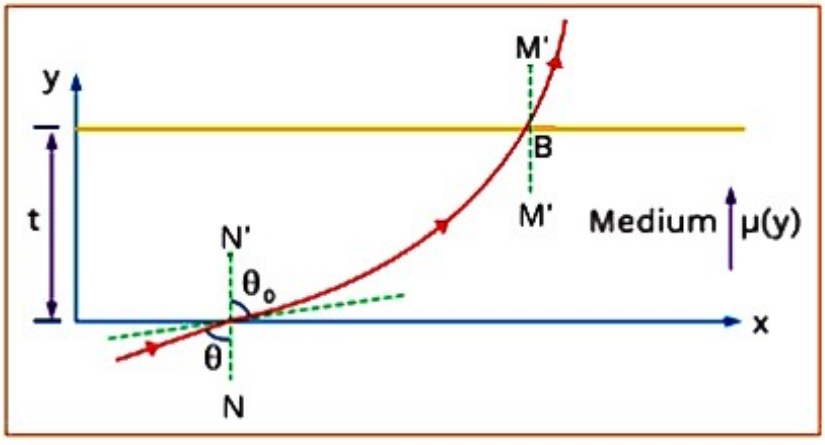


$$\tan\theta = \frac{dy}{dx} = \tan(90 - i)$$

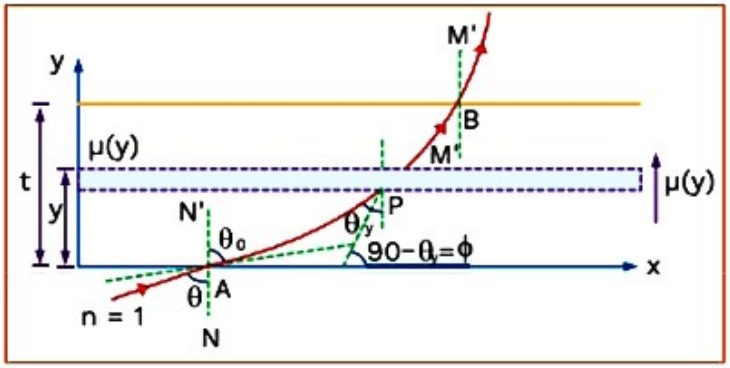
Substitute for i from Eq. (i) and determine dy/dx as a function of x and y.
Integrate and obtain an expression for y as a function of x. Let us discuss a case where the refractive index is changing in the y-direction.

As 1. $\mu = f(y)$

Consider a slab with a medium where the refractive index varies from μ_1 to μ_2 in relation to y. Assume that the outside medium is air. Let light be incident on the glass slab at an angle θ. The light ray strikes the glass slab at the points. The normal at this point is NN' as shown in figure. Once the ray enters the medium, the refractive index varies continuously as if there are infinite number of successive glass slabs. Therefore, the angle of incidence also varies continuously, and the path of the ray is curved. The ray finally reaches the second surface at the point B before exiting the slab, once again refracting at the exit point.



How can we derive a mathematical expression for the equation of the ray in the medium? Consider refraction at some height y (figure). Here, the angle of incidence is θ_y and the refractive index of the medium is μ_y.



The product of the refractive index and the sine of the angle of incidence is constant. Therefore, at the height y, we can say that

$$\mu_y \sin\theta_y = \text{constant} \qquad \ldots(i)$$

Furthermore, applying the law of refraction at point A which is the interface between air and the medium, we have.

$$1 \times \sin(\theta) = \mu_y \sin\theta \qquad \ldots (ii)$$

$$\therefore \quad \sin\theta_y = \frac{\sin\theta}{\mu_y}$$

$$\cos\theta_y = \sqrt{\frac{\mu_y^2 - \sin^2\theta}{\mu_y^2}}$$

$$\text{Hence, } \cot\theta_y = \frac{\sqrt{\mu_y^2 - \sin^2\theta}}{\sin\theta}$$

Now, the slope of the curve at point P is given by.

$$\tan\phi = \frac{dy}{dx} \qquad \ldots(iii)$$

But $\phi = 90 - \theta_y$

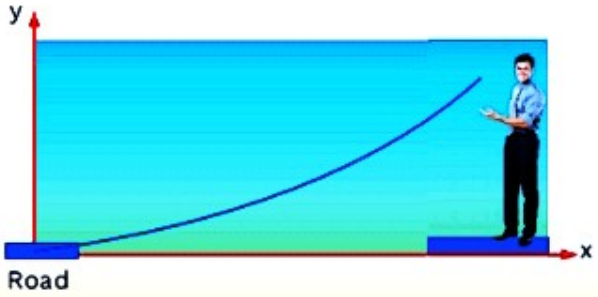
$$\text{Therefore, } \frac{dy}{dx} = \cot\left(\theta_y\right) = \frac{\sqrt{\mu_y^2 - \sin^2\theta}}{\sin\theta} \qquad \ldots..(iv)$$

$$\frac{\sin\theta dy}{\sqrt{\mu_y^2 - \sin^2\theta}} = dx$$

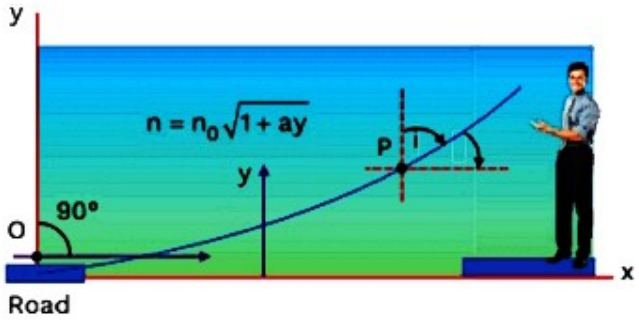
We are integrating both sides with proper limits.
If the functional dependence of " μ_y" is known, we can easily solve the equation of the curve.

$$\sin\theta \int_{y_1}^{y_2} \frac{dy}{\sqrt{\mu_y^2 - \sin^2\theta}} = \int_{x_1}^{x_2} dx$$

Q **Due to variation in temperature refractive index of the atmosphere changes along the y-axis. Suppose the index of refraction varies as $n = n_0\sqrt{1+ay}$, where n_0 is the index of refraction at the earth's surface and $a = 2.0 \times 10^{-6}\ m^{-1}$. Suppose A man of height 2 meters stands on a level surface. Find beyond what distance he cannot see the runway?**



Sol: We can assume a number of thin layers of air placed parallel to the x-axis. Suppose 'O' be the distant object which is just visible to the man. Let P be a point on the trajectory of the light ray.



From Snell's law, n.sin i = constant
At the surface $n = n_0$ and $i = 90°$

$$n_0 \sin 90° = n \sin i = (n_0\sqrt{(1+ay)}) \sin (90° - \theta)$$

$$\Rightarrow \quad \cos\theta = \frac{1}{\sqrt{1+ay)}}$$

$$\Rightarrow \quad \tan\theta = \sqrt{ay}$$

$$\tan\theta = \sqrt{ay} = \frac{dy}{dx} \Rightarrow dx = \frac{dy}{\sqrt{ay}}$$

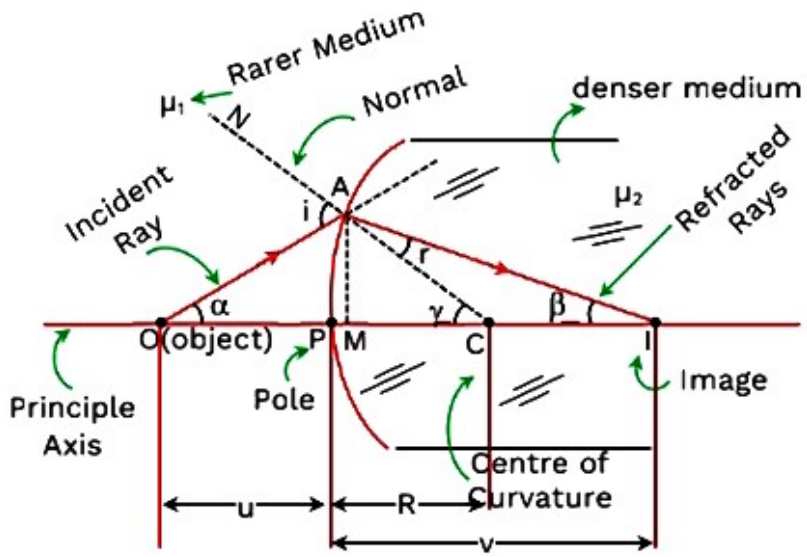
$$\int_0^x dx = \int_0^y \frac{dy}{\sqrt{ay}} \Rightarrow x = \frac{1}{\sqrt{a}}\left[\frac{y^{\left(-\frac{1}{2}+1\right)}}{\left(-\frac{1}{2}+1\right)}\right]_0^y = 2\frac{y^{\frac{1}{2}}}{\sqrt{a}} = 2\sqrt{\frac{y}{a}}$$

Hence equation of trajectory $x = 2\sqrt{\frac{y}{a}}$ or $y = \frac{ax^2}{4}$

On substituting y = 2.0m and $a_0 = 2 \times 10^{-6} m^{-1}$,

$$x_{max} = 2\sqrt{\frac{2}{2 \times 10^{-6}}} = 2000m$$

Refraction of Light at Spherical Surfaces



∠ i = Angle of incidence
∠r = Angle of refraction

(Fig. –a)

A refracting surface that forms a part of a sphere of a transparent medium is called a spherical refracting surface.

The following terms are related to the study of refraction from spherical refracting surfaces.

Pole The Centre of the spherical surface is called its pole (P).

Centre of Curvature

The center of the sphere of which this refracting surface is part is called the center of curvature of the refracting surface. It is denoted by 'C.'

Radius of Curvature

It is the radius of the Spherical surface.

Aperture

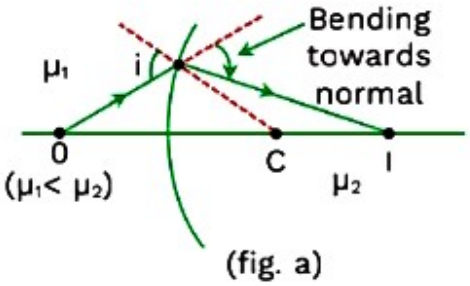
The diameter of the spherical refracting surface is called its aperture.

Principle Axis

The line PC produced on either side of the refracting surface is called its principal axis.

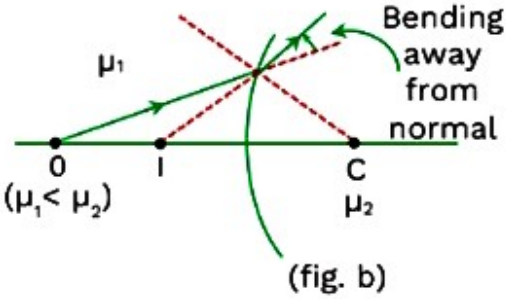
Bending of Light Ray

When goes from rarer to denser; Bends toward normal



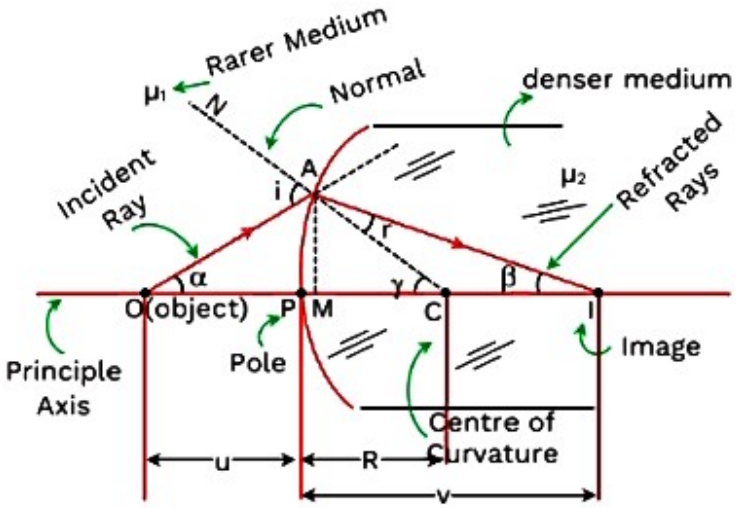
(fig. a)

When goes from denser to rarer bends away from normal



(fig. b)

Derivation of Formula of Spherical Refraction



∠ i = Angle of incidence
∠ r = Angle of refraction

fig. -(a)

Assumption

(1) The aperture of the refracting surface is very small so that distance of any point from the surface is almost equal to the distance of that point from the pole of the surface.

(2) The angle of incidence and consequently the angles of refraction are small.
Since Aperture is very small, rays are paraxial, so angle ∠i and ∠r are very small; this implies that angles α, β and γ are also very small.

Let O be a point object lying at a distance u from P on the principal axis of the spherical surface. A ray of light OA (incident ray) starting from O meets the refracting surface at A. After refraction, it bends toward the normal CAN and meets the principal axis again at I. In another ray of light, OP is normally falling on the refracting surface along the principal axis, suffers no deviation, and passes along PC.

The refracted rays PC and AI actually meet at I, which is, therefore, the real image of the point object O.

Let $\angle OAN = i$ $\angle CAI = r$

Applying Snell law at 'A'

$$\mu_1 \sin i = \mu_2 \sin r$$

As angles I and 'r' are small,

$\sin i \approx i$ and $\sin r \cong r$

$$\mu_1 i = \mu_2 r \quad \ldots(1)$$

In ΔOAC, $i = \gamma + \alpha$ and in ΔIAC, $r = \gamma - \beta$

From Eq. (1) $\mu_1(\gamma + \alpha) = \mu_2(\gamma - \beta)$

$$\mu_1\alpha + \mu_2\beta = (\mu_2 - \mu_1)\gamma \quad \ldots(2)$$

As the aperture is small, the angles α, β and γ are also called and as such these can be replaced by their Tangents.

Thus, $$\alpha \approx \tan\alpha = \frac{AM}{MO} \approx \frac{AM}{PO}$$

$$\beta \approx \tan\beta = \frac{AM}{MI} \approx \frac{AM}{PI}$$

$$\gamma \approx \tan\gamma = \frac{AM}{MC} \approx \frac{AM}{PC}$$

From Equation –(2)

$$\mu_1 \frac{AM}{PO} + \mu_2 \frac{AM}{PI} = (\mu_2 - \mu_1)\frac{AM}{PC}$$

or $$\frac{\mu_1}{PO} + \frac{\mu_2}{PI} = \frac{(\mu_2 - \mu_1)}{PC} \quad \ldots(3)$$

From Eq. - (2)

$$\left.\begin{matrix} PO = -u \\ PI = v \\ PC = R \end{matrix}\right\} \quad \frac{\mu_1}{-u} + \frac{\mu_2}{v} = \frac{\mu_2 - \mu_1}{R}$$

$$\boxed{\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}}$$

So; Formula of Spherical Refraction

$$\boxed{\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}}$$

Important Points -

(a) Put u, v, R with the proper sign.

(b) Sign convection same as a mirror.

(c) $\mu_2 = \mu_2$ is the refractive index of that medium in which rays are going.

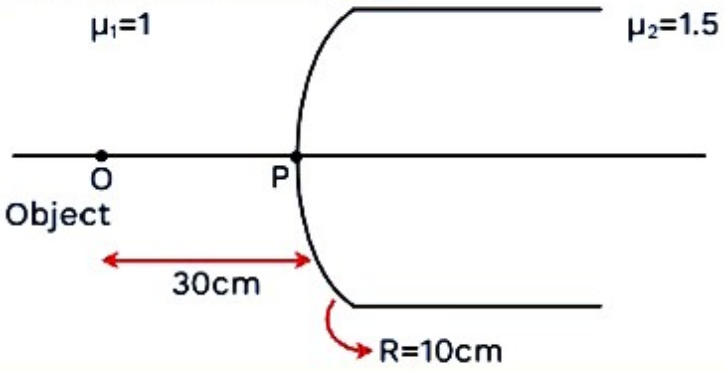
(d) $\mu_1 = \mu_1$ is the refractive index that medium from where rays are incoming.

Note - This formula

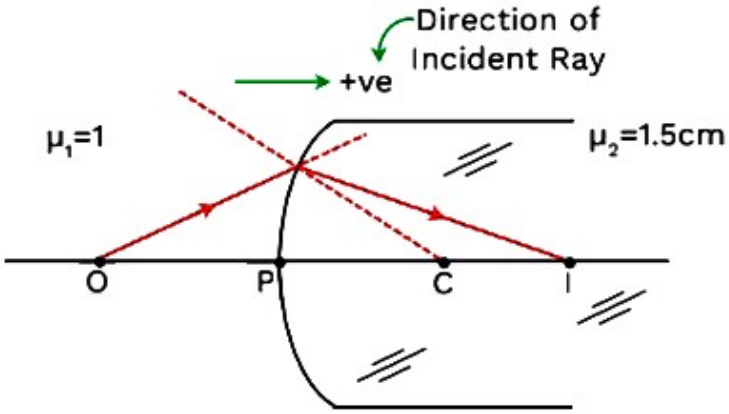
$$\boxed{\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}}$$

is valid for both concave and convex spherical surfaces.

Q **For given figure find location of image.**



Sol:



Use the formula of spherical refraction

$$\boxed{\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}}$$

Use sign convection

v = ? (Unknow variable ; do not use sign)
u = −30 cm
R = + 10 cm
μ_2 = 1.5 (where ray is going)
μ_1 = 1 (From where ray is incoming)

$$\frac{1.5}{V} - \frac{1}{-30} = \frac{(1.5-1)}{+10}$$

$$\frac{1.5}{v} + \frac{1}{30} = \frac{0.5}{10}$$

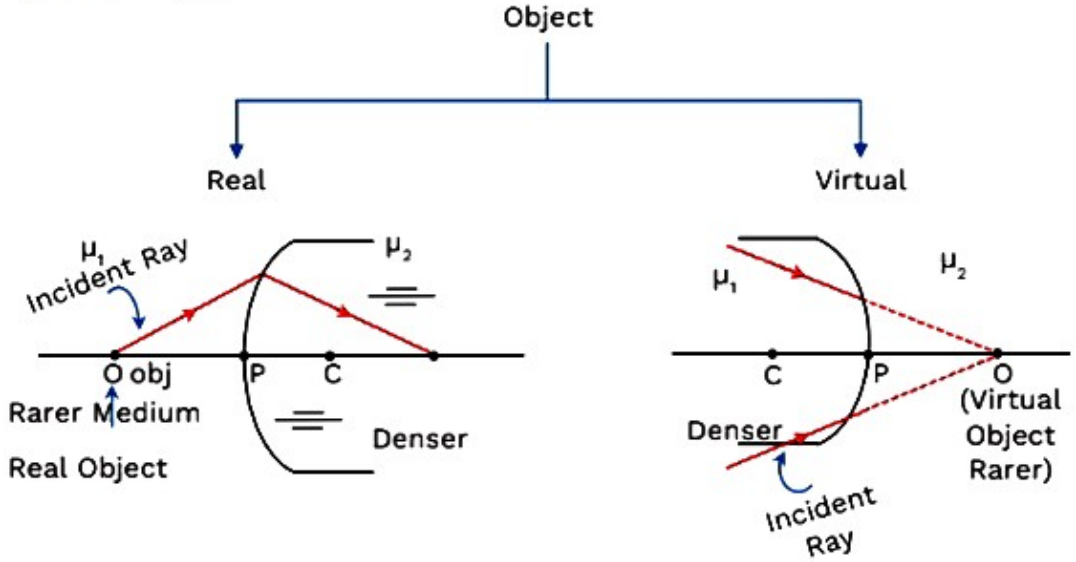
$$\frac{1.5}{V} = \frac{0.5}{10} - \frac{1}{30}$$

$$= \frac{1.5-1}{30} = \frac{0.5}{30}$$

$$\boxed{V = +90\,cm}$$

- Plus, sing indicate that it is formed in the direction of incident ray.
- It is real image.

Type of Image and Object



- Incident Rays responsible for object formation.

- Incident Rays responsible for object formation.

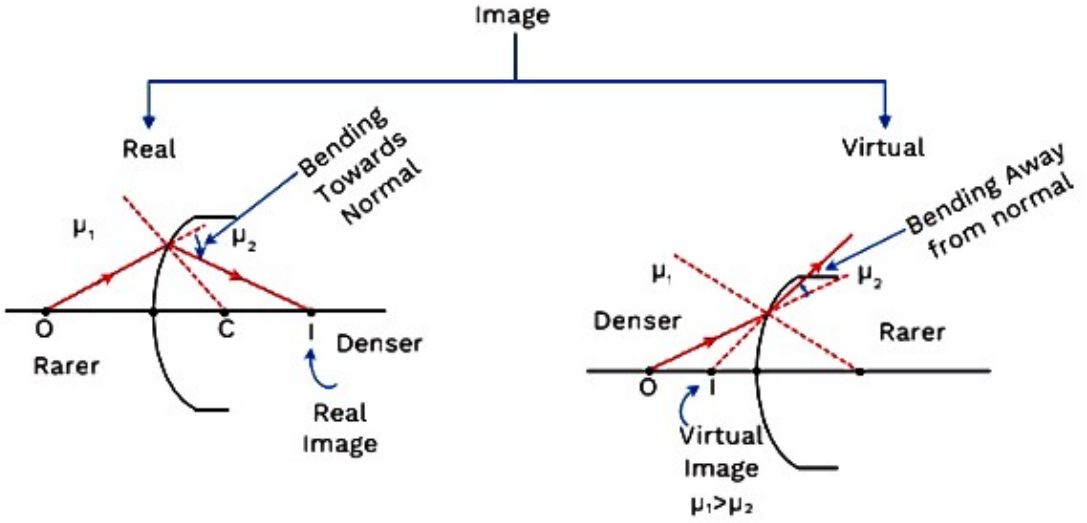
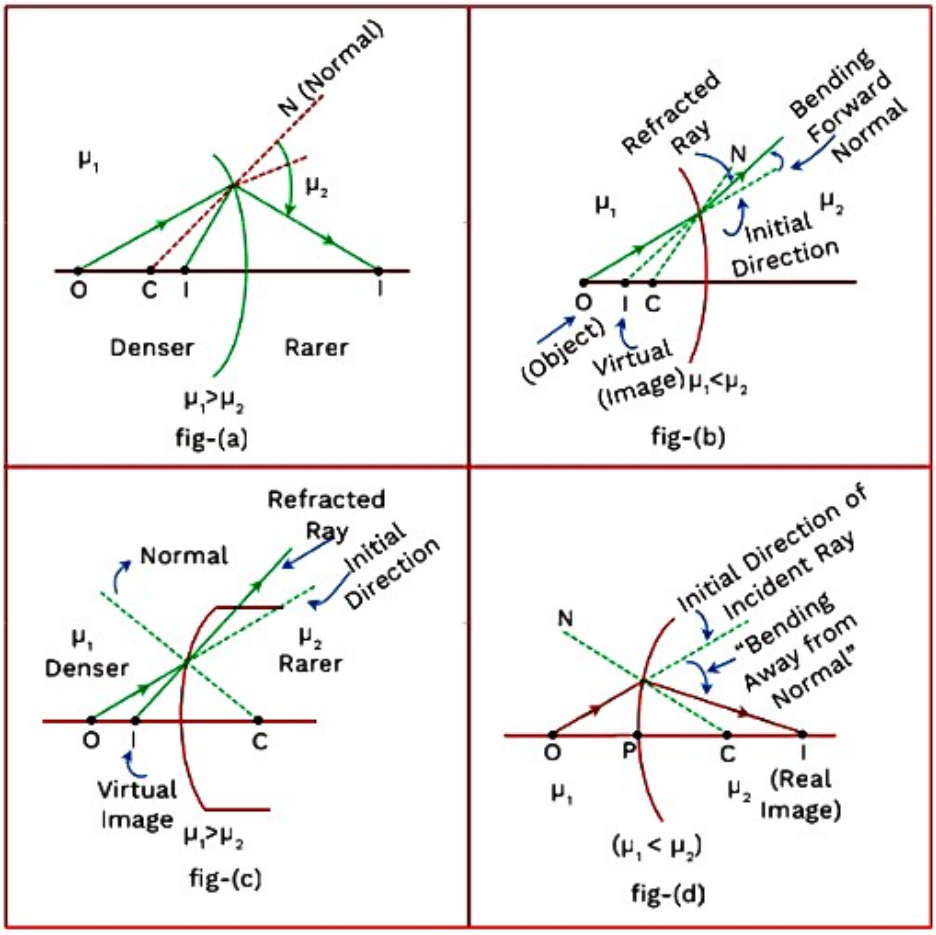


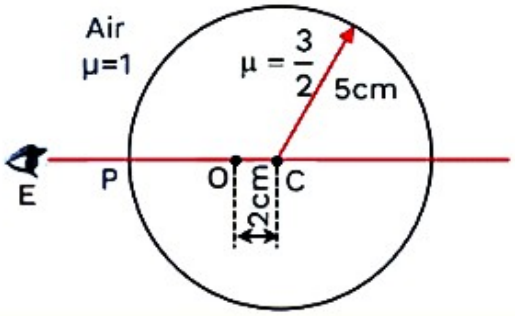
fig-(a)

fig-(b)

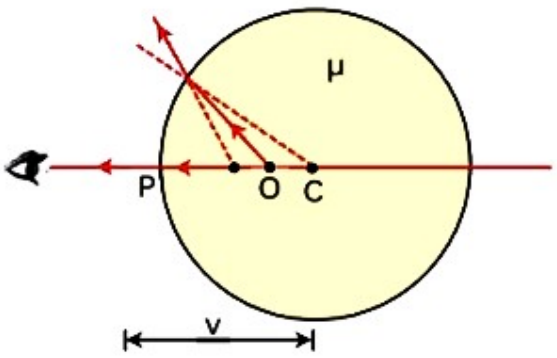
Summary



Q **The figure shows a solid glass sphere of radius 5 cm that has a small air bubble O trapped at a distance of 2 cm from the centre C. The refractive index of the material of glass is 1.5. Find the apparent position of the bubble where it will appear when seen through the surface of the sphere from an outside point E.**



Sol: Here rays are striking on the concave side of the sphere



Using, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{(\mu_2 - \mu_1)}{R}$

Here, $u = -PO = -(PC - OC) = -(5-2)\text{cm} = -3\text{cm}$

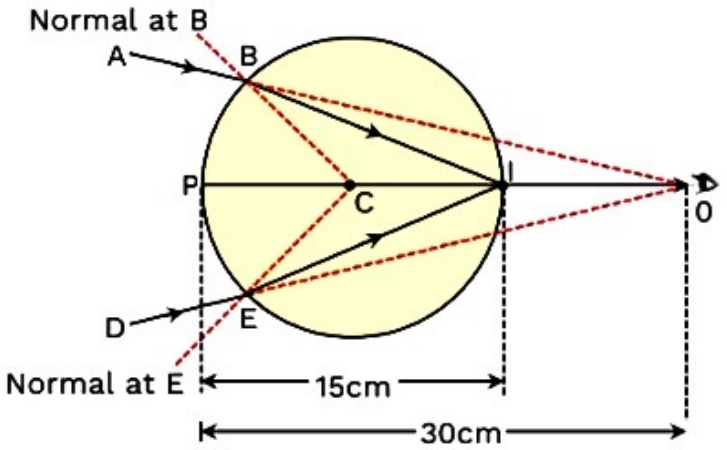
$R = -5\text{cm};$ and $\mu_1 = \frac{3}{2}$ and $\mu_2 = 1$

$$\frac{1}{v} - \frac{3/2}{(-3)} = \frac{(1-3/2)}{(-5)} \Rightarrow v = -\frac{5}{2}\text{cm} = -2.5\,\text{cm}$$

The course of rays giving the apparent position I of the air bubble O trapped inside the glass sphere will be as shown in the figure.

Q **A beam of light strikes a glass sphere of diameter 15 cm converging towards a point 30 cm behind the pole of the spherical surface. Find the position of the image if the refractive index of glass is 1.5.**

Sol: Let the incident rays on the sphere from left. In the absence of the spherical surface, they converge at O, 30 cm from the pole P. Thus, O serves as the virtual object for the first refraction at left surface.



Here, $u = PO = +30\,cm$, $R = \frac{15}{2}\,cm, \mu_1 = 1$ and $\mu_2 = \frac{3}{2}$

Let the distance of the image I from P be v.

Using, $\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$

or $\frac{3/2}{v} - \frac{1}{(+30)} = \frac{(3/2 - 1)}{(+15/2)} \Rightarrow v = +15\ cm$

Thus, the image formed is real.

Q **There are two objects O_1 and O_2 at an identical distance of 20 cm on the two sides of the pole of a spherical concave refracting boundary of radius 60 cm. The indices of refraction of the media on two sides of the boundary are 1 and $\left(\frac{4}{3}\right)$ respectively. Find the location of the object**

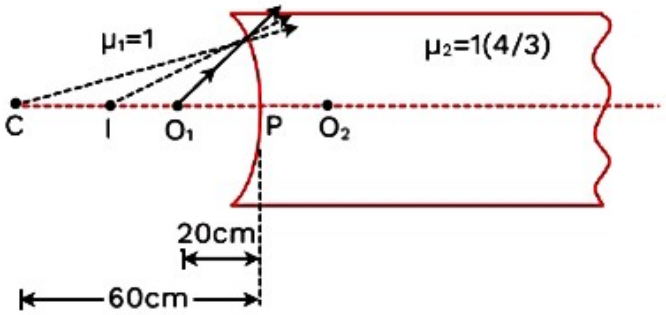
(a) O_1 when seen from O_2
(b) O_2 when seen from O_1

Sol: The formula for refraction from a curved boundary is

$$\frac{\mu_2}{v} - \frac{\mu_1}{u} = \frac{\mu_2 - \mu_1}{R}$$

(a) From the ray diagram drawn, we get

$u_1 = -20\,cm$, $R = -60\,cm, \mu_1 = 1, \mu_2 = \frac{4}{3}$.



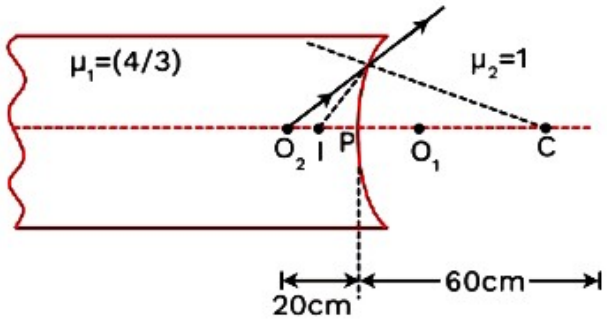
$$\Rightarrow \quad \frac{\left(\frac{4}{3}\right)}{v} - \frac{1}{(-20)} = \frac{\left(\frac{4}{3}\right) - 1}{(-60)}$$

$\Rightarrow$ $v = -24$ cm

Thus, the object O_1 will appear at a distance of 24 cm from P towards C.

(b) Keeping the object O_2 on the left of the pole P as shown, here, we get

$$u_1 = -20\,\text{cm}, R = +60\,\text{cm}, \mu_1 = \frac{4}{3}, \mu_2 = 1$$



$$\Rightarrow \quad \frac{1}{v} - \frac{\left(\frac{4}{3}\right)}{(-20)} = \frac{1 - \left(\frac{4}{3}\right)}{60} \Rightarrow v = -\frac{180}{13} = -13.84$$

Thus, the object O_2 will appear at a distance of 13.84 cm from P towards O_2.